11/29/2025

COMPLETED PROJECT DOCUMENTATION “Cowabunga Pizza”

This project implements a centralized Windows Server architecture for The Cowabunga Pizza Project, including Active Directory Domain Services, GPO enforcement, and structured access controls. The goal is to deliver a secure, scalable infrastructure that improves authentication workflows, hardens data security, and simplifies network administration.

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About

- **Company Background**
  The Cowabunga Pizza Project is an expanding organization that works in both creative and operational areas, including marketing, design, kitchen production, and administration. As the company grows, the number of users, devices, and shared resources also increases. This creates a need for a more structured IT environment where user accounts, permissions, and data can be managed securely and from one central place.
- **Current Situation**
  As the company expands, managing users, securing internal data, and maintaining consistency across multiple workstations has become increasingly challenging. The Cowabunga Pizza Project requires a centralized Windows Server environment to streamline authentication, enforce security policies, and ensure smooth communication across creative and operational teams.
- **The Mission**
  Deploy a complete Windows Server infrastructure including Active Directory, Group Policies, and access controls, to provide a secure, scalable, and efficient environment that supports both daily operations and future growth.

Diagram Section

Network Diagram

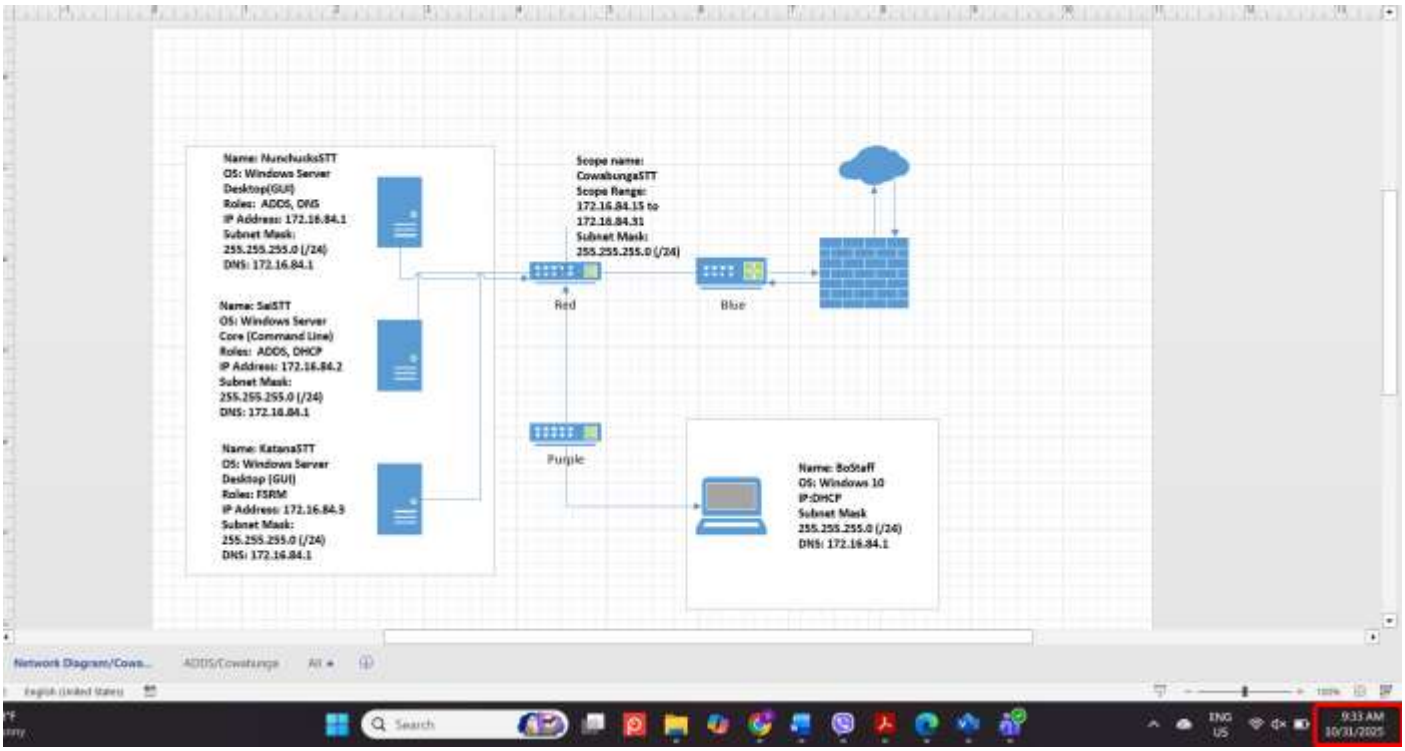


Figure 1 of 3 Network Diagram

ADDS Diagram

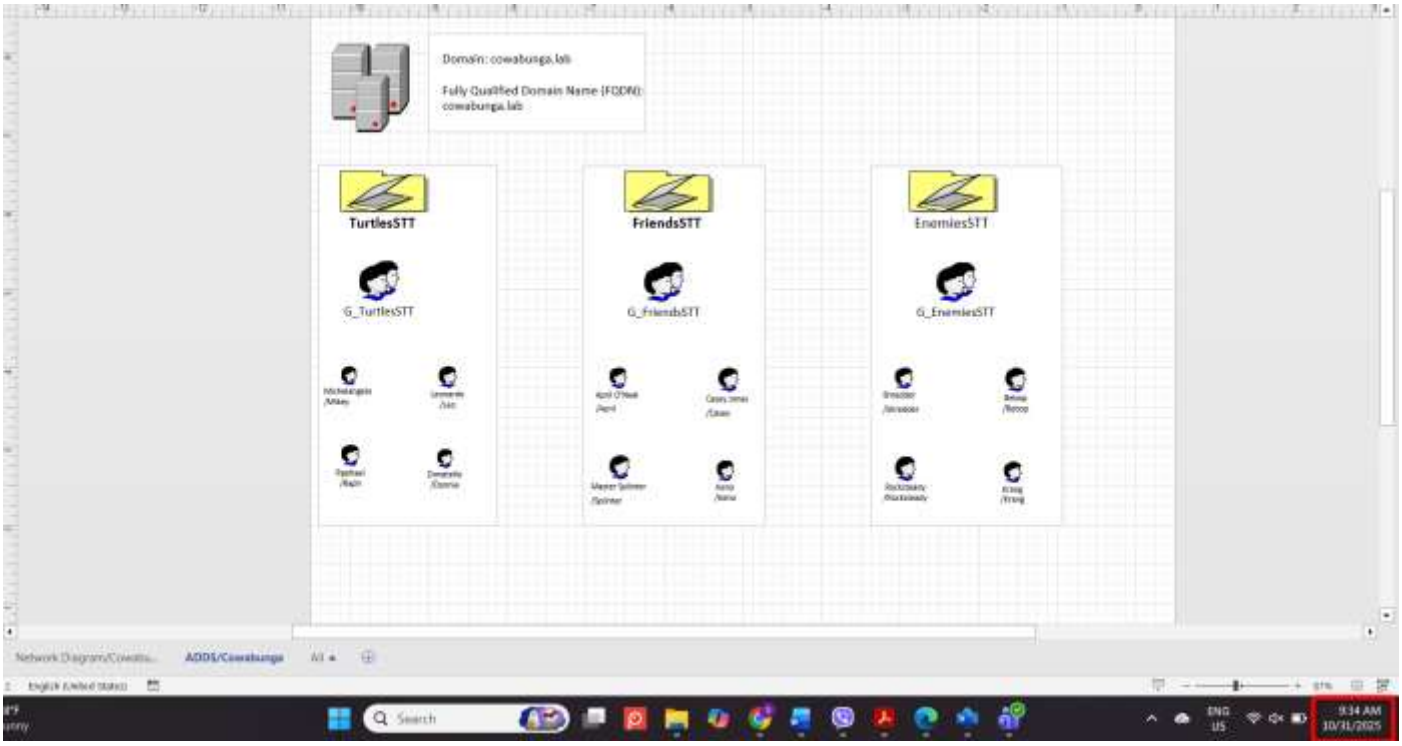


Figure 2 of 3 ADDS Diagram

Shared Folders Diagram

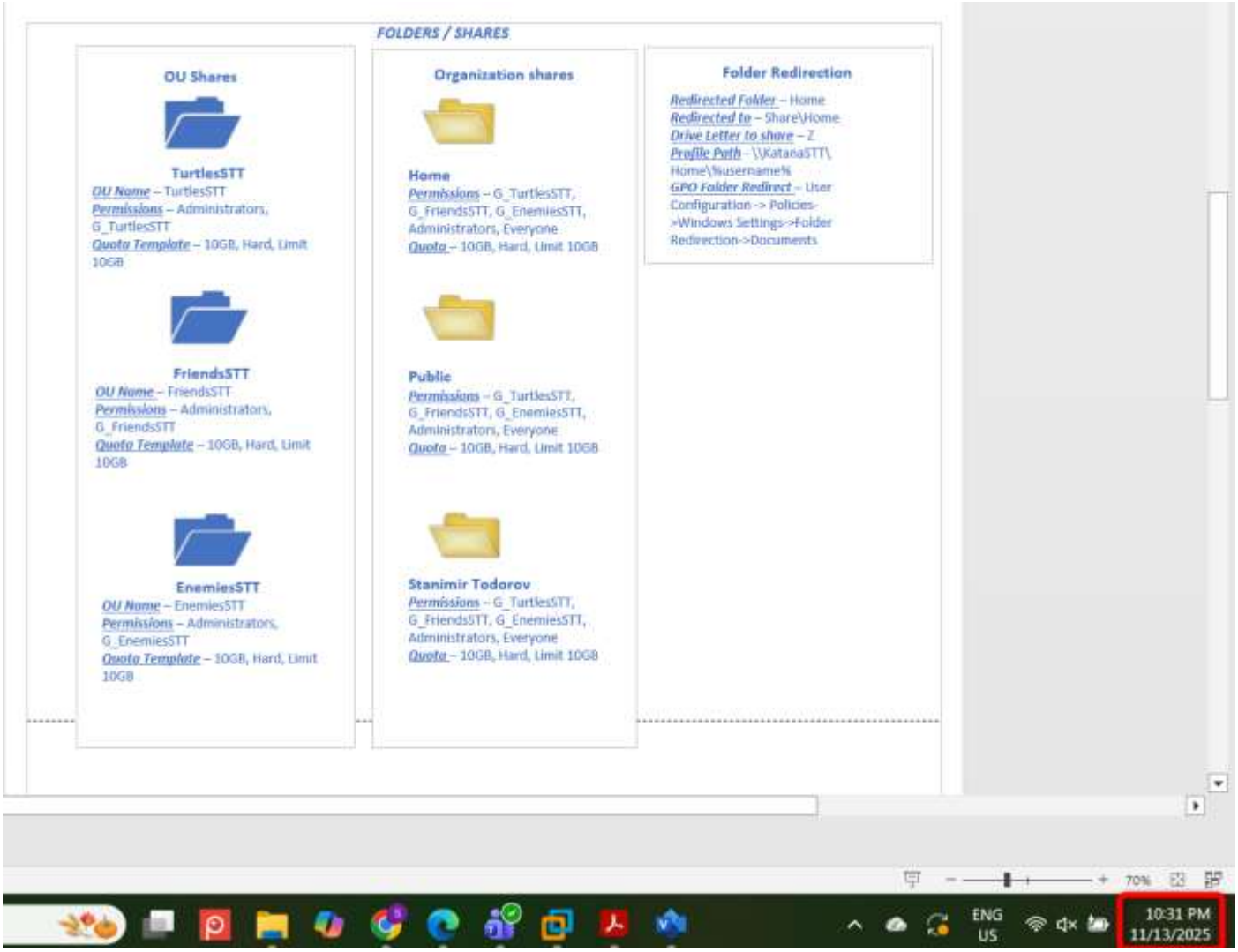
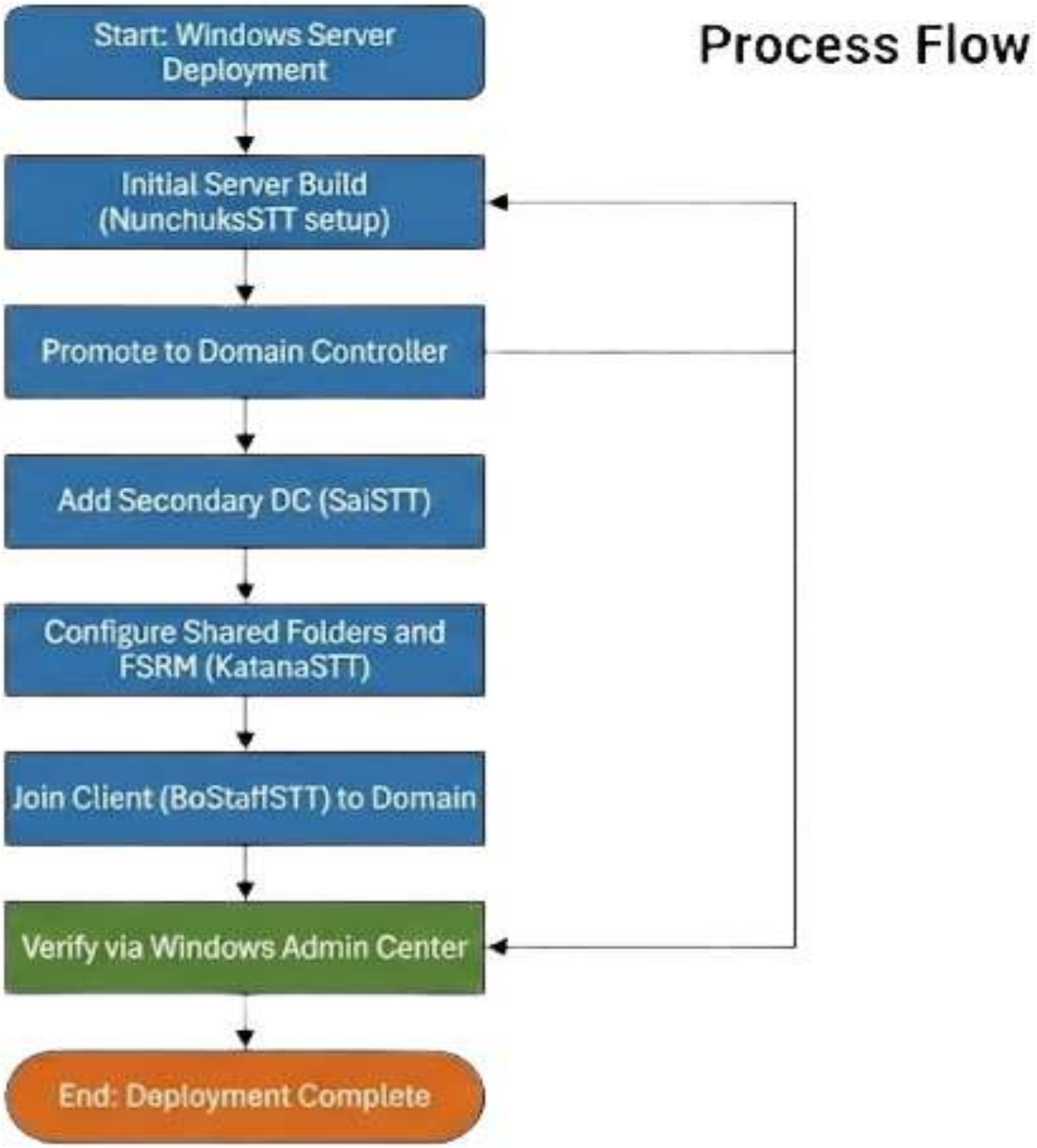


Figure 3 of 3 Shared Folders Diagram

Process Diagram/Flow Chart



Build Section

Initial Build

NunchuksSTT - Server 1

Setting up the server as follows:

1. **Install VMware Tools** (normal method)

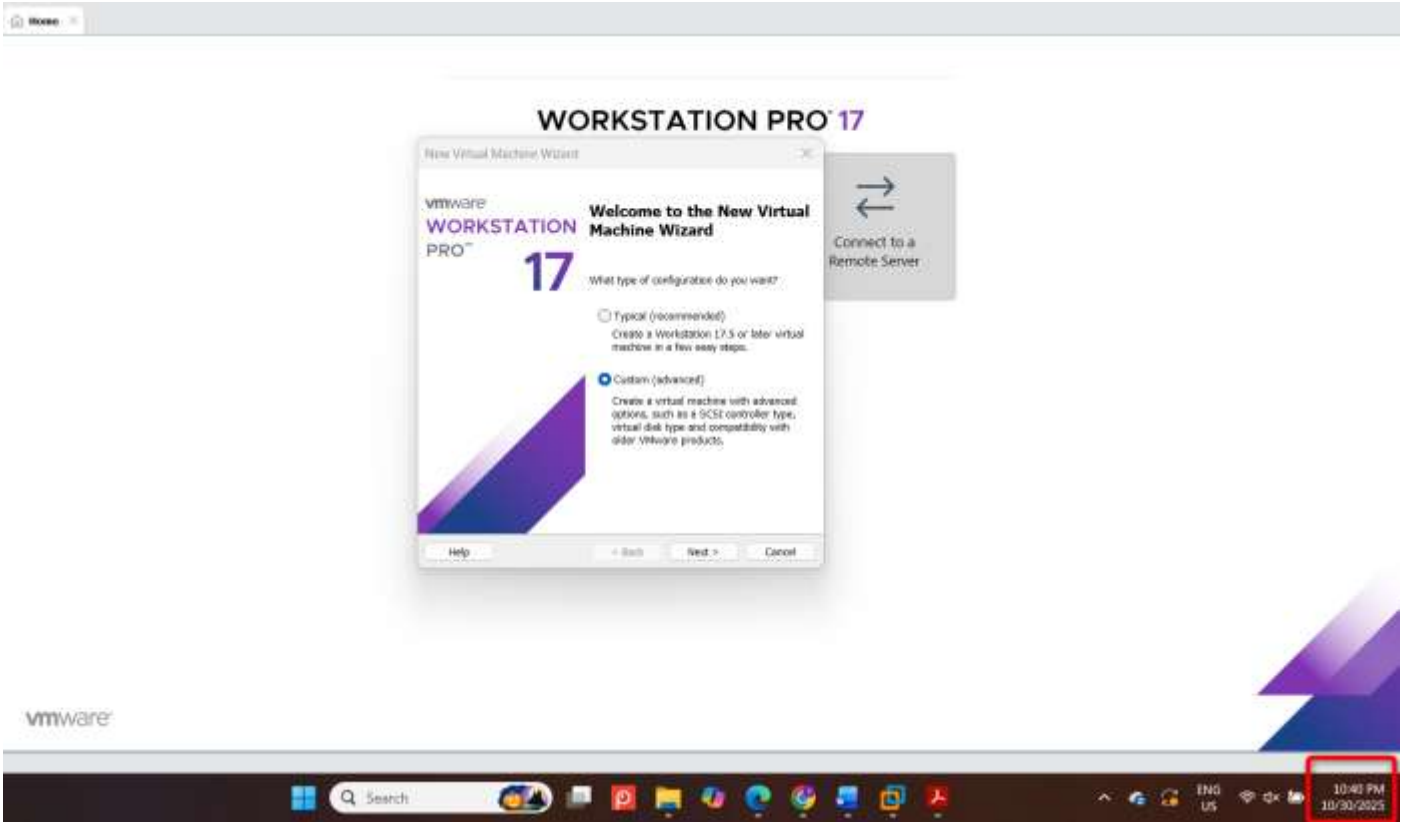
   2 GB (2048MB) RAM
   2CPU's, 2 Cores
   Hard Drive - 60GB
   Admin Password: Password 01

2. **Set Time Zone:**
   - Set-Timezone -ID "Eastern Standard Time"
3. **Rename Server:**
   - Rename-Computer -NewName "NunchuksSTT"
4. **Prevent Screen from Turning Off:**
   Use the PowerCFG tool:
   - Get power scheme:
   - PowerCFG /L
   - Query power settings and modify timeout values to 0:
   - PowerCFG /change monitor-timeout-ac 0
   - PowerCFG /change monitor-timeout-dc 0
5. **Assign Static IP & DNS:**
6. Get-NetAdapter
7. Set-NetIPInterface -InterfaceAlias "Ethernet0" -DHCP Disabled -Passthru
8. New-NetIPAddress -AddressFamily IPv4 -InterfaceAlias "[Ethernet0]" -IPAddress [IP address] -PrefixLength [Subnet Length]

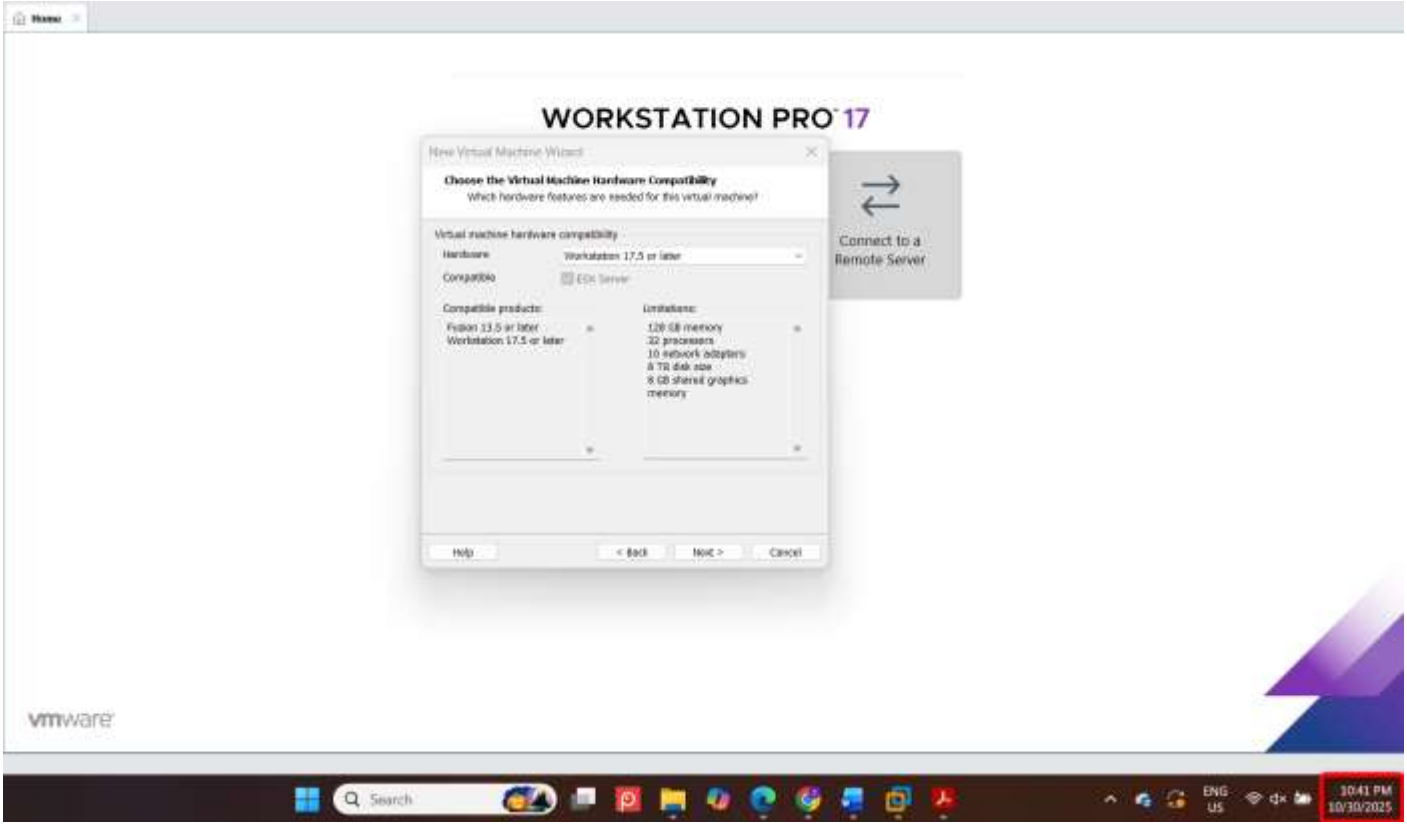
9. Set-DNSClientServerAddress -InterfaceAlias "[Ethernet0]" -ServerAddresses [Server 1 Address]
10. **Set UAC to Lowest:**
11. Set-ItemProperty -Path "HKLM:\SOFTWARE\Microsoft\Windows\CurrentVersion\Policies\System" -Name "EnableLUA" -Value 0
12. **Disable IE Enhanced Security Configuration:**
13. Set-ItemProperty -Path "HKLM:\SOFTWARE\Microsoft\Active Setup\Installed Components\{A509B1A8-...}" -Name "IsInstalled" -Value 0

Set-ItemProperty -Path "HKLM:\SOFTWARE\Microsoft\Active Setup\Installed Components\{A509B1A7-...}" -Name "IsInstalled" -Value 0

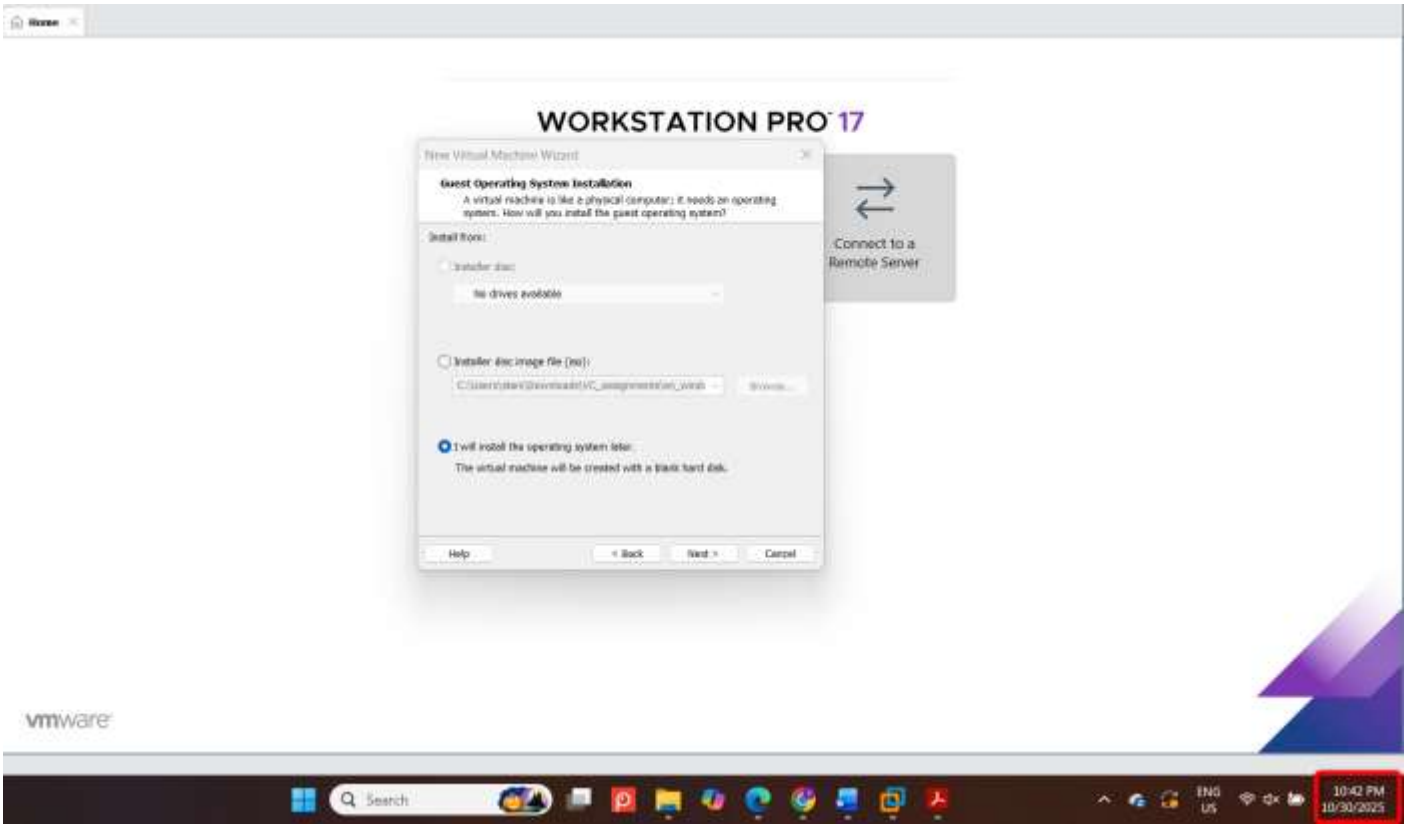
Install VMware Tools (normal method)



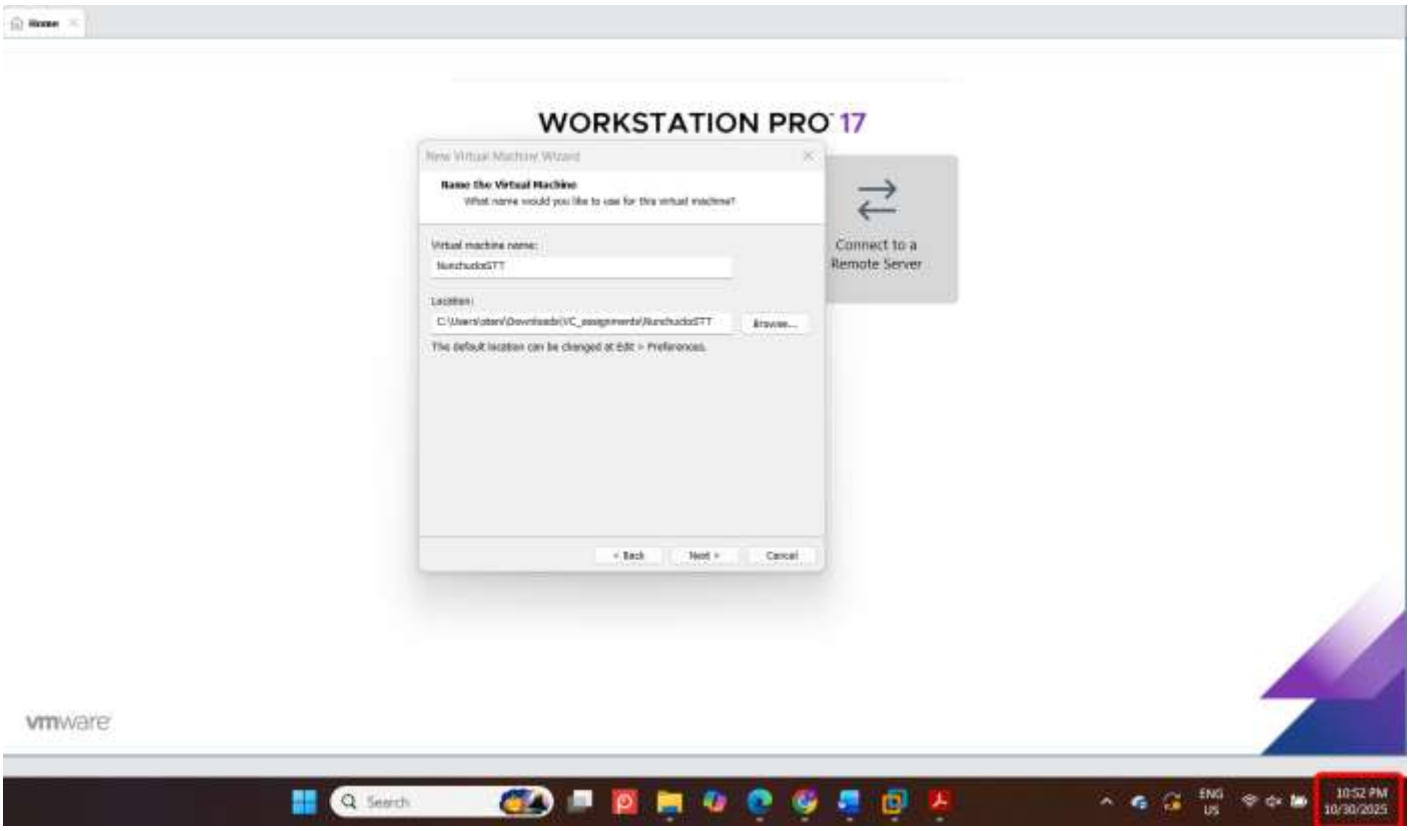
1. Install VMware Tools (normal method)



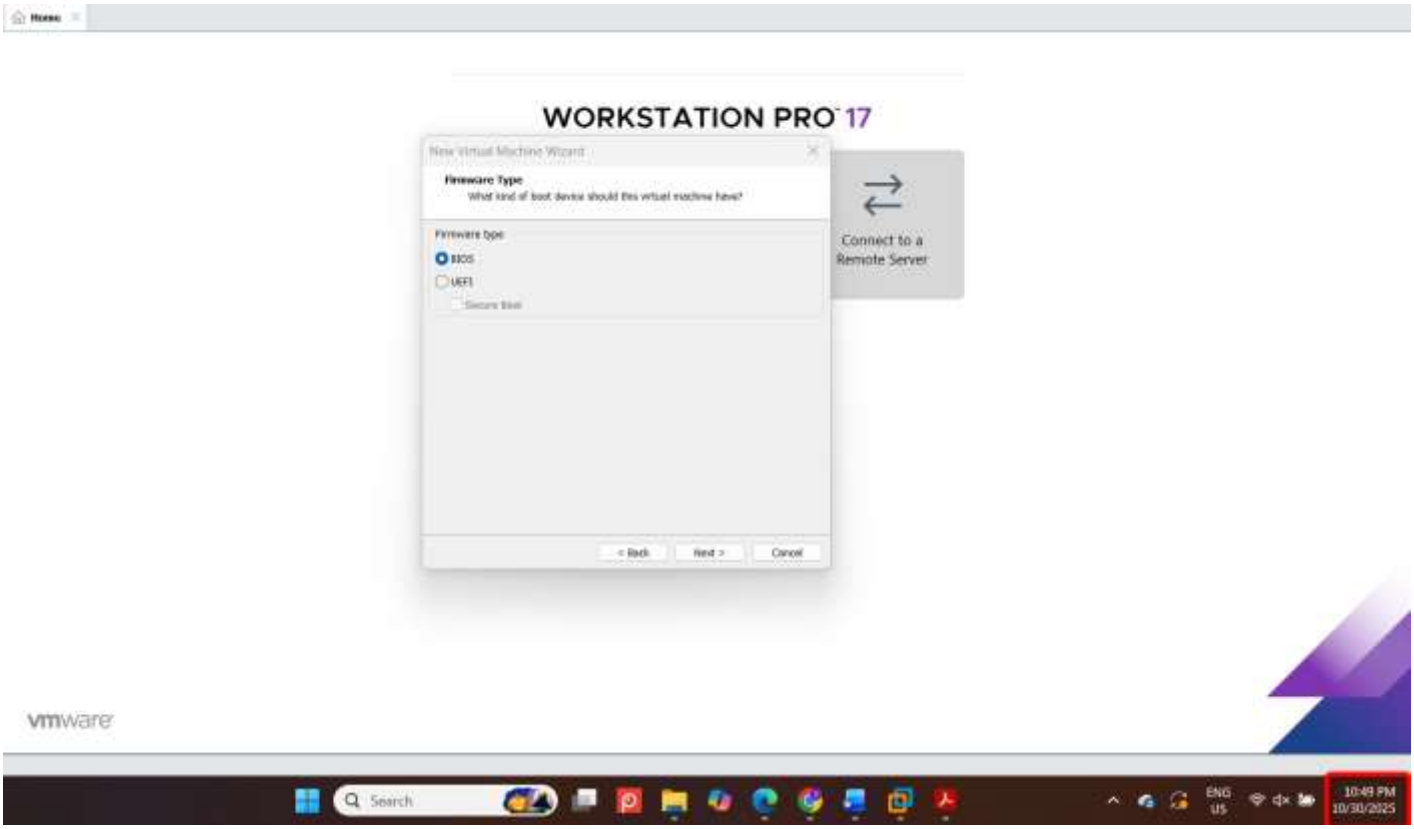
2. Install VMware Tools (normal method)



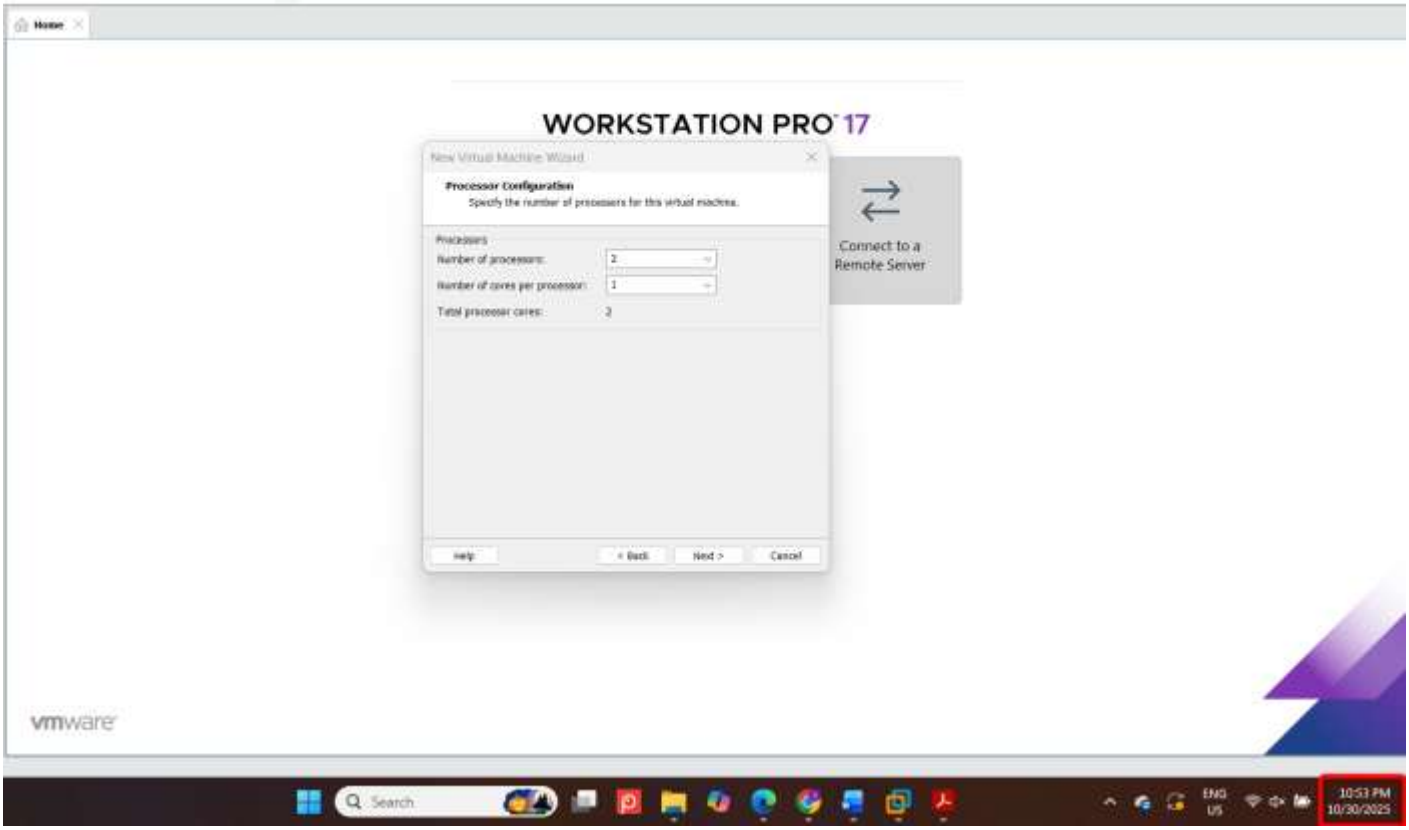
3.Install VMware Tools (normal method)



4. Install VMware Tools (normal method)



5. Install VMware Tools (normal method)



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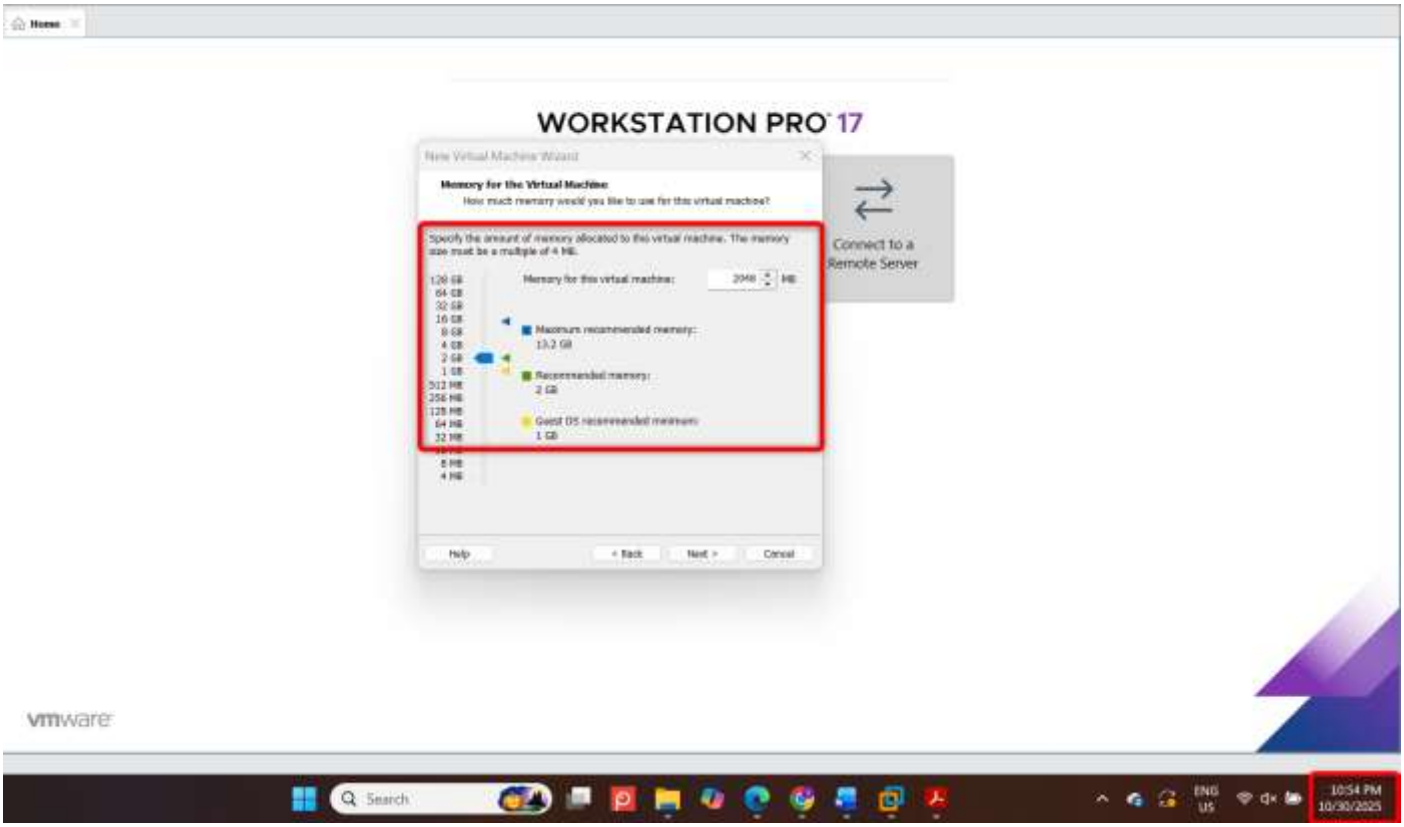
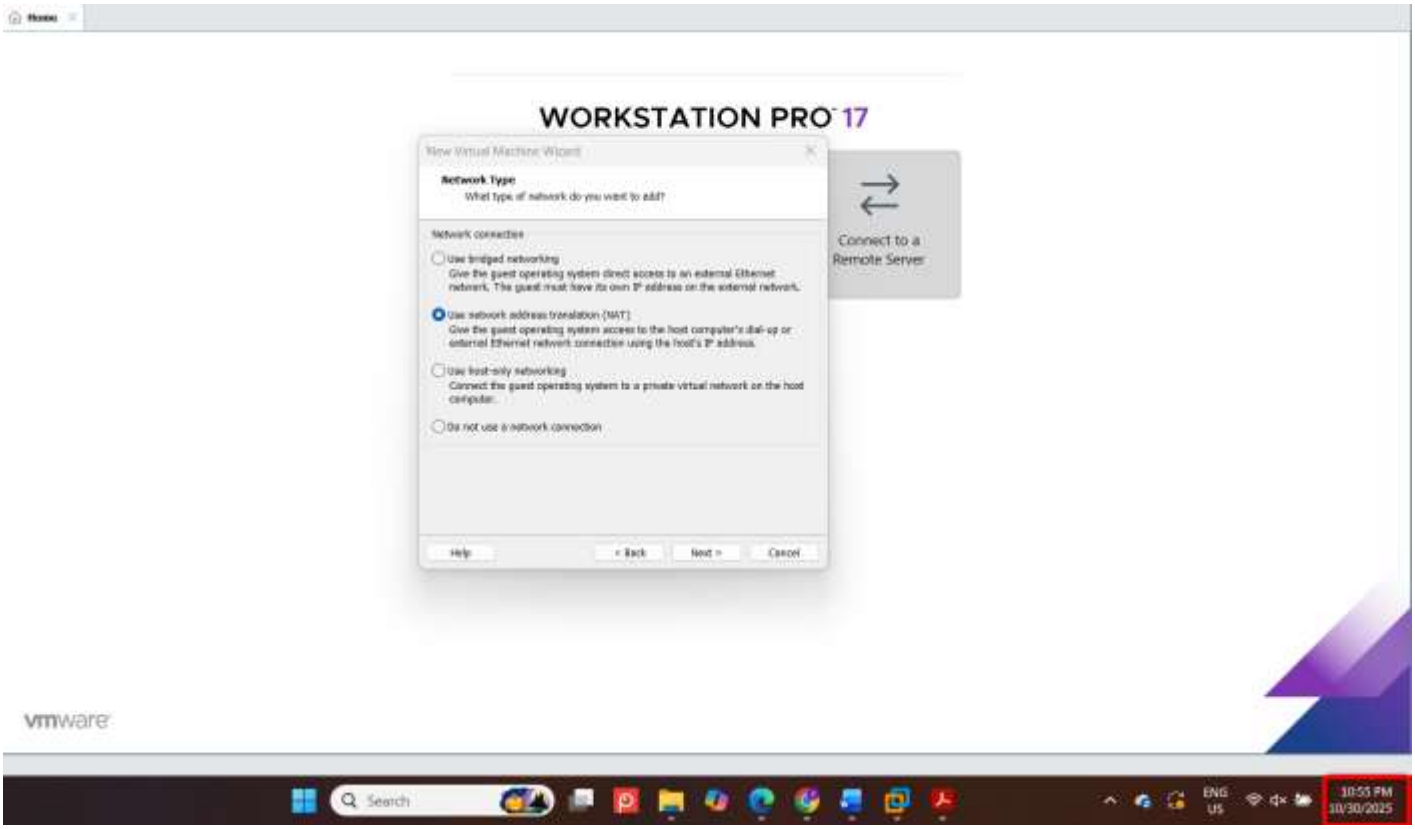
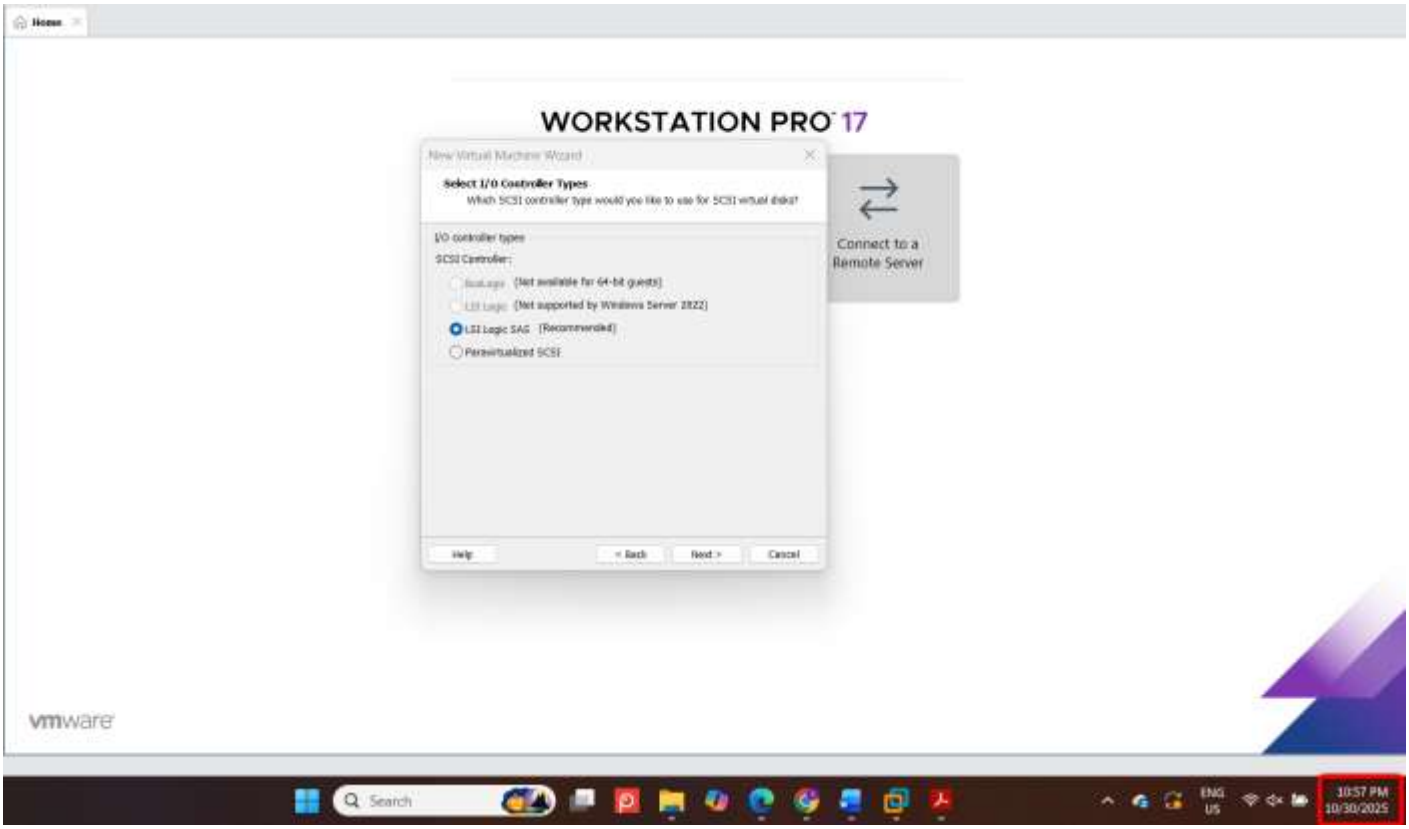


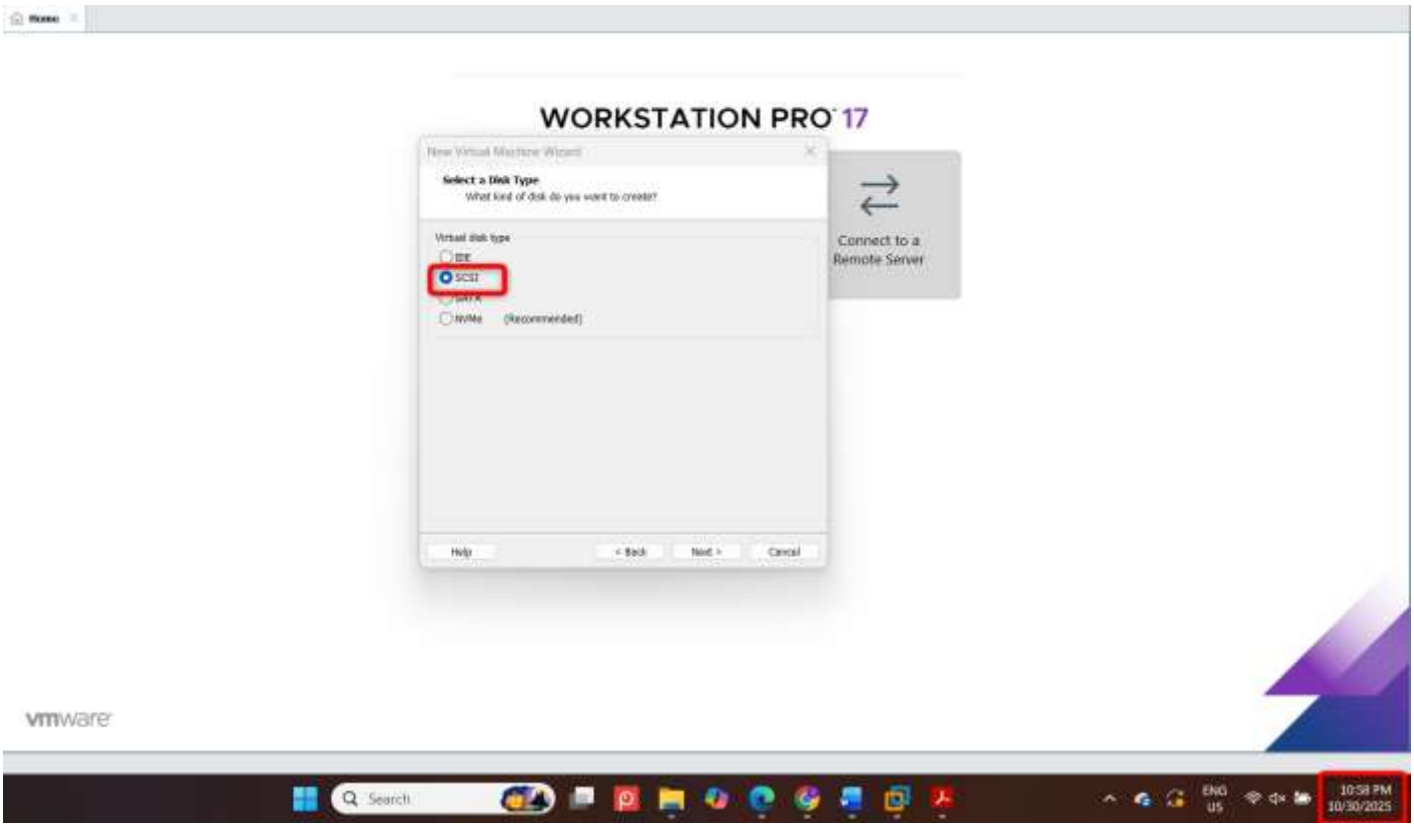
7. Install VMware Tools (normal method)



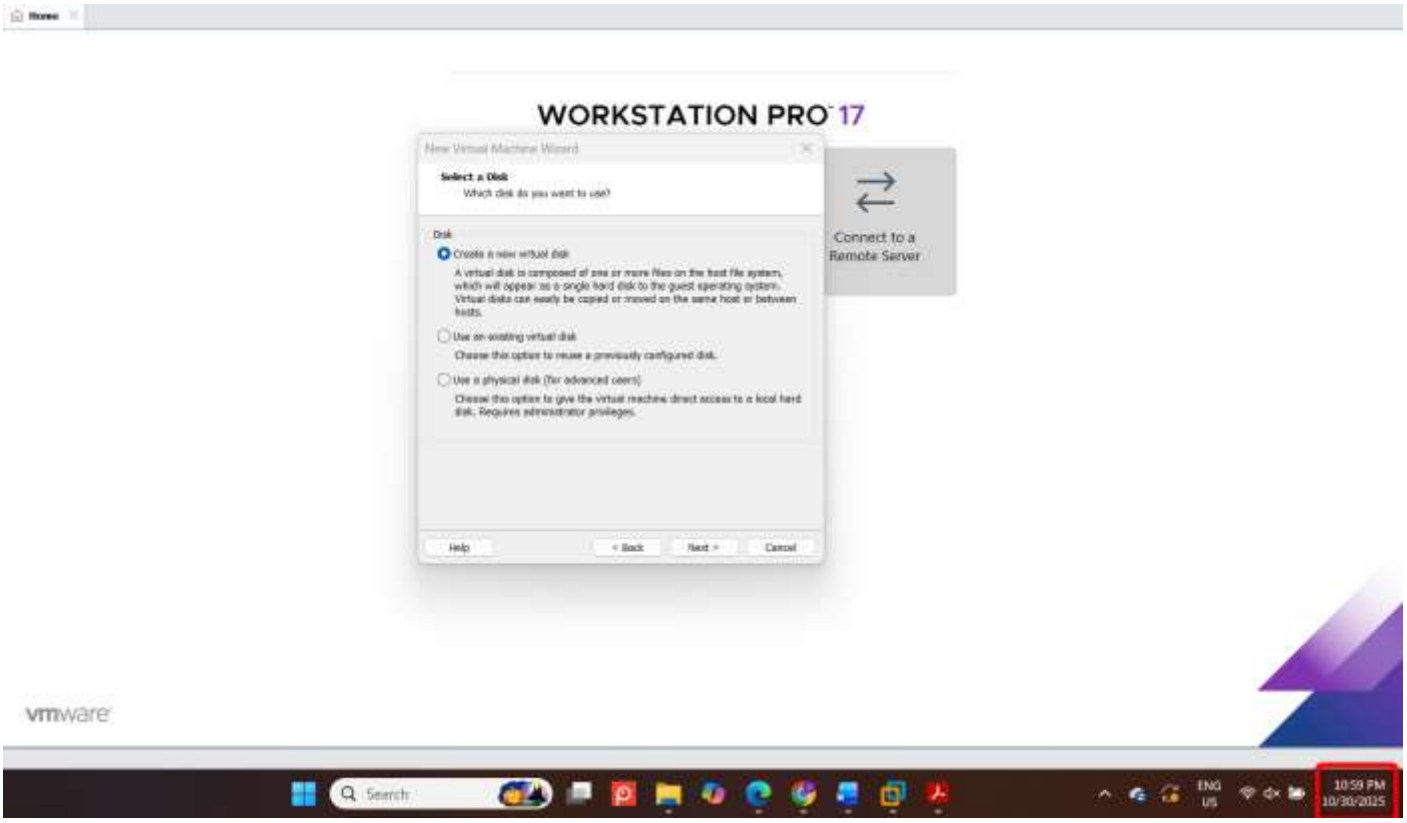
8. Install VMware Tools (normal method)



9. Install VMware Tools (normal method)



10. Install VMware Tools (normal method)



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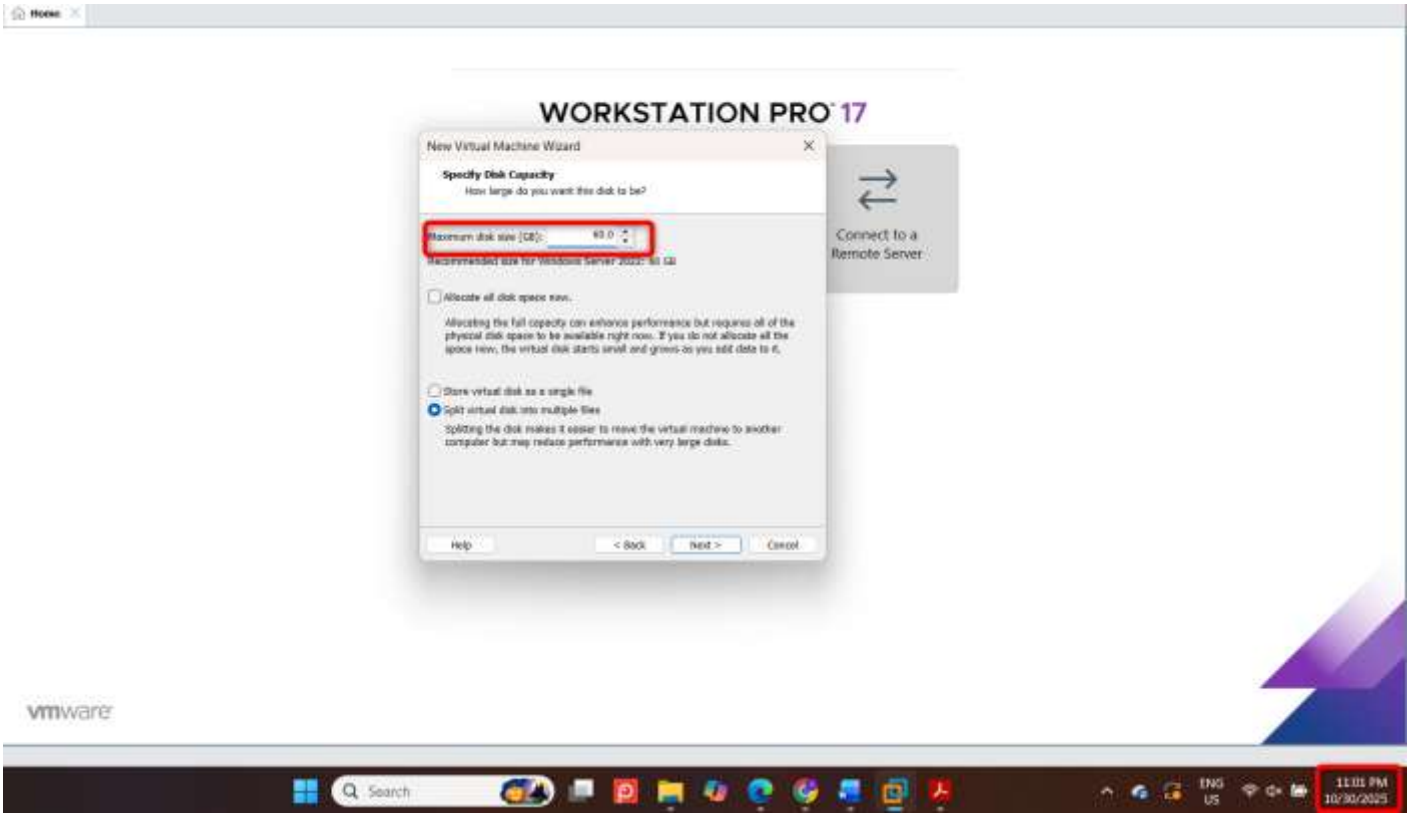
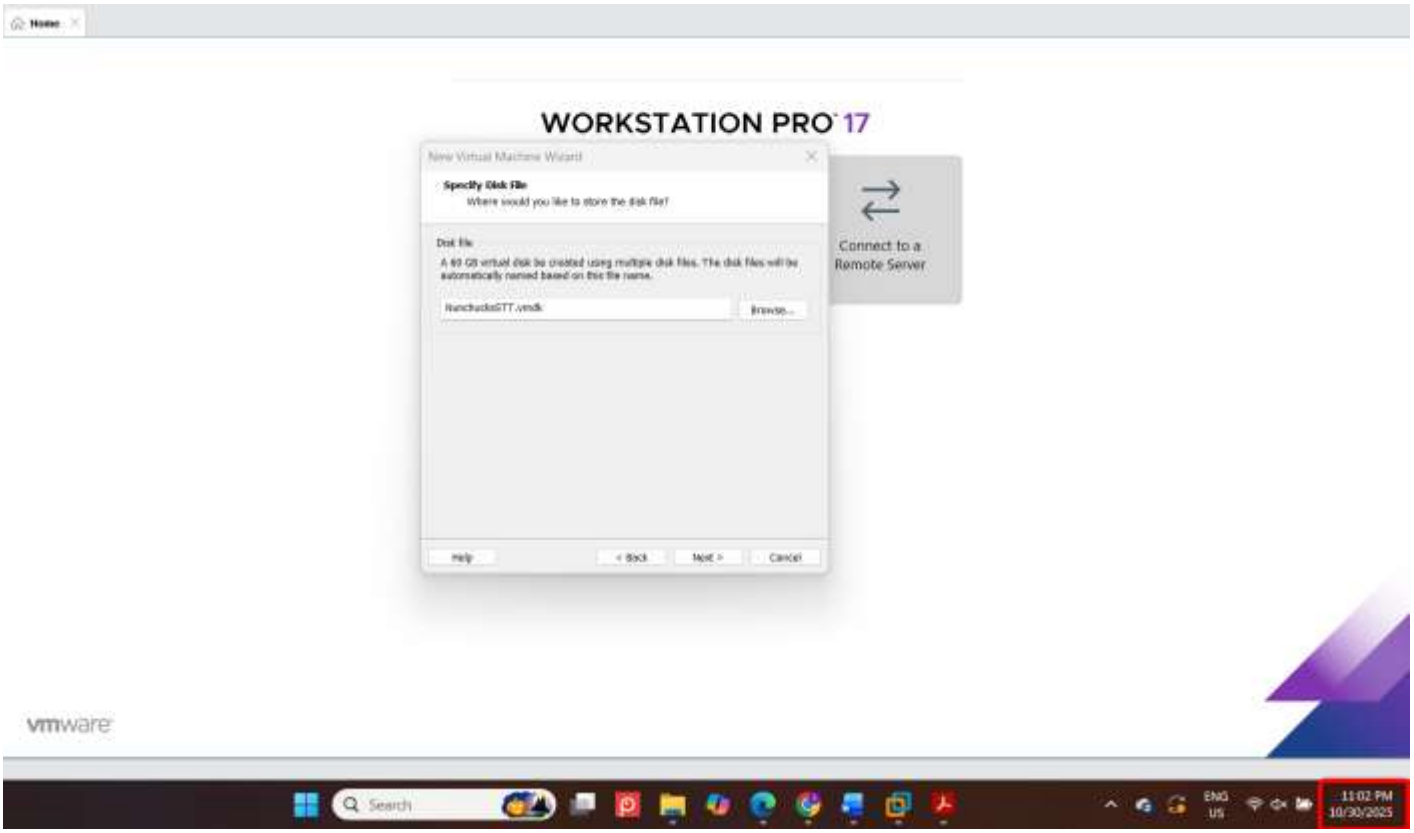
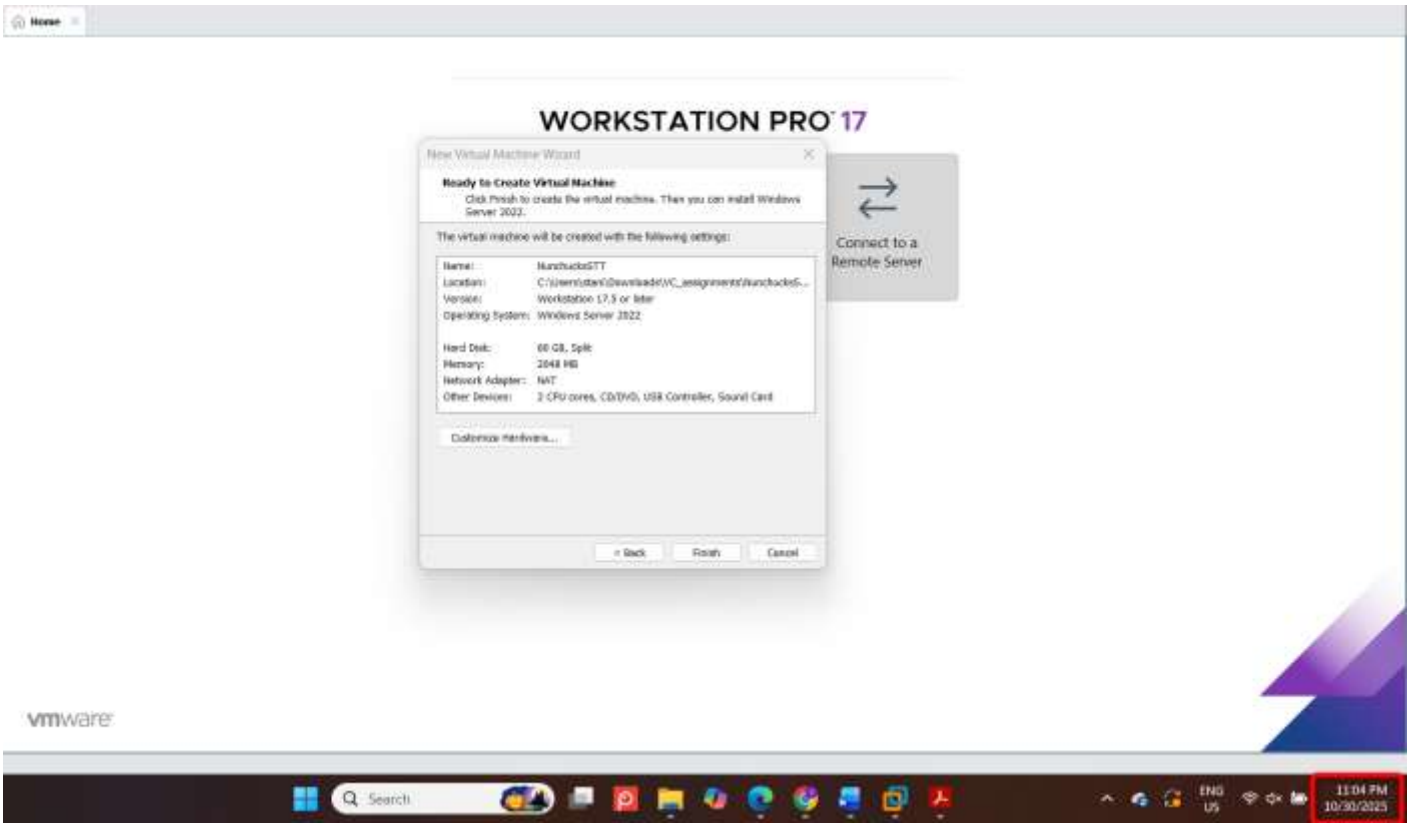


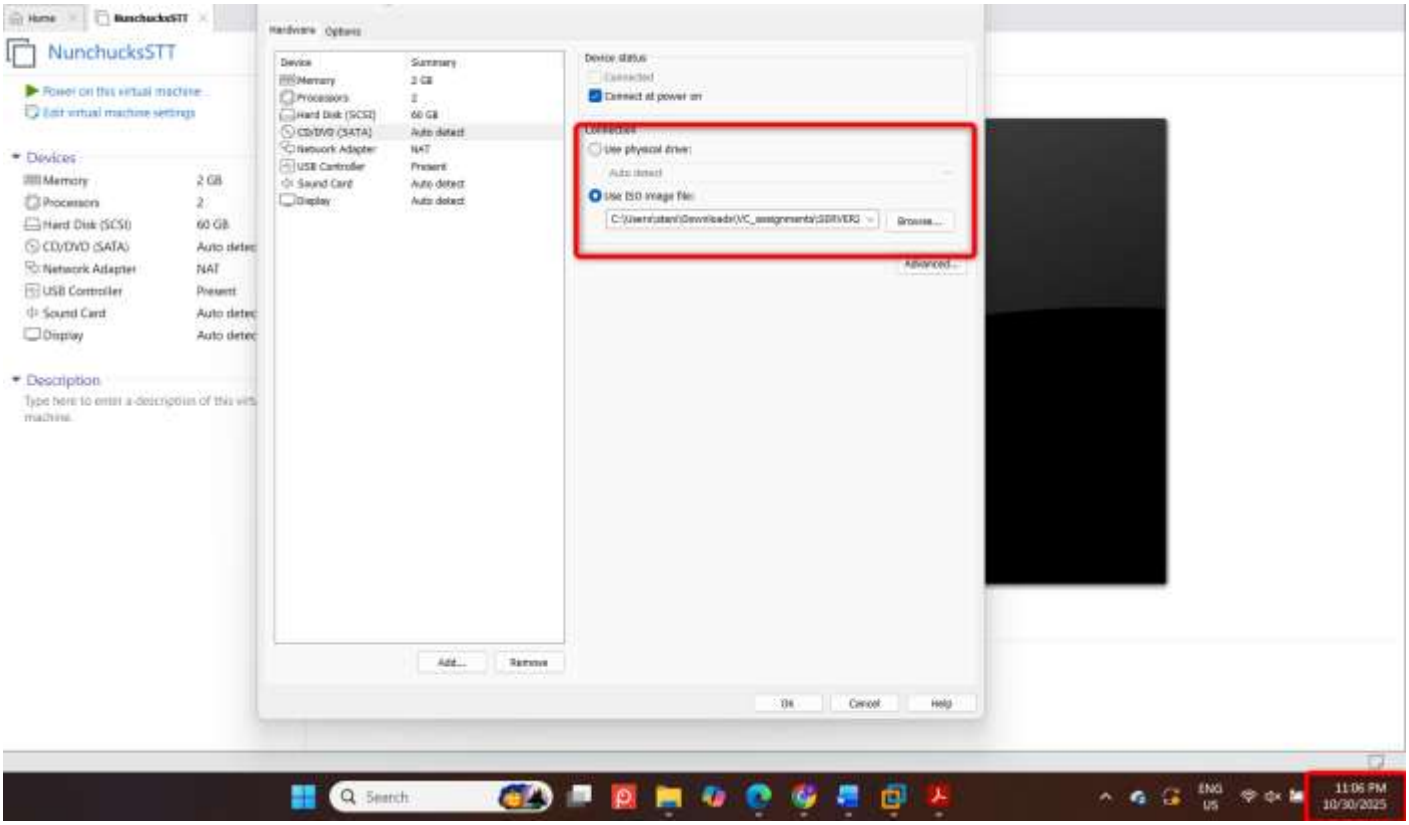
13. Install VMware Tools (normal method)



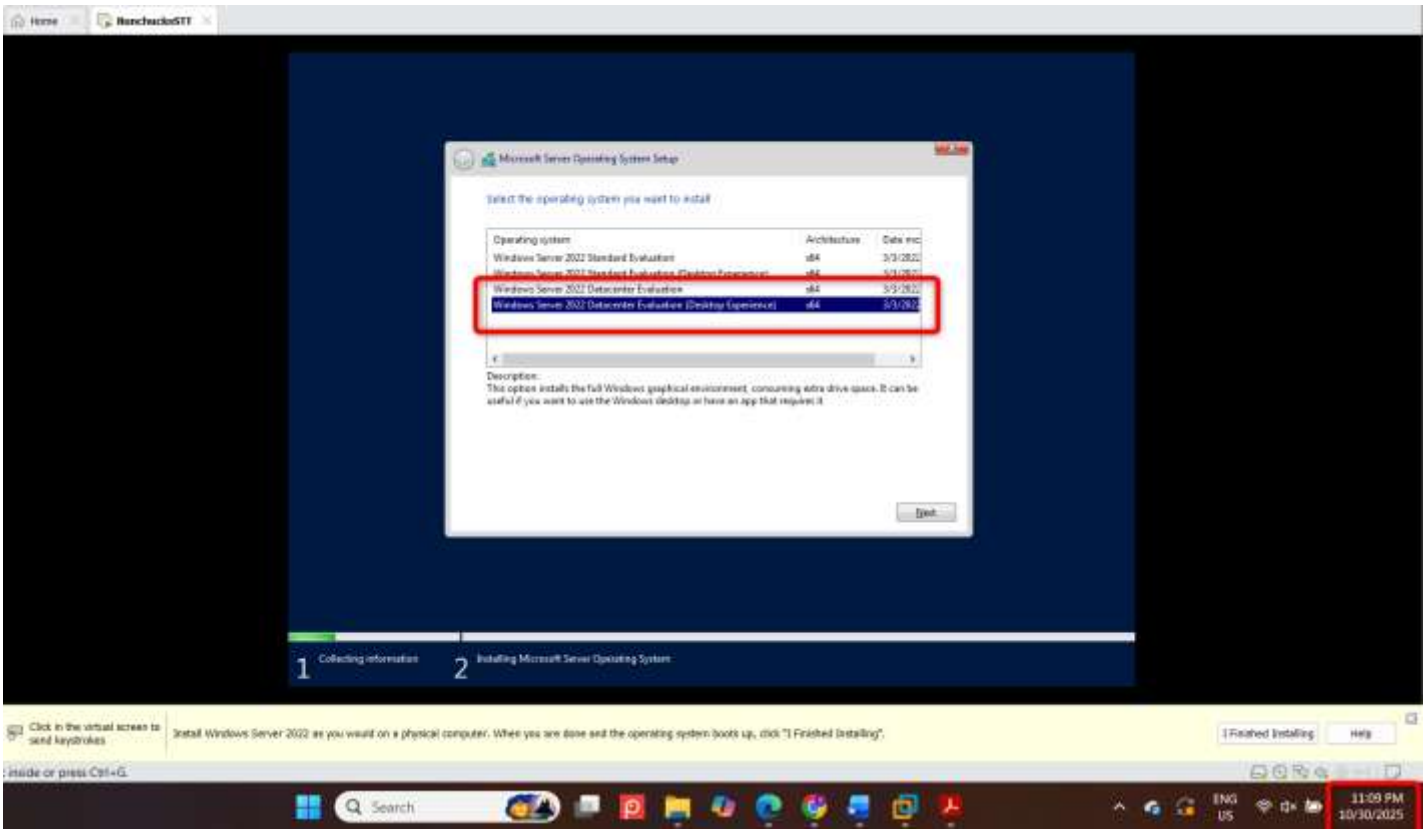
14. Install VMware Tools (normal method)



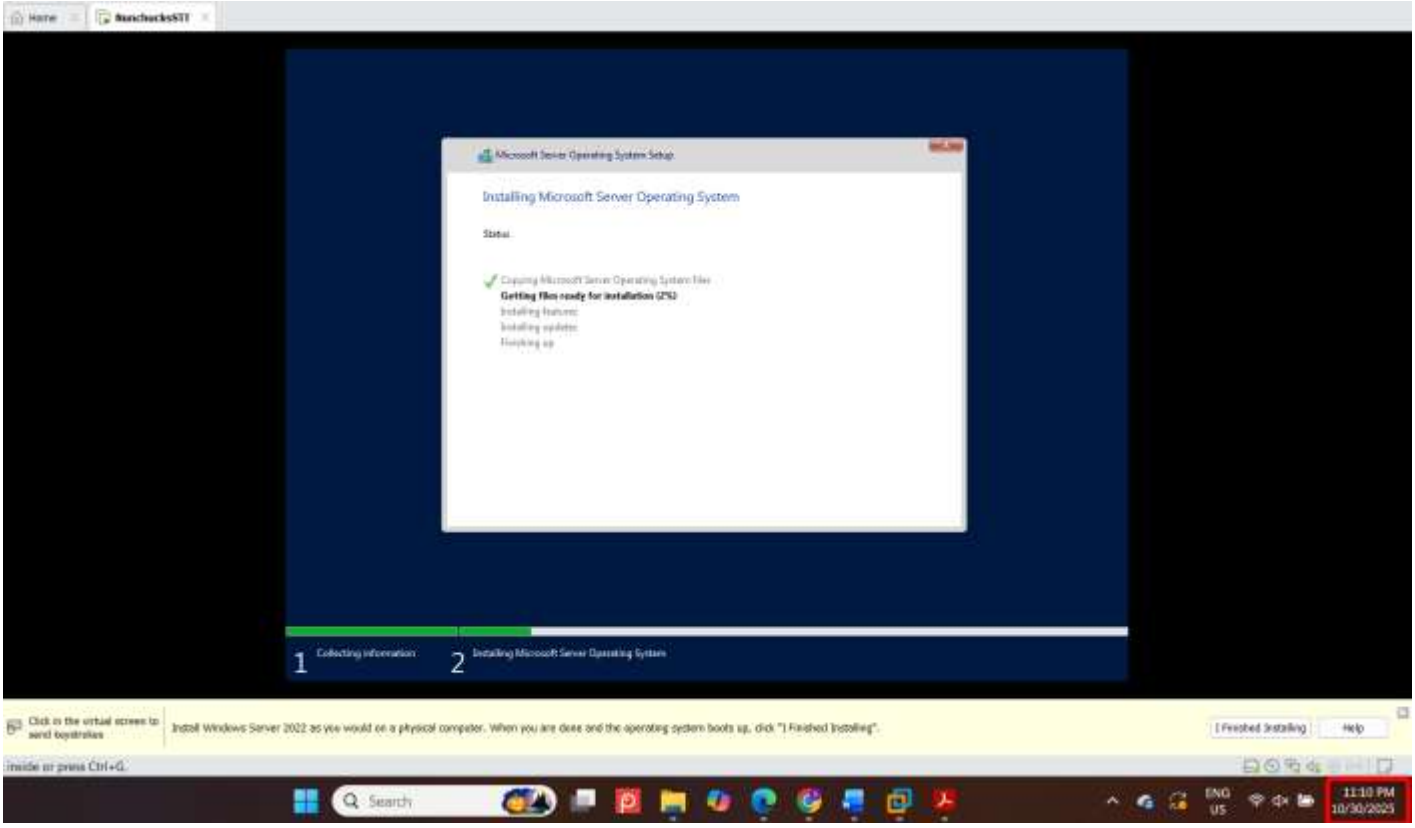
1. *Install VMware Tools (normal method)*



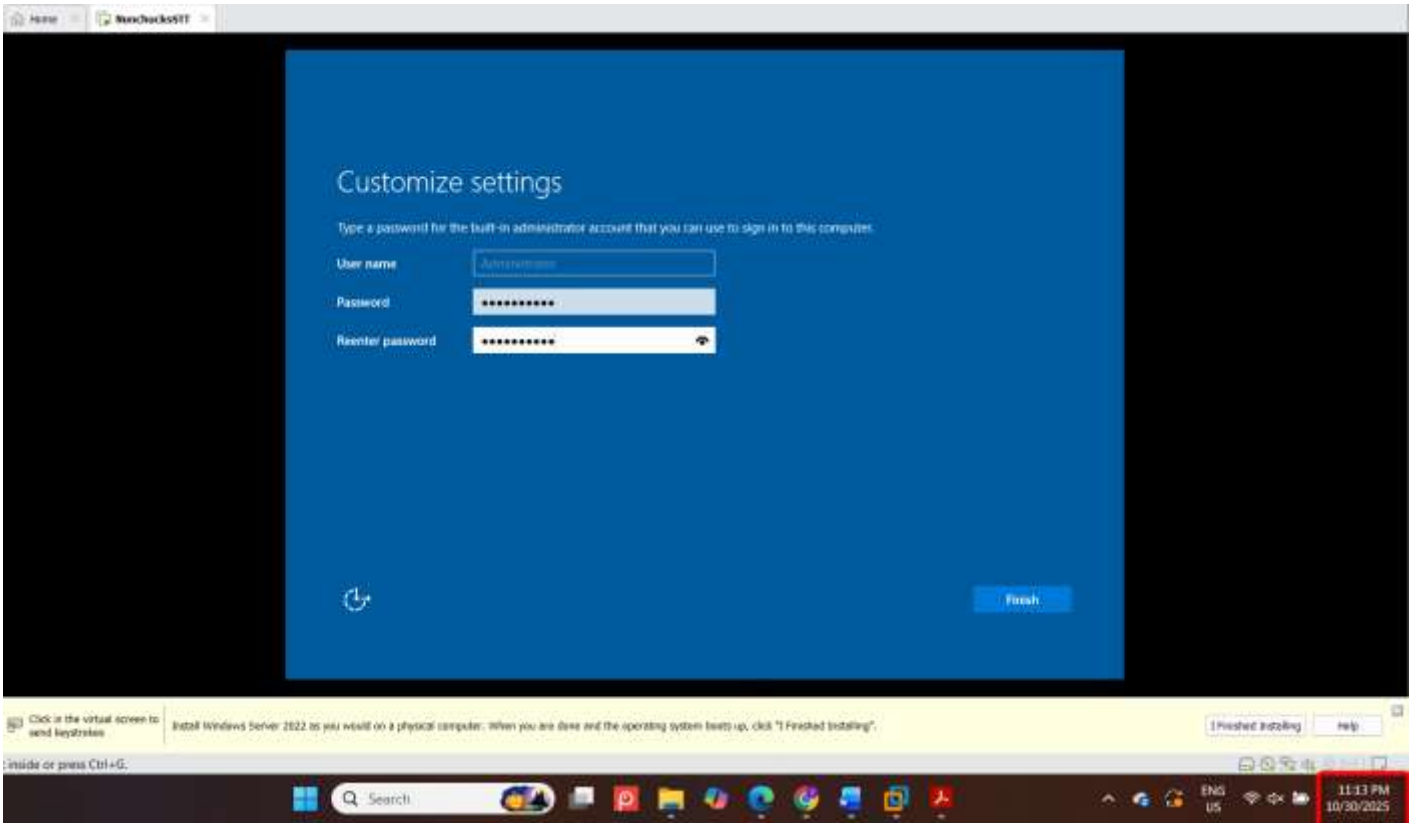
15. Install VMware Tools (normal method)



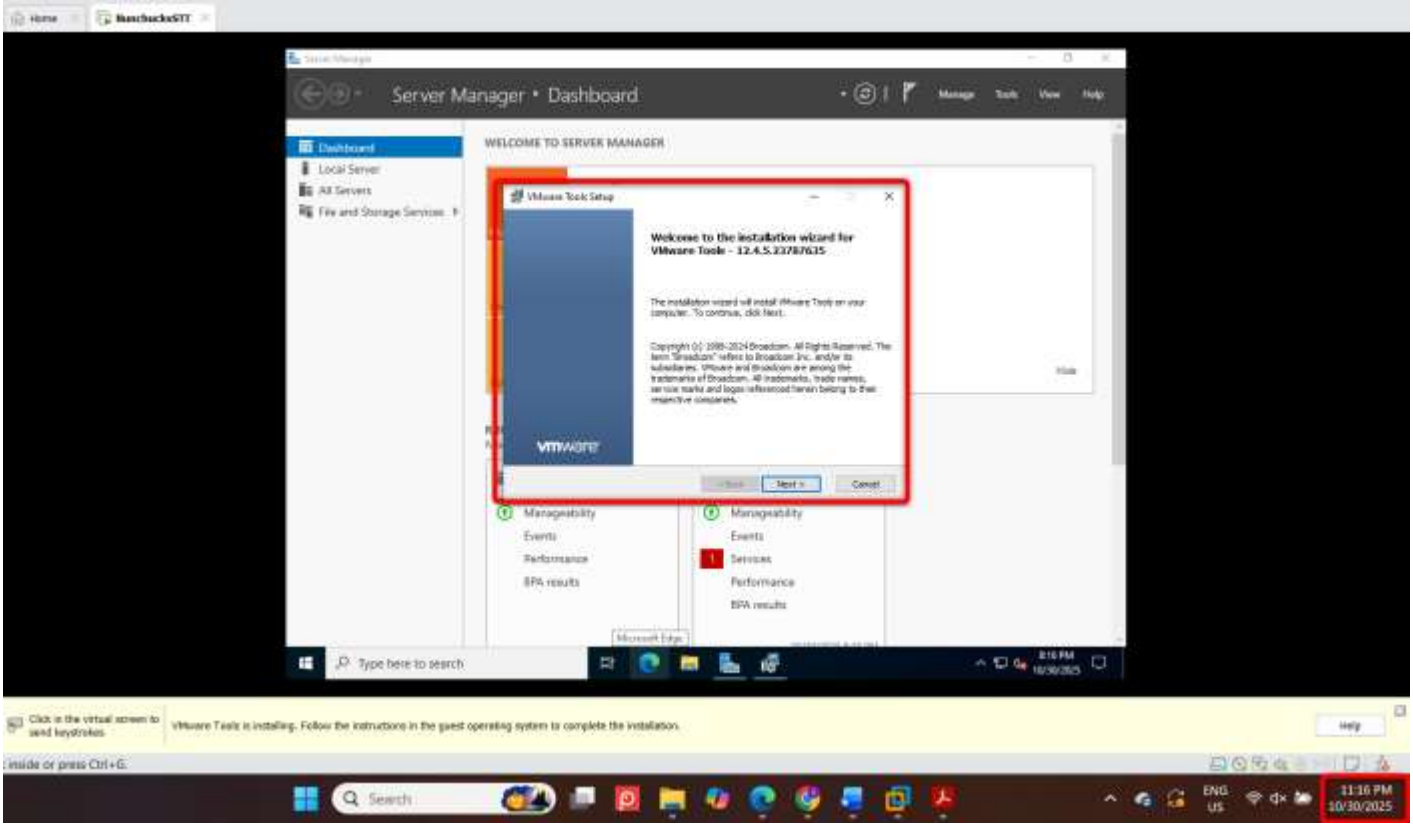
16. Install VMware Tools (normal method)



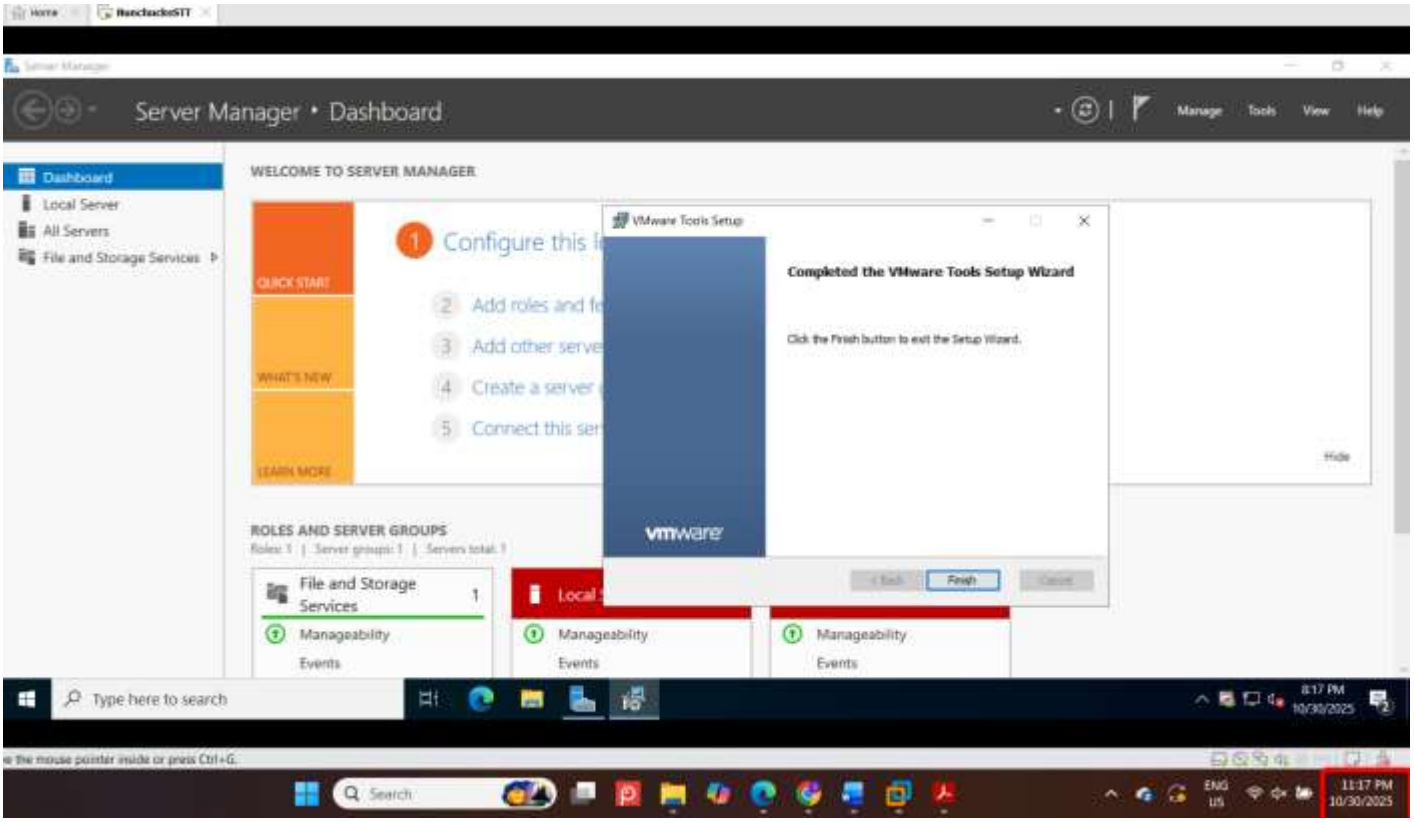
17. Install VMware Tools (normal method)



18. Install VMware Tools (normal method)

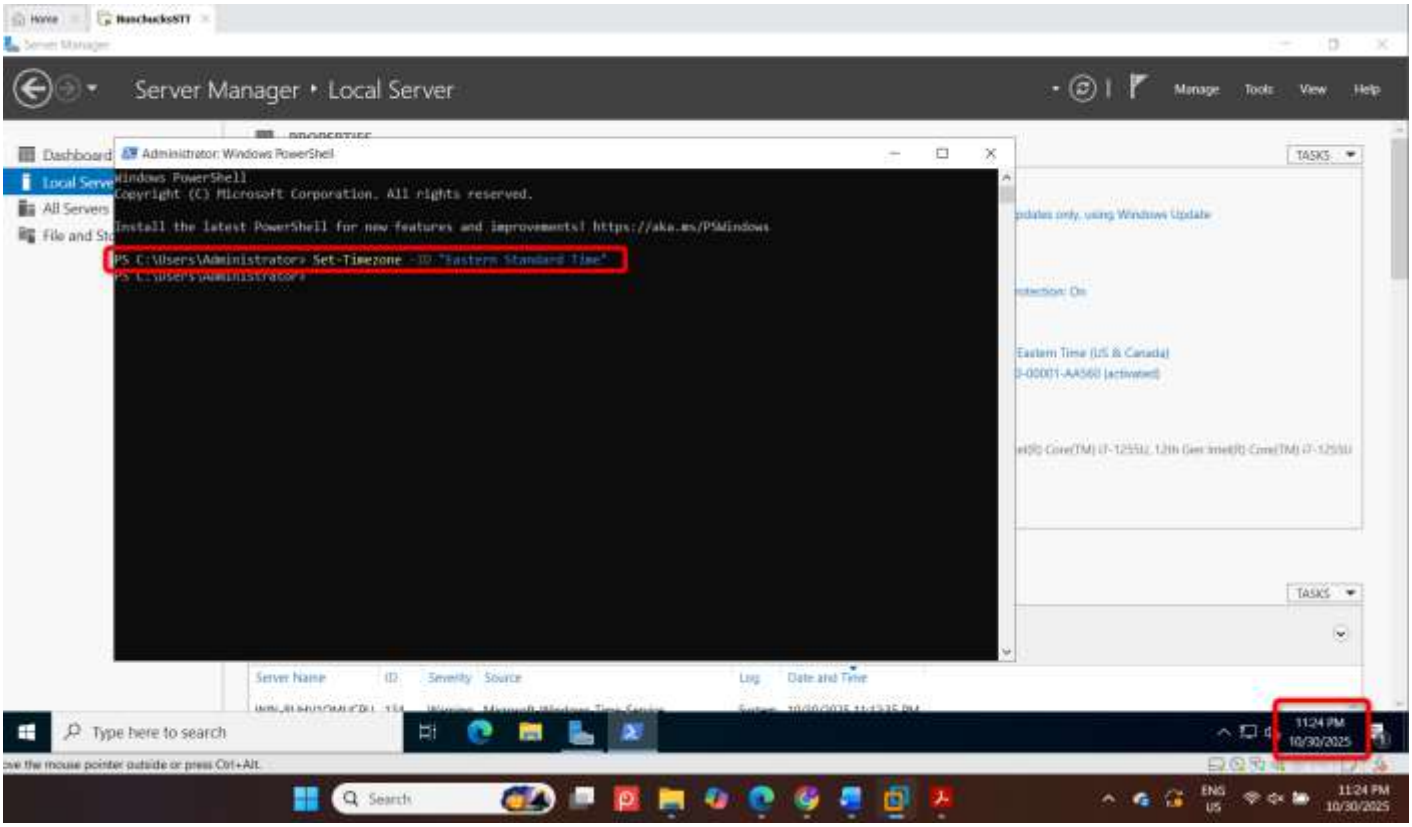


19. Install VMware Tools (normal method)



20. Install VMware Tools (normal method)

Set Time Zone



Rename Server

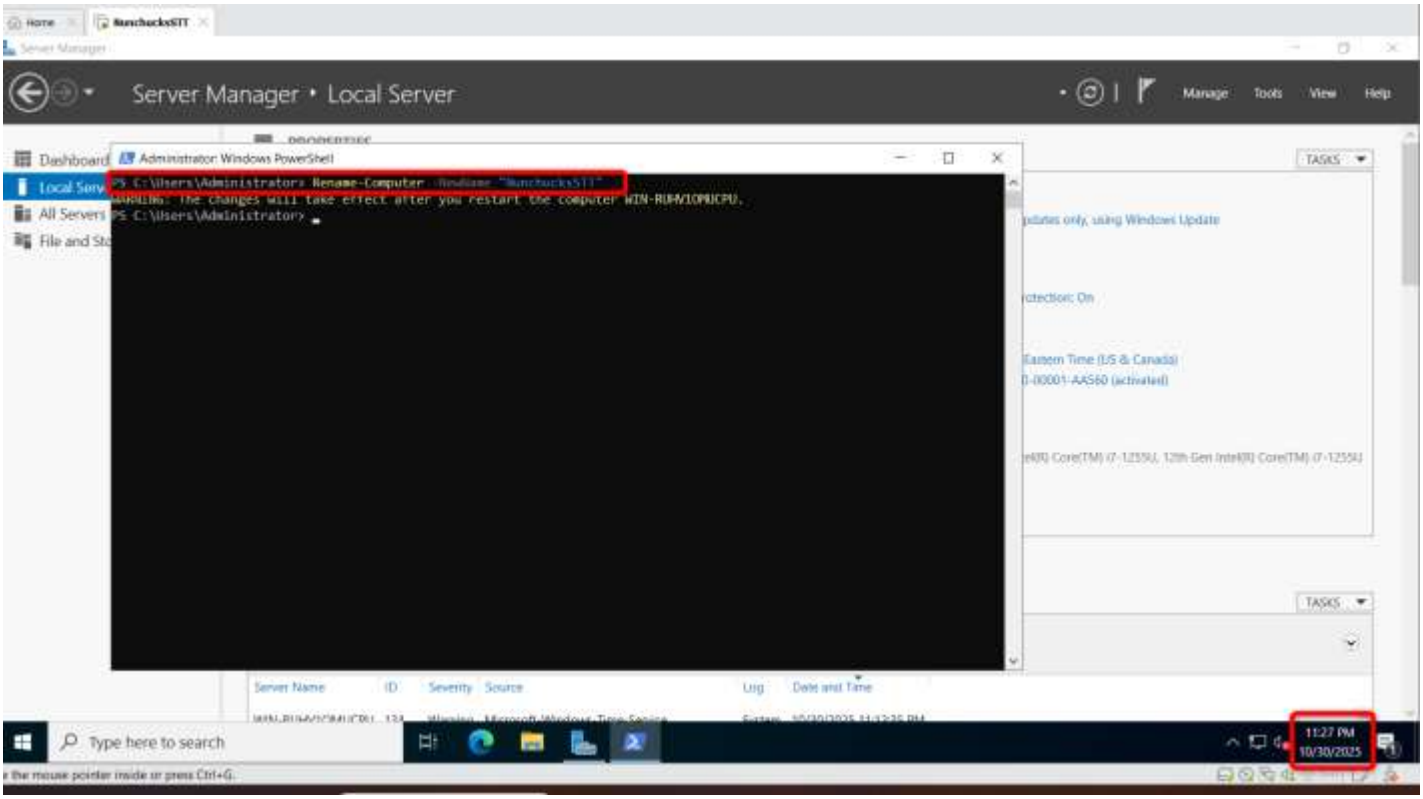


Figure 1 of 2 Rename Server

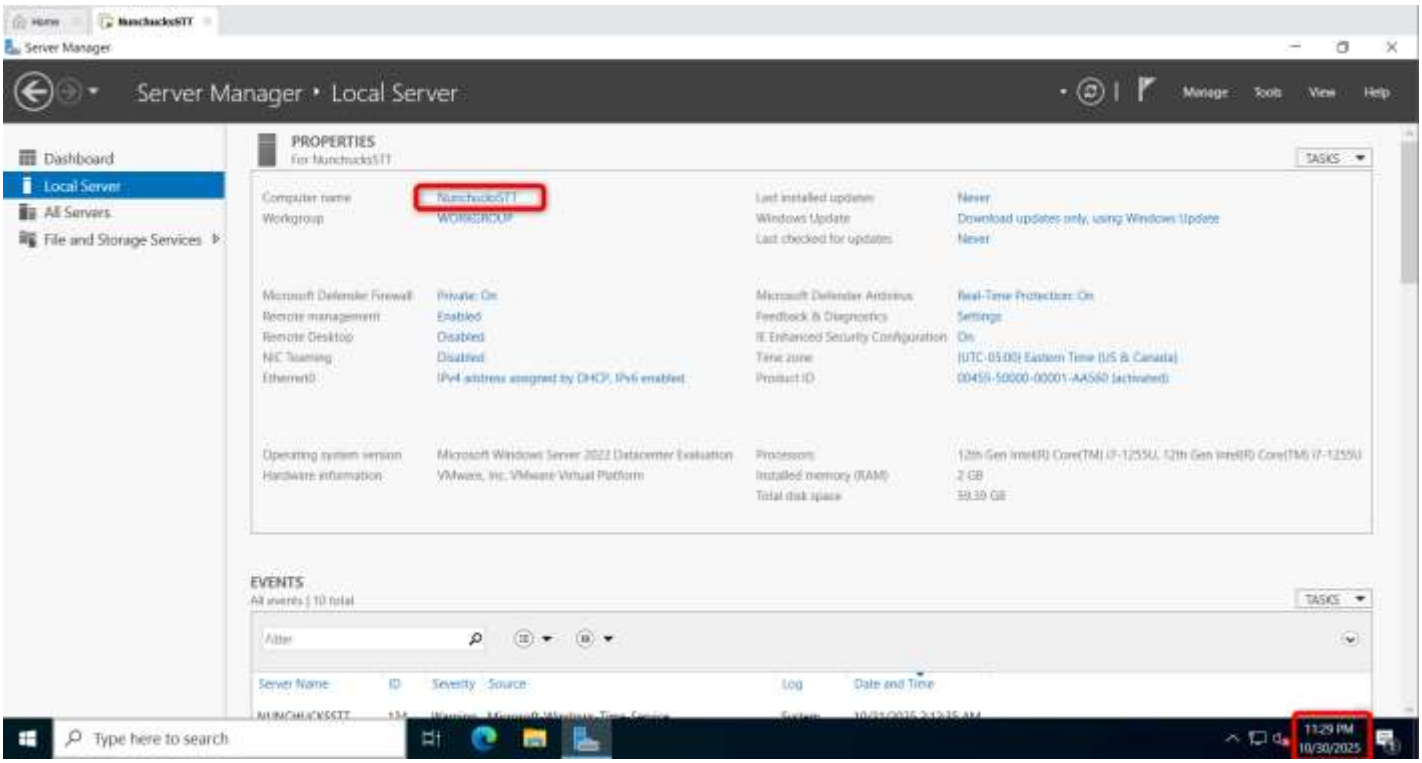
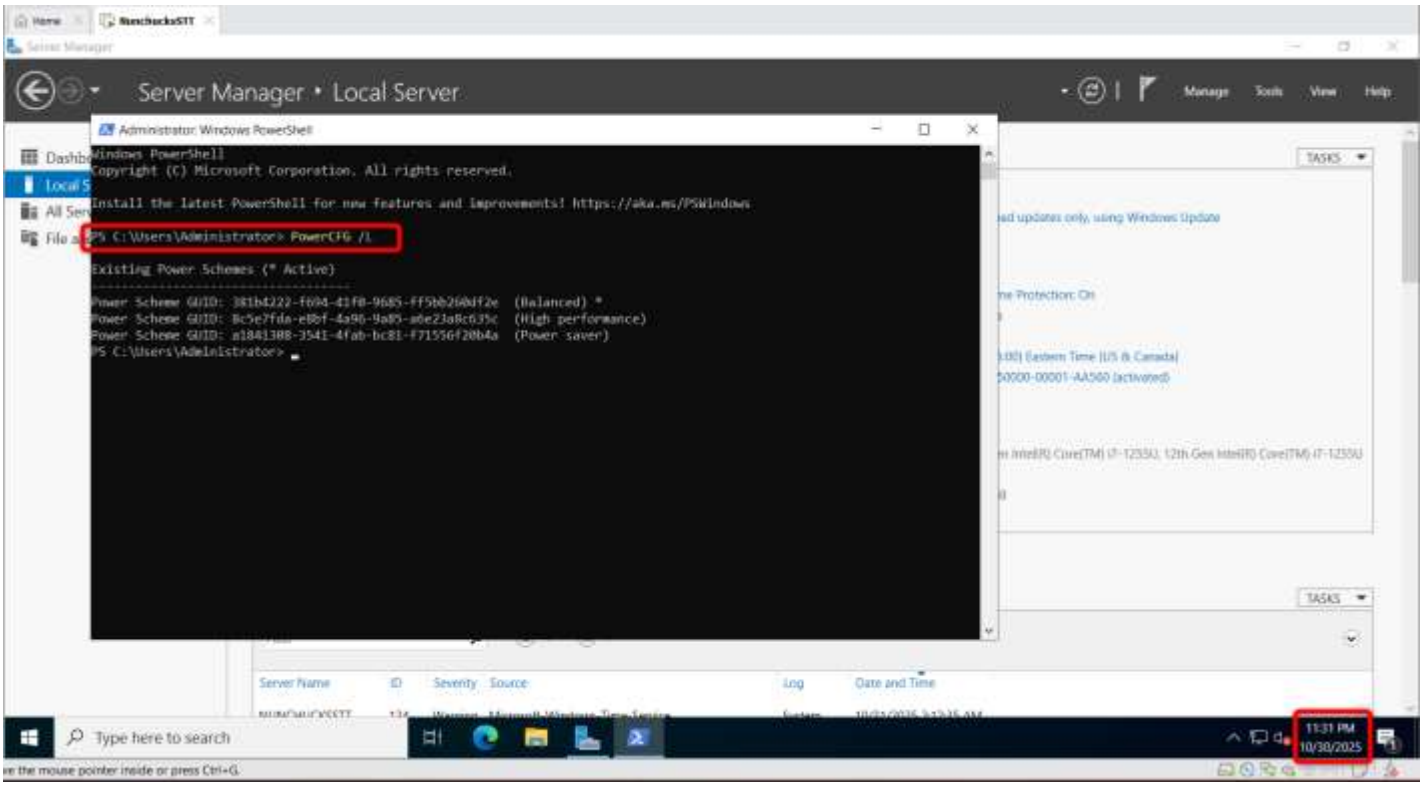


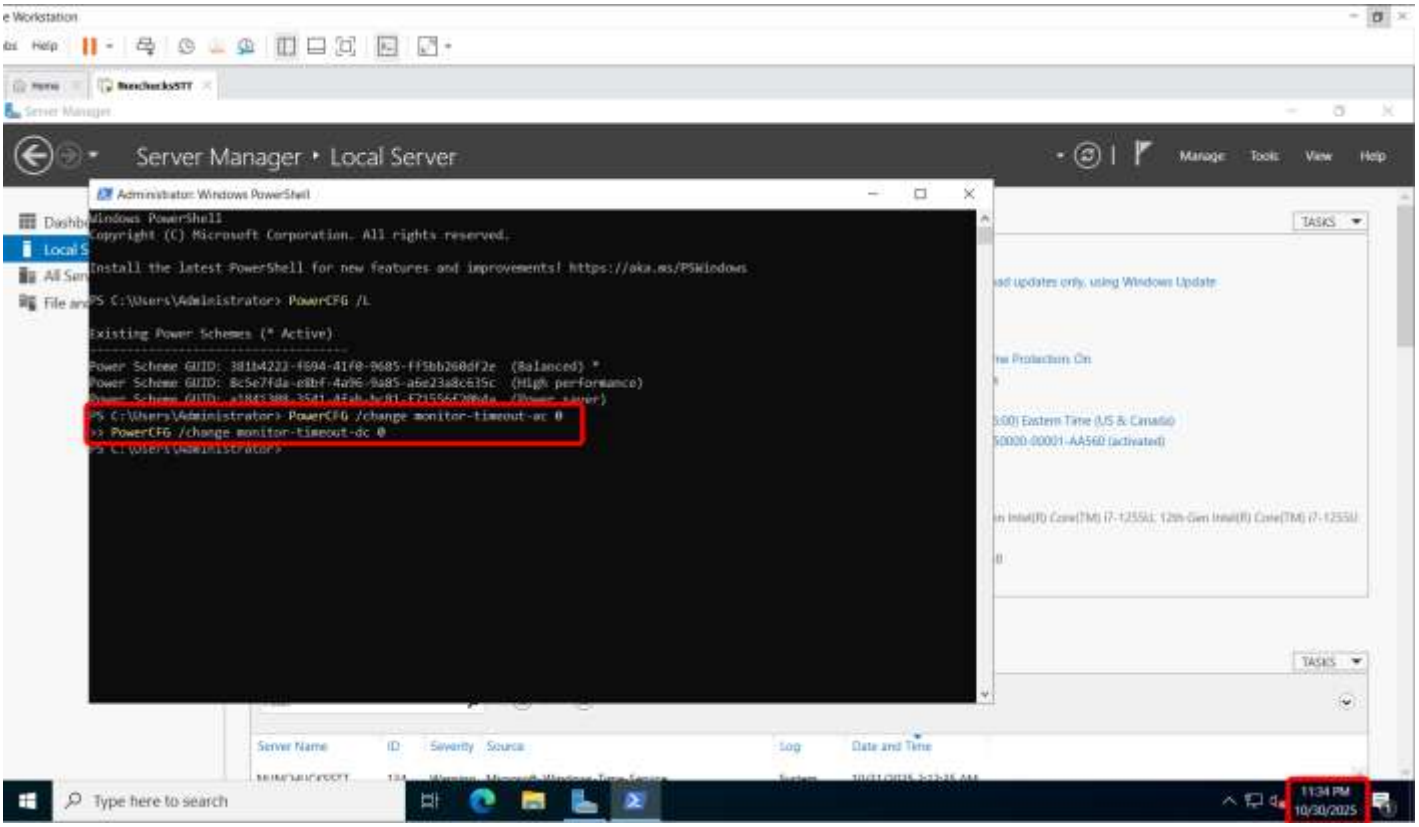
Figure 2 of 2 Rename Server

Prevent Screen from Turning Off

Get power scheme



Query power settings and modify timeout values to 0



Assign Static IP & DNS

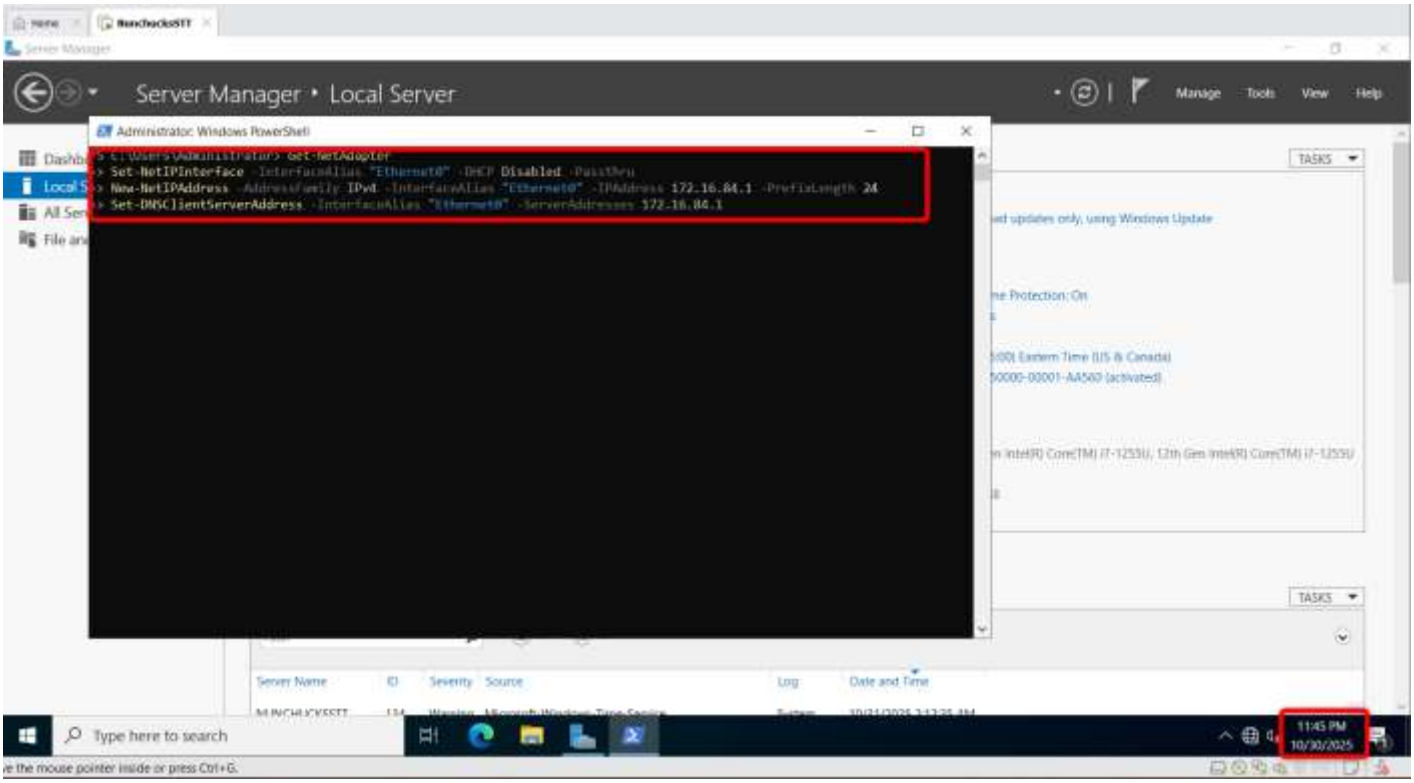


Figure 1 of 2 Assign Static IP & DNS

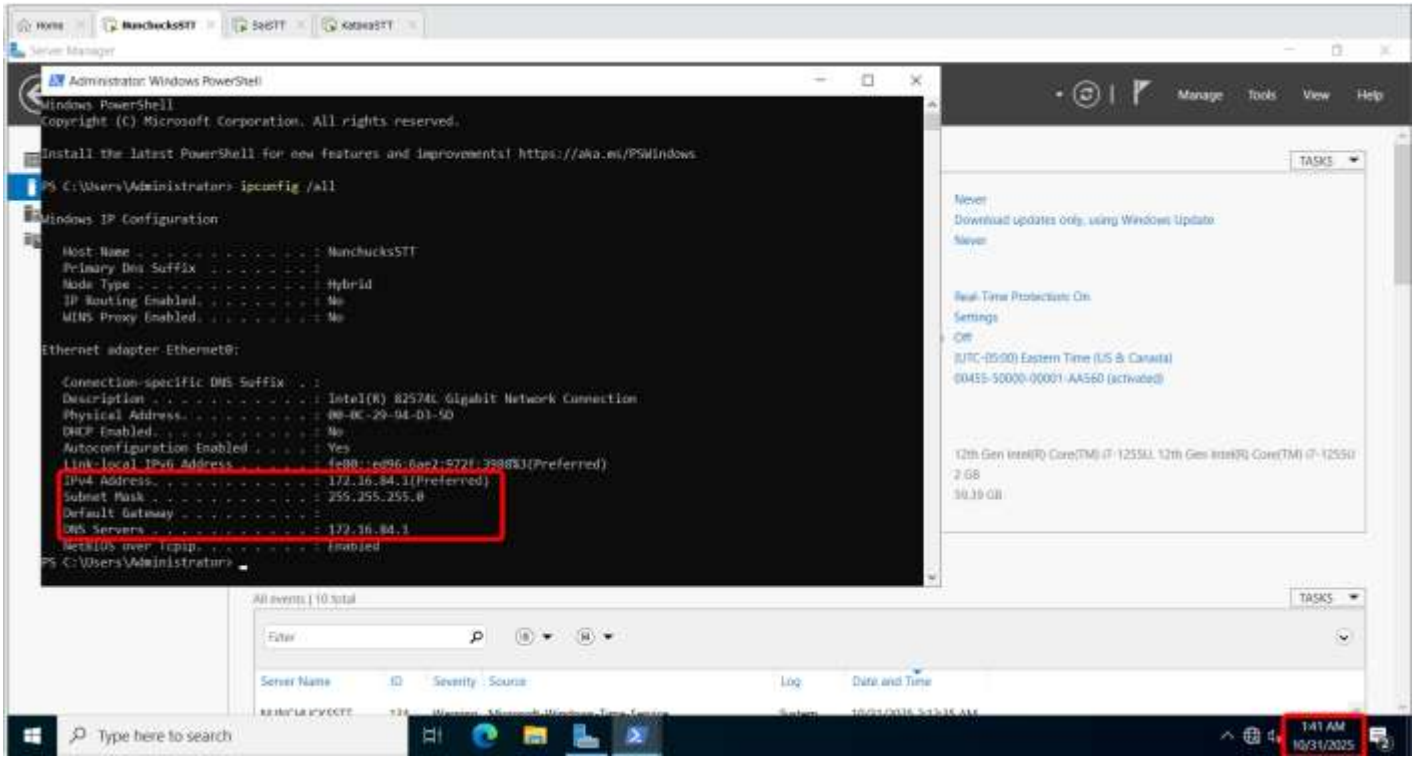
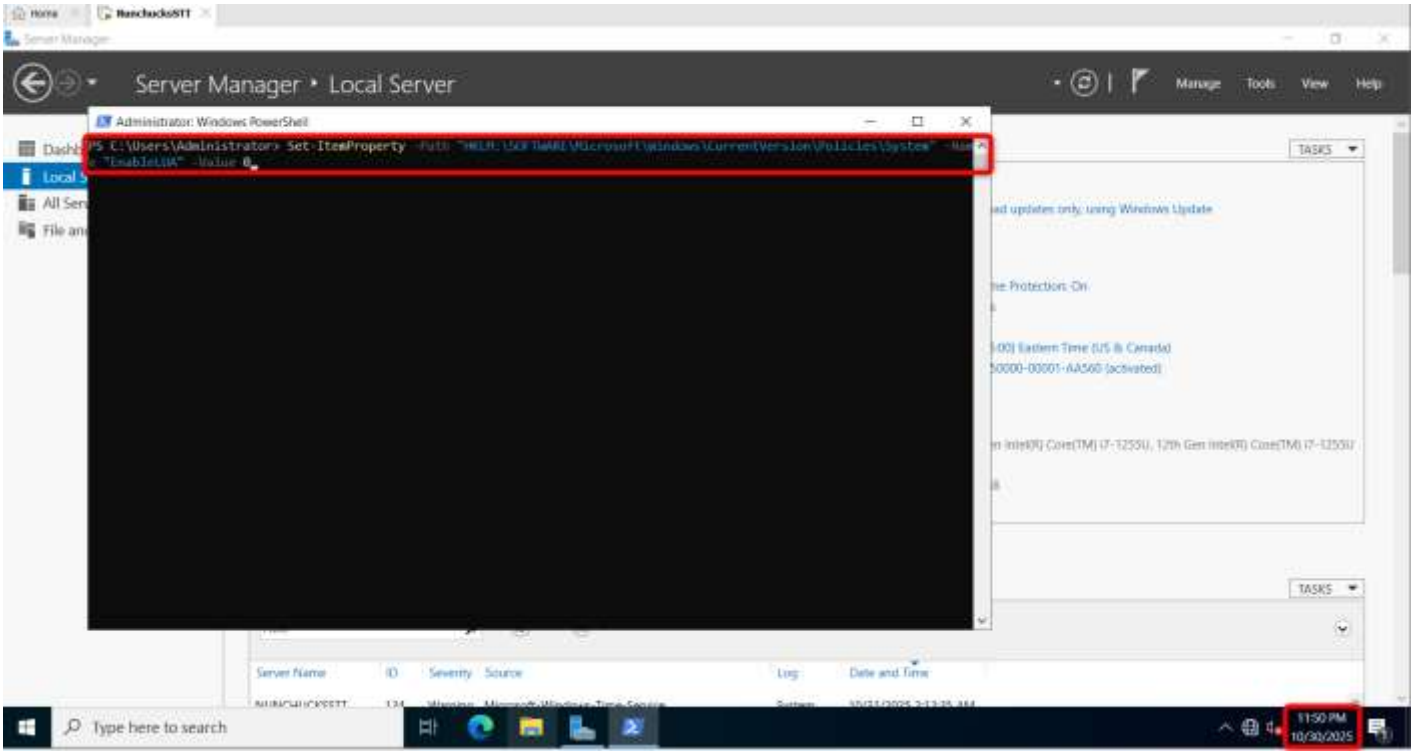
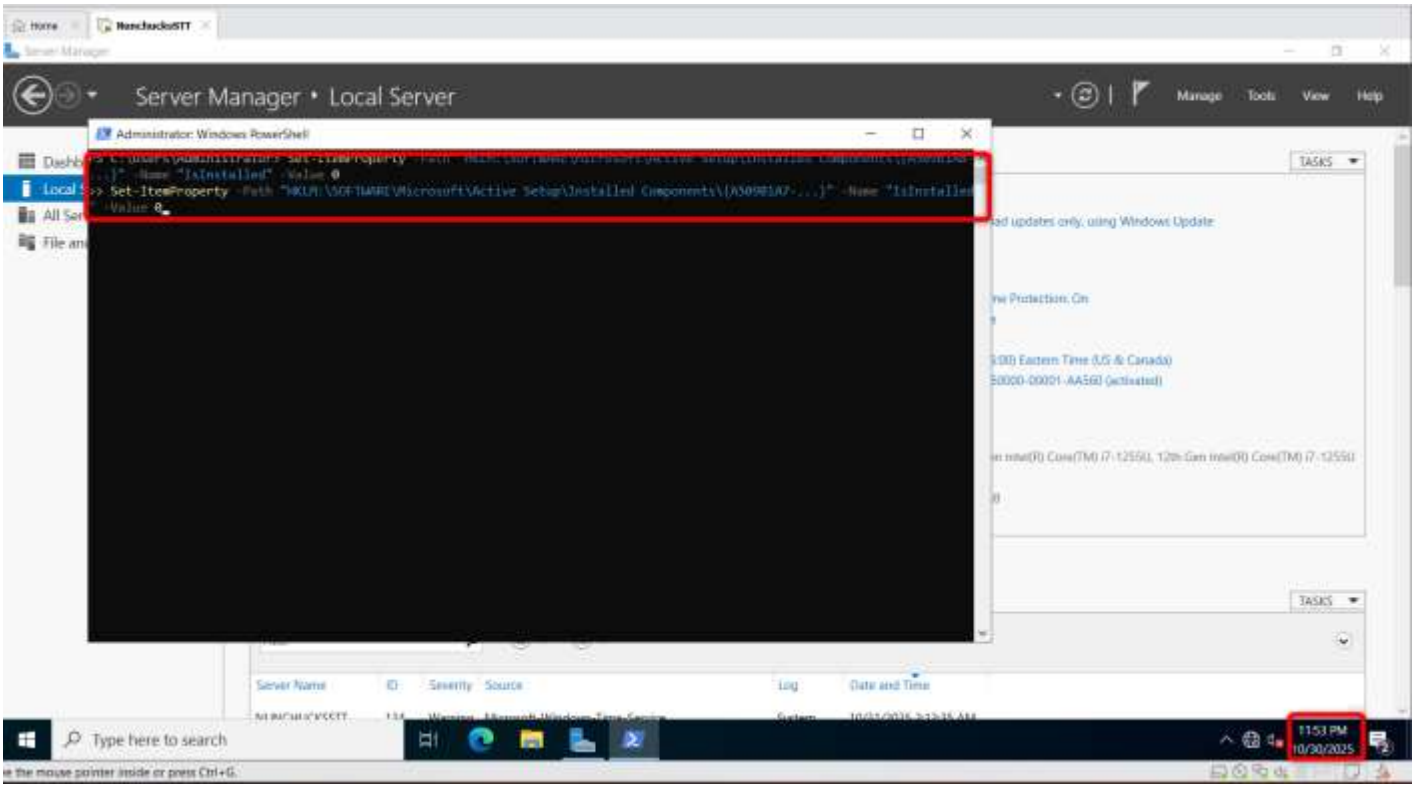


Figure 2 of 2 Assign Static IP & DNS

Set UAC to lowest

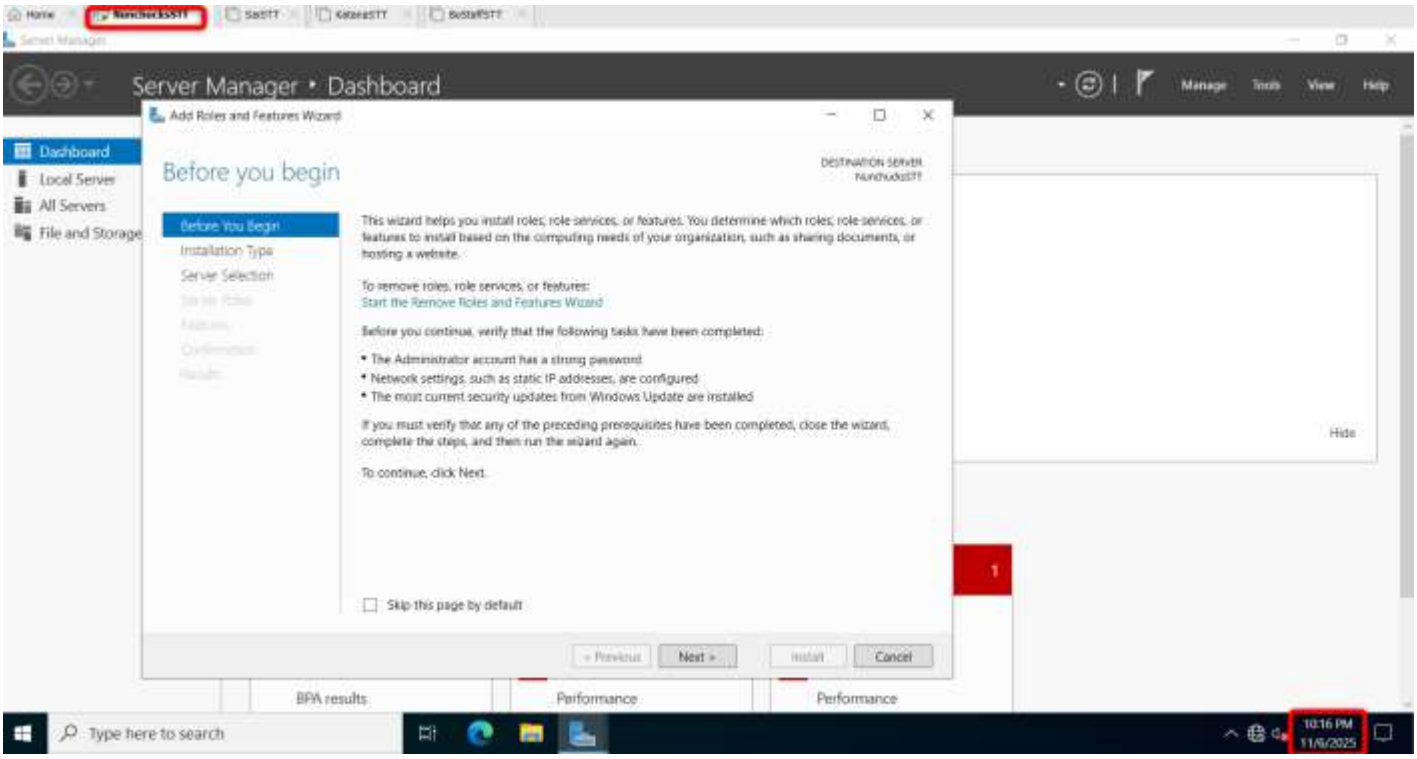


Disable IE Enhanced Security Configuration

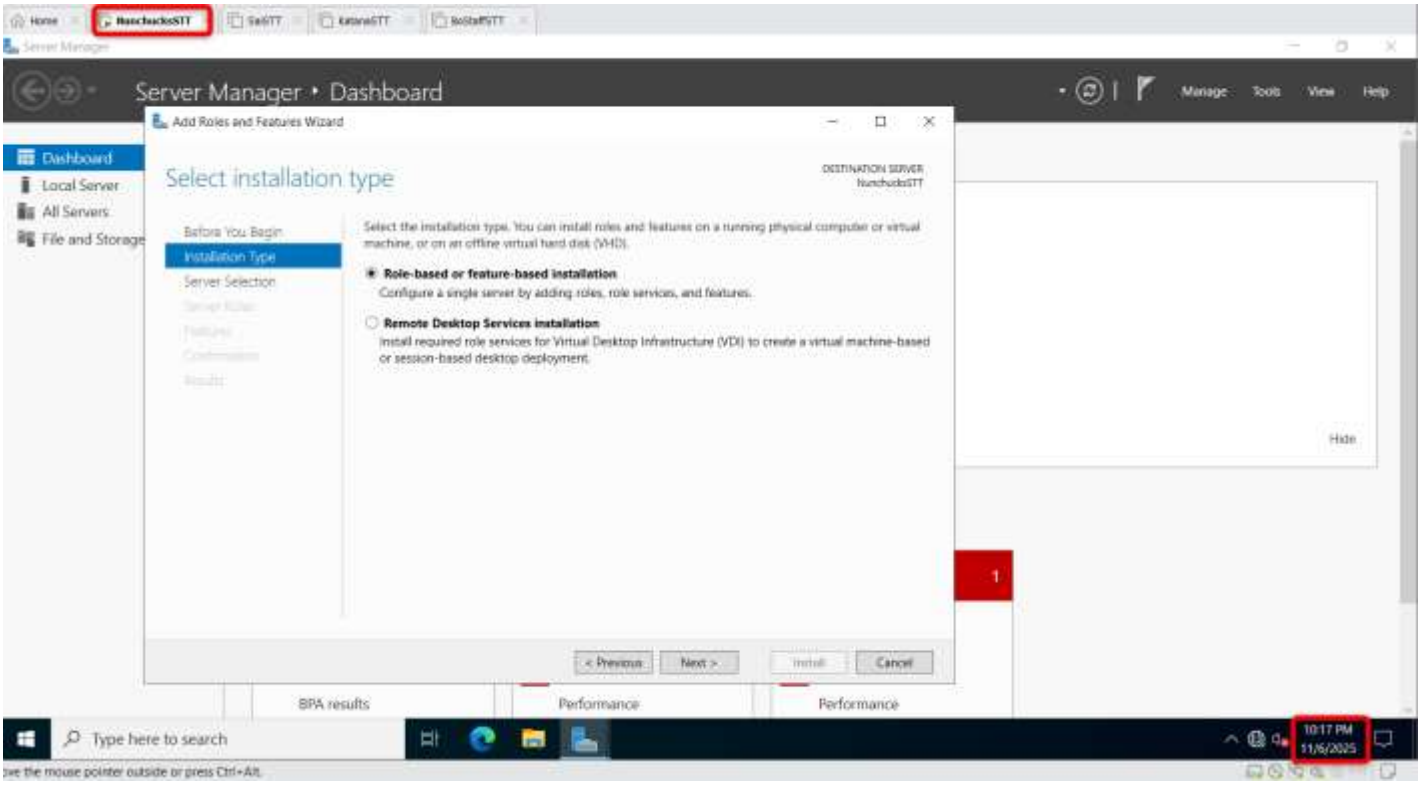


Promote a Domain Controller Build

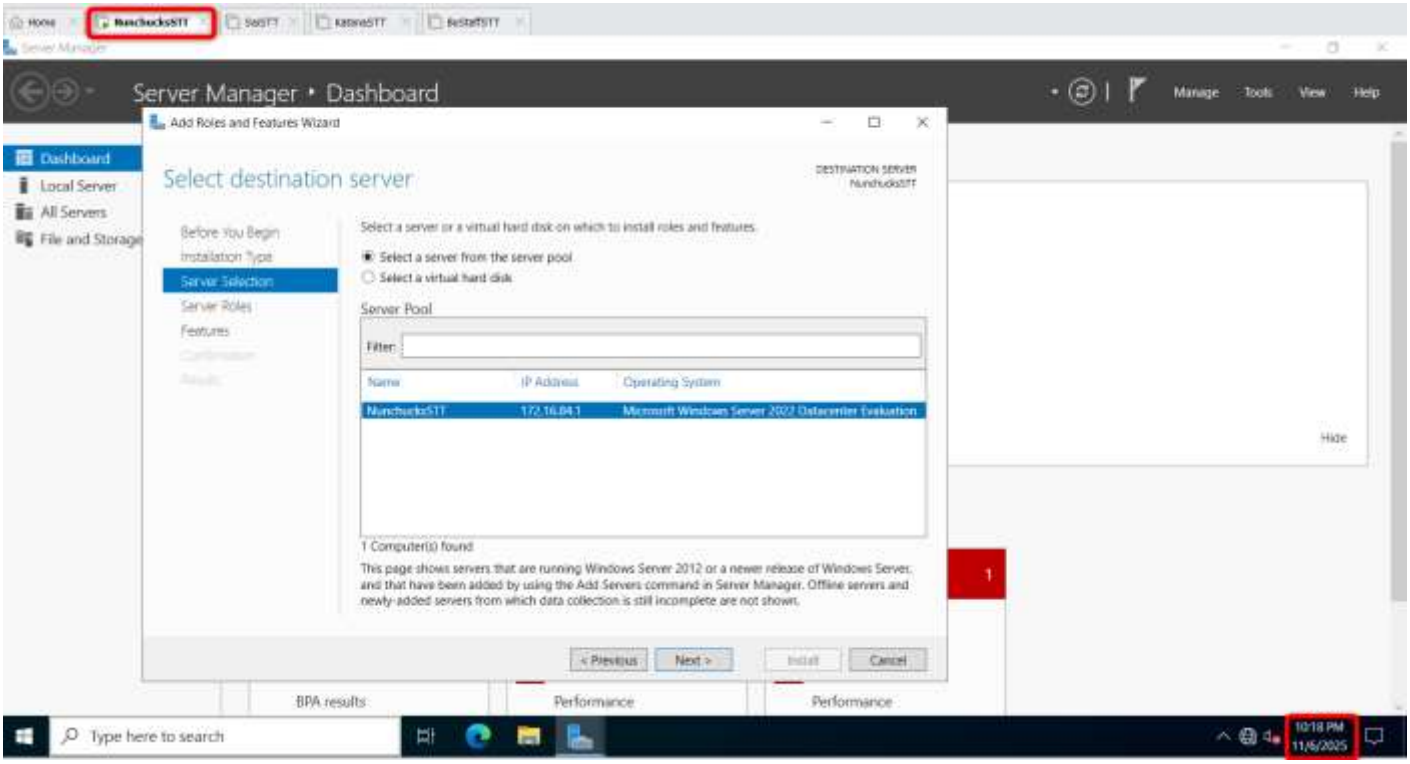
Install Active Directory Domain Services (ADDS)



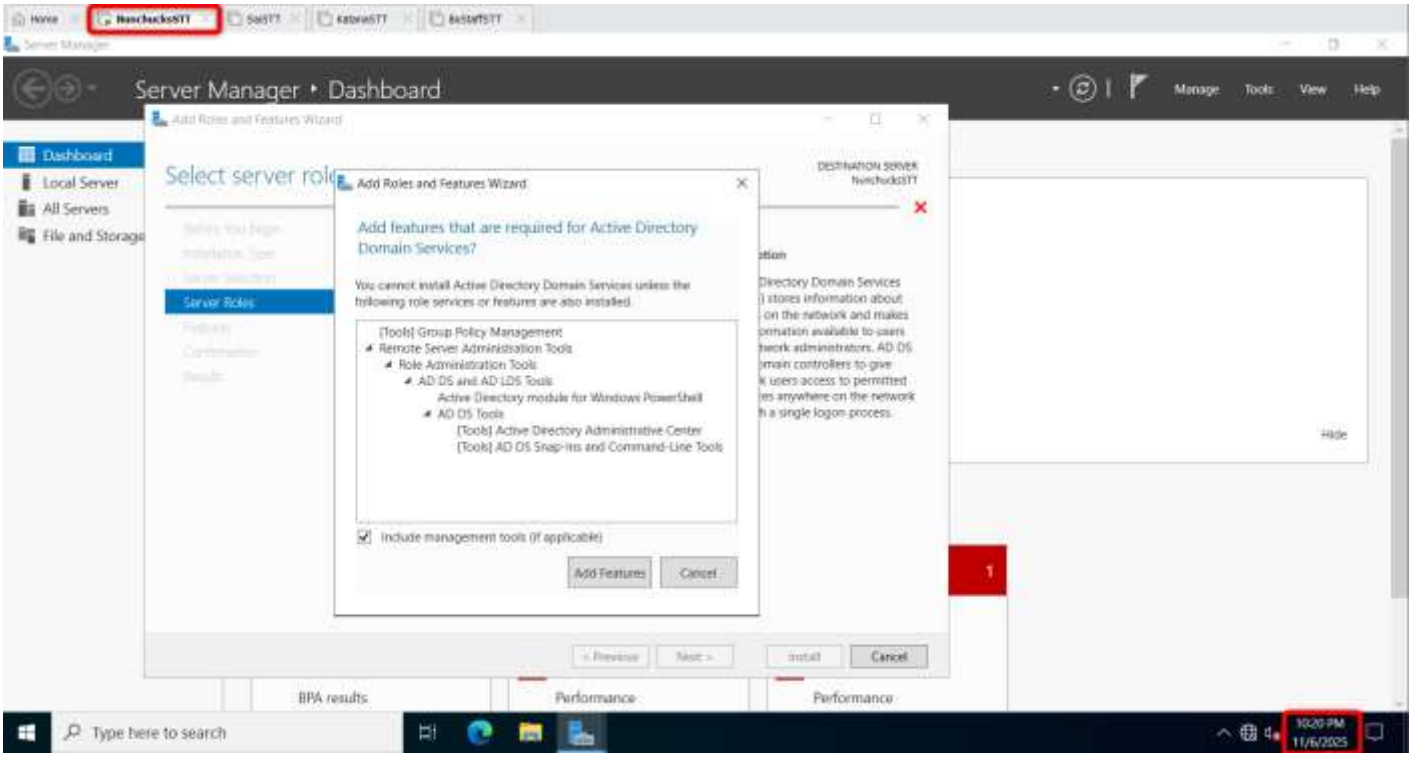
1. *Install AD DS*



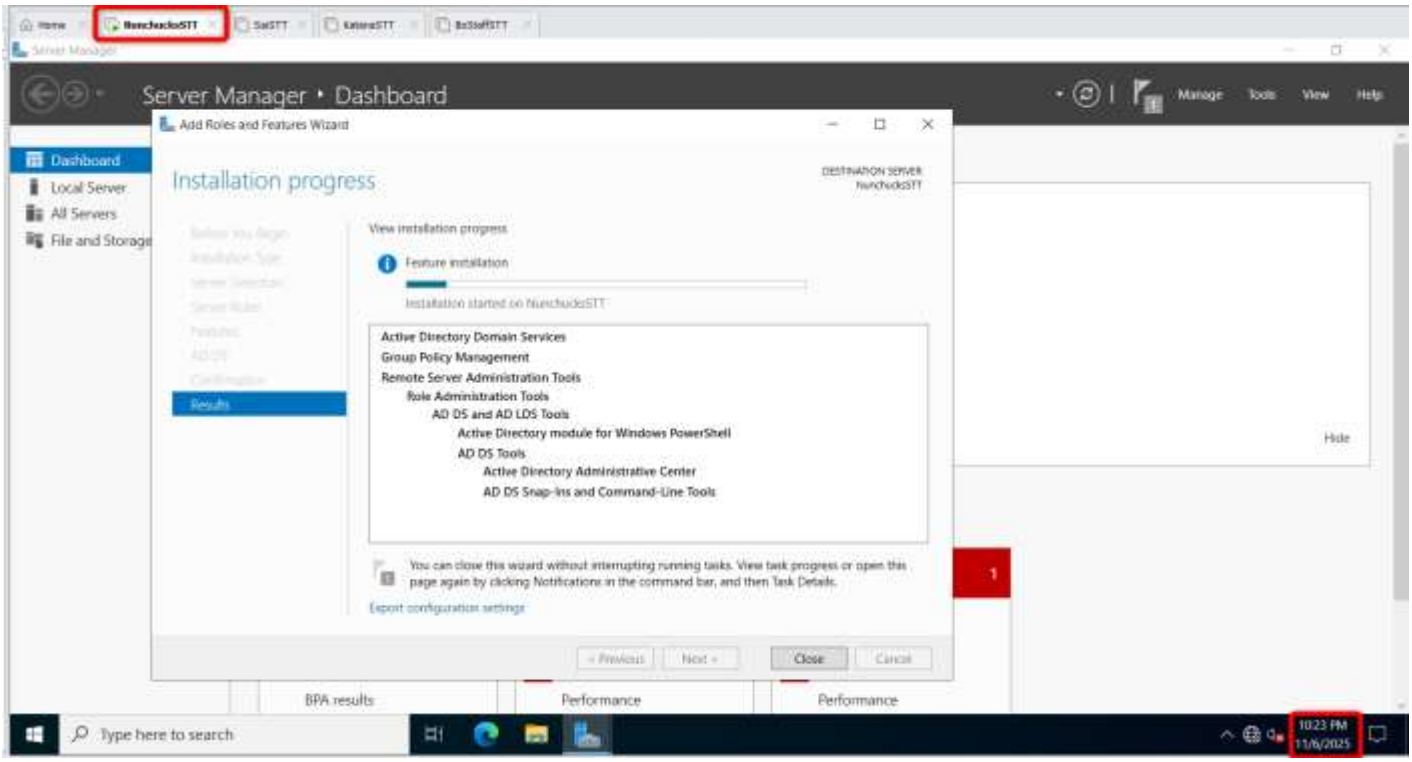
2. *Install AD DS*



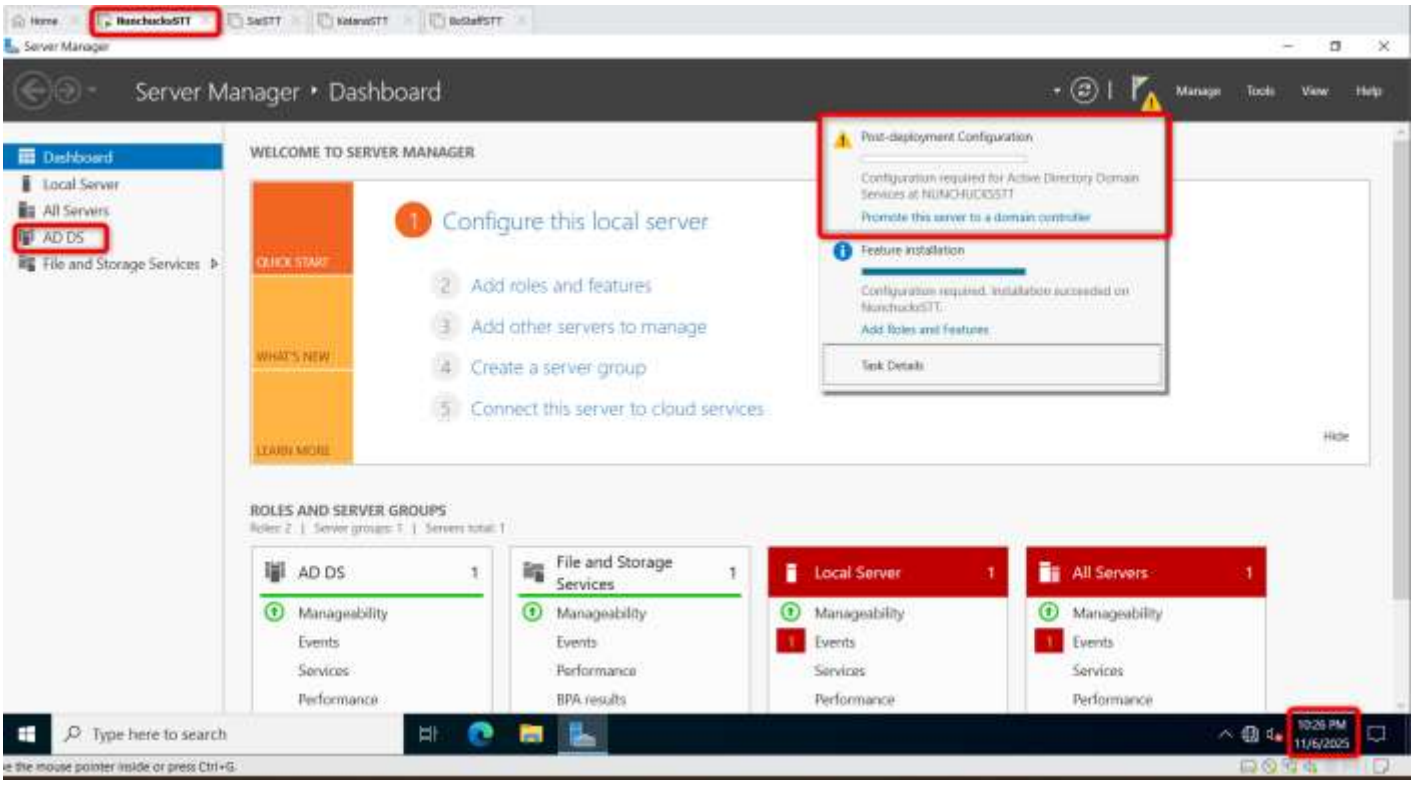
3. *Install AD DS*



4. *Install AD DS*

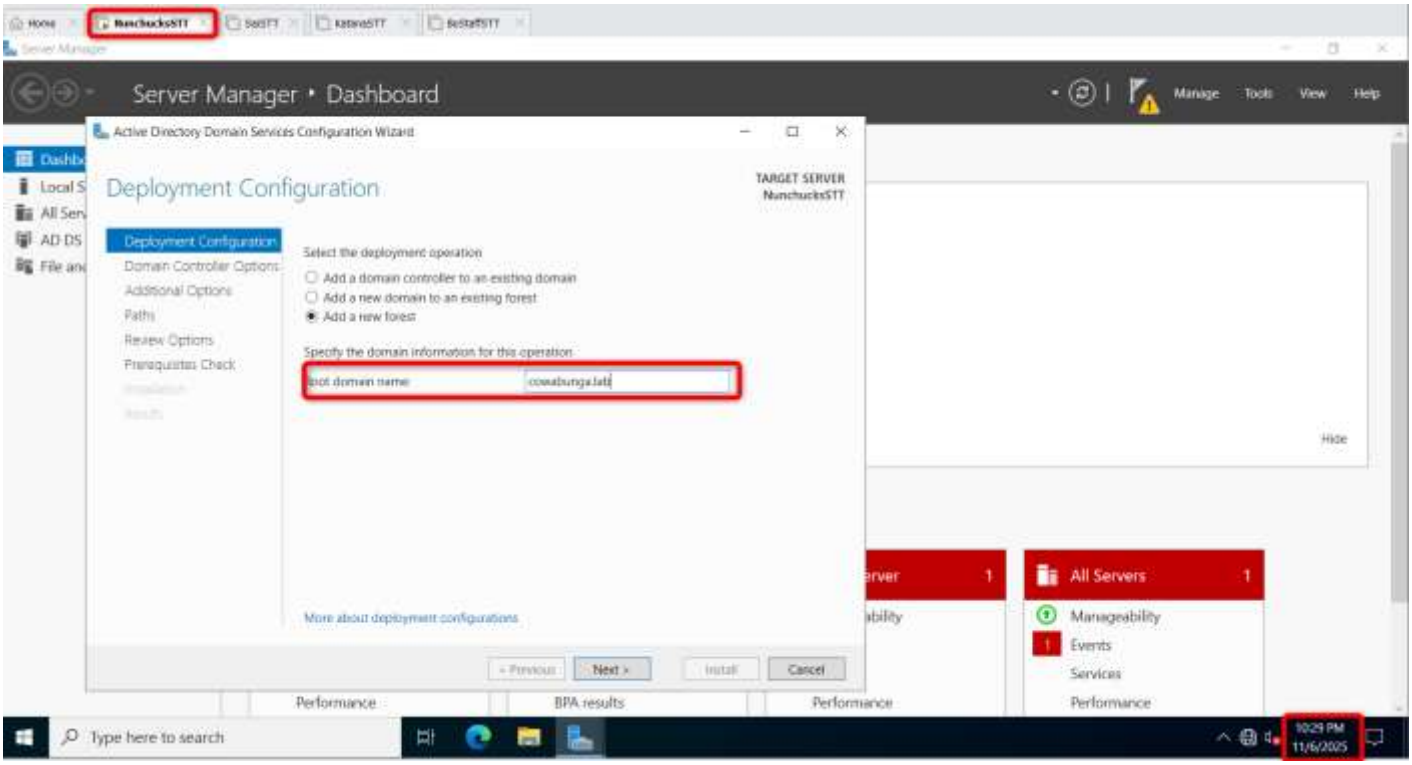


5. *Install AD DS*

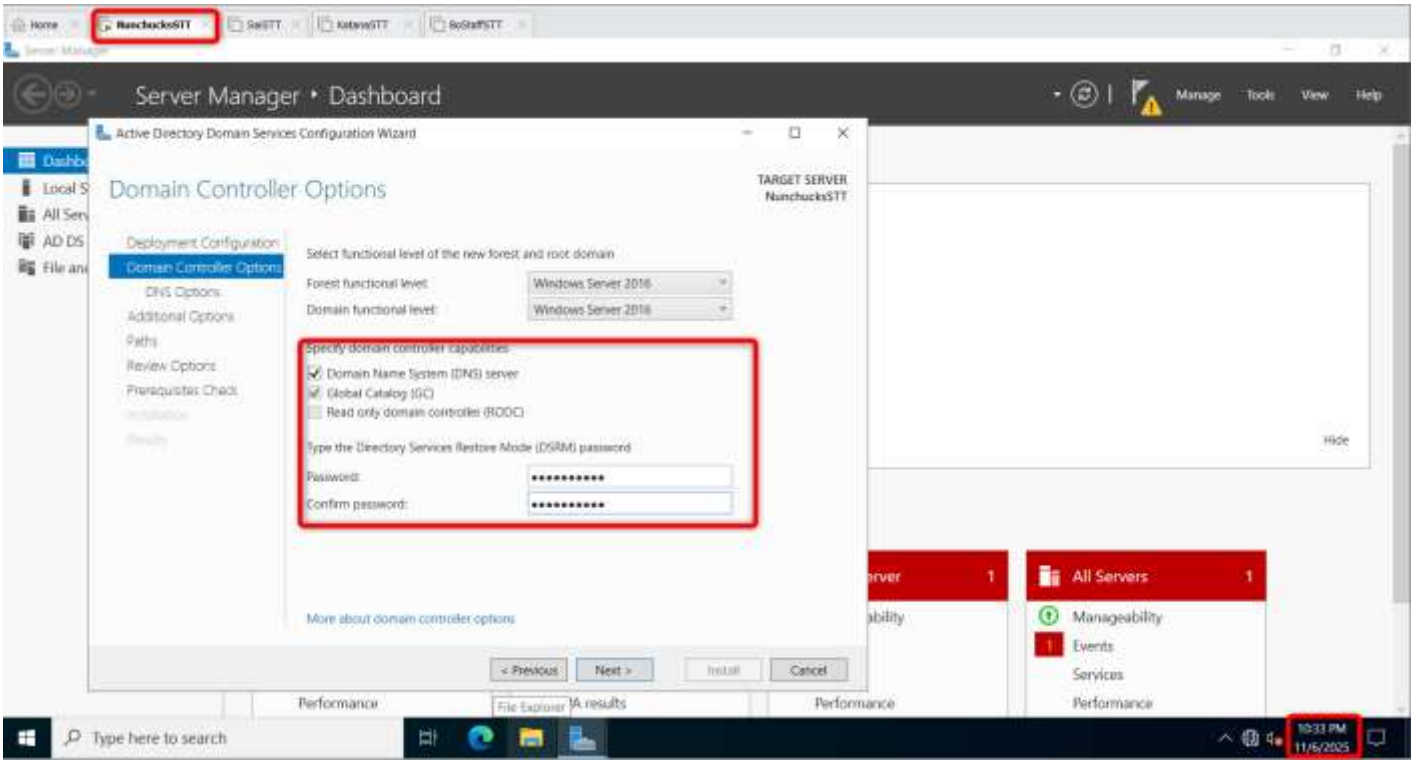


6. *Install AD DS*

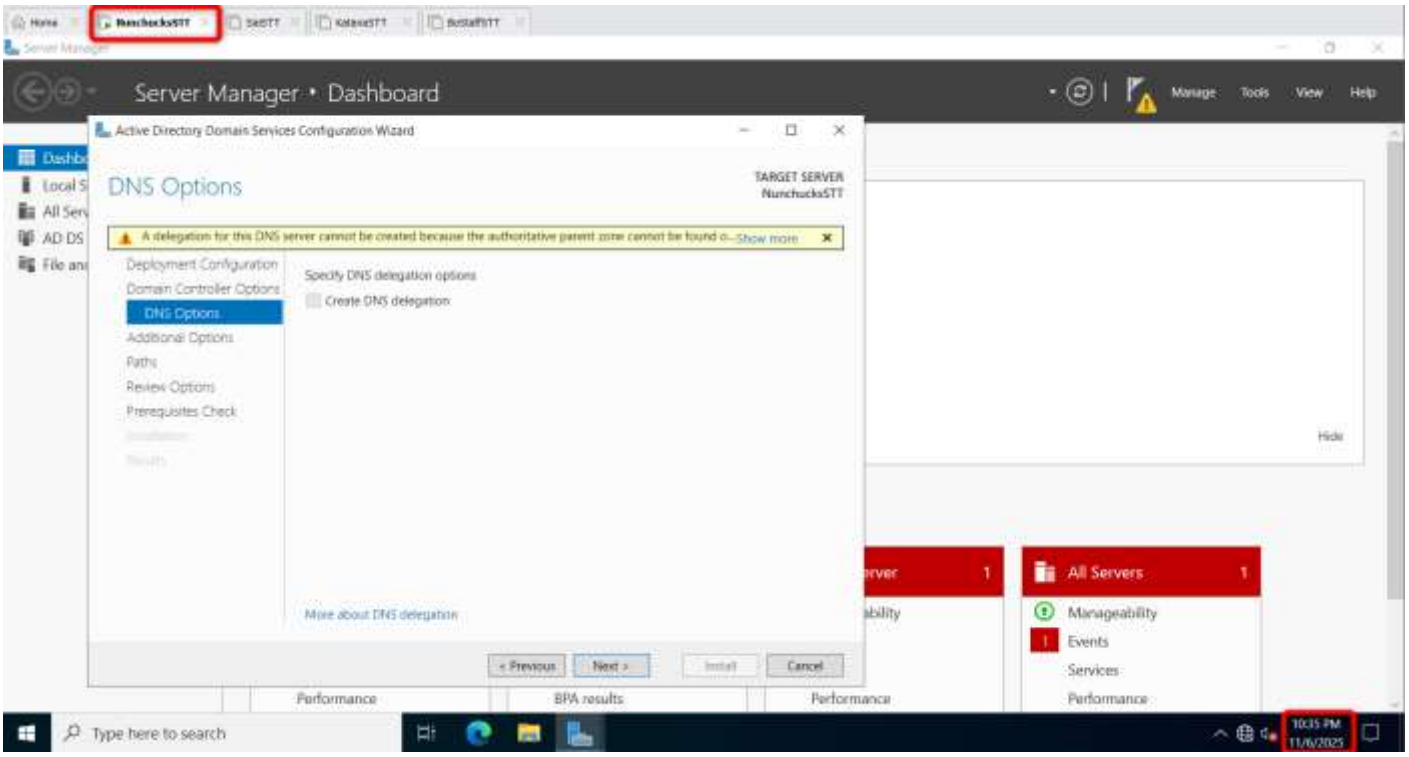
Configure the Domain Name



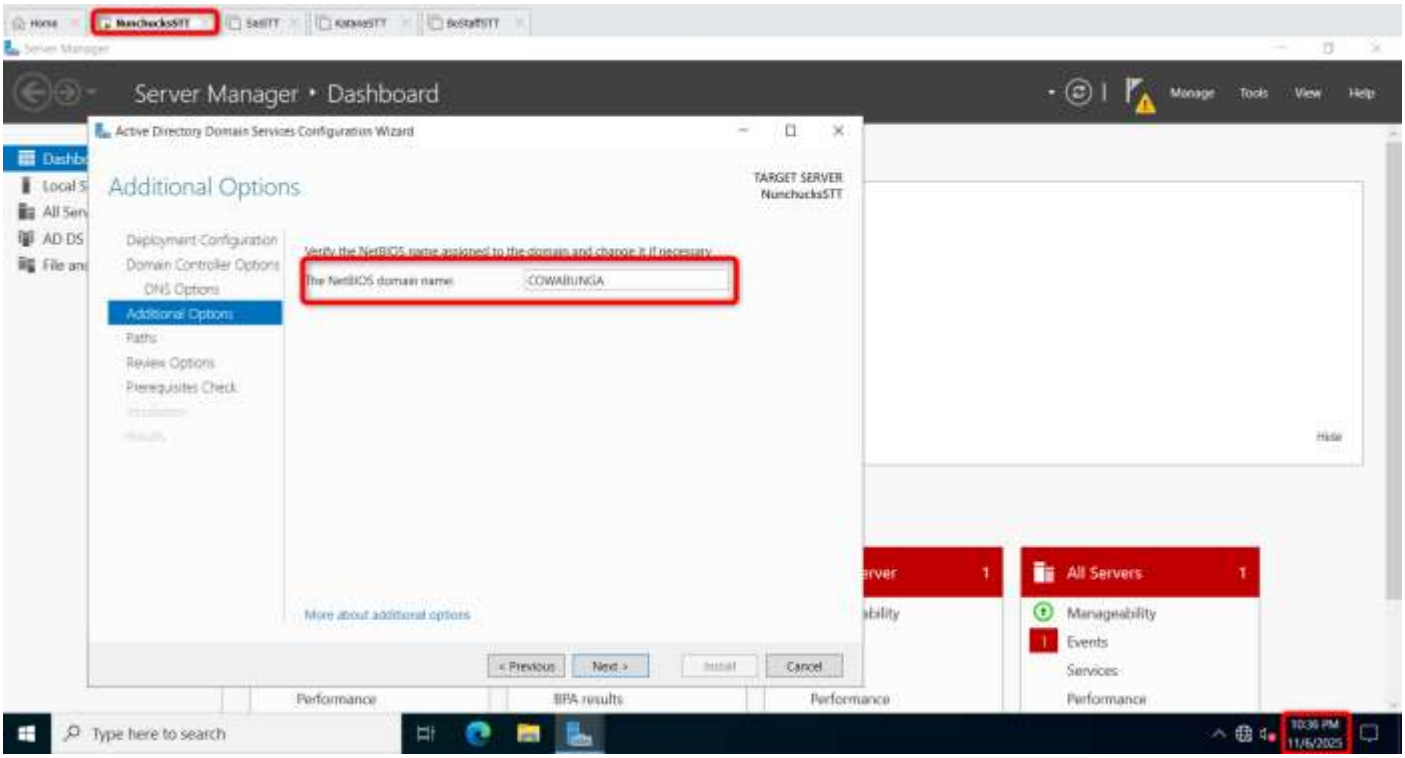
1. *Configure the Domain Name*



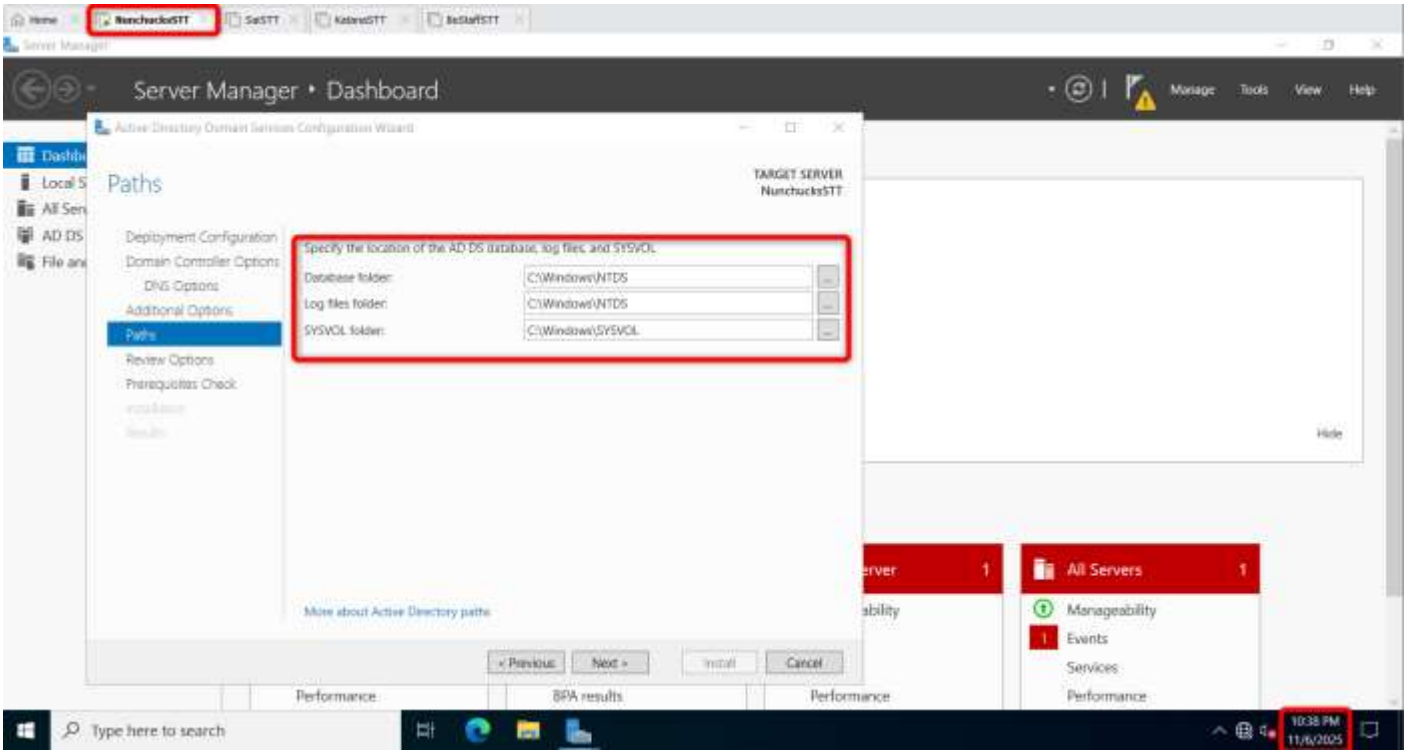
2. *Configure the Domain Name*



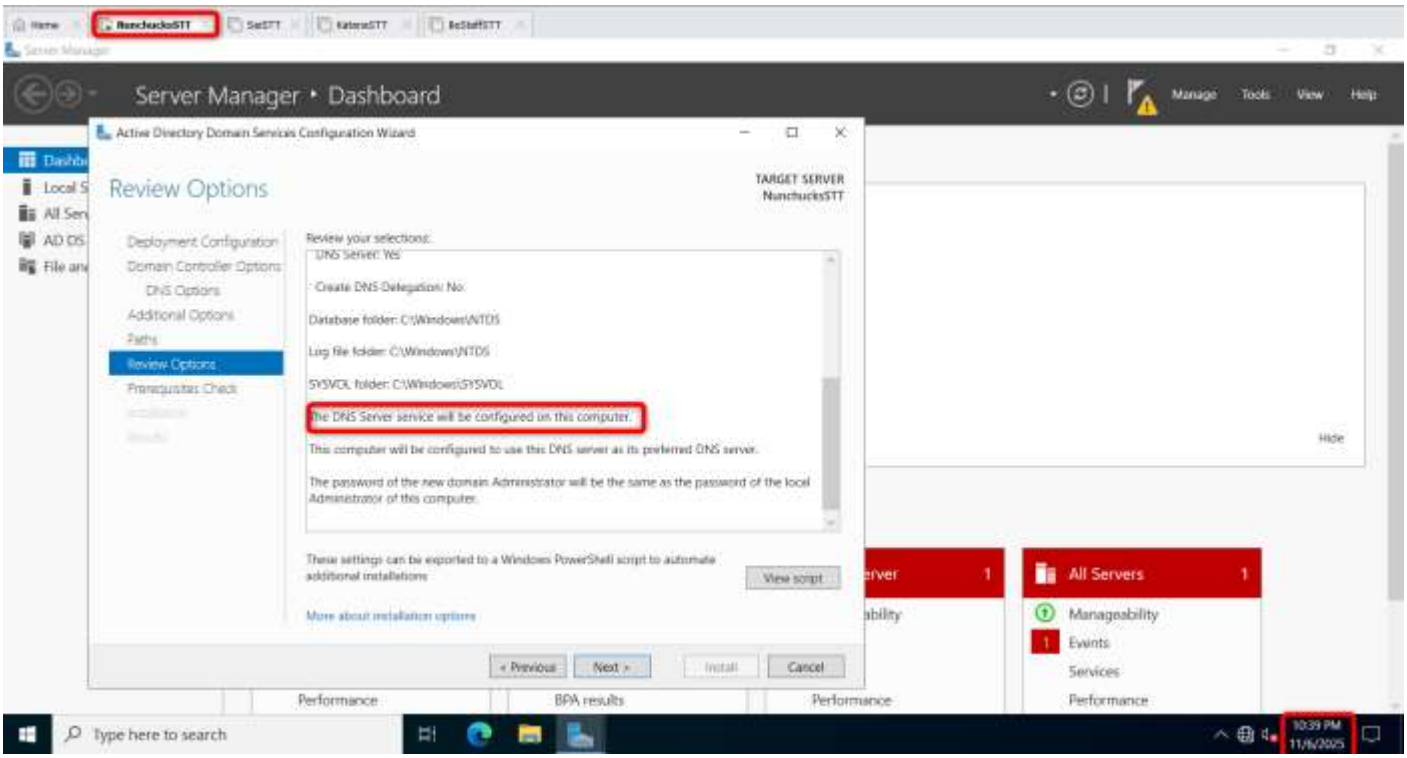
3. *Configure the Domain Name*



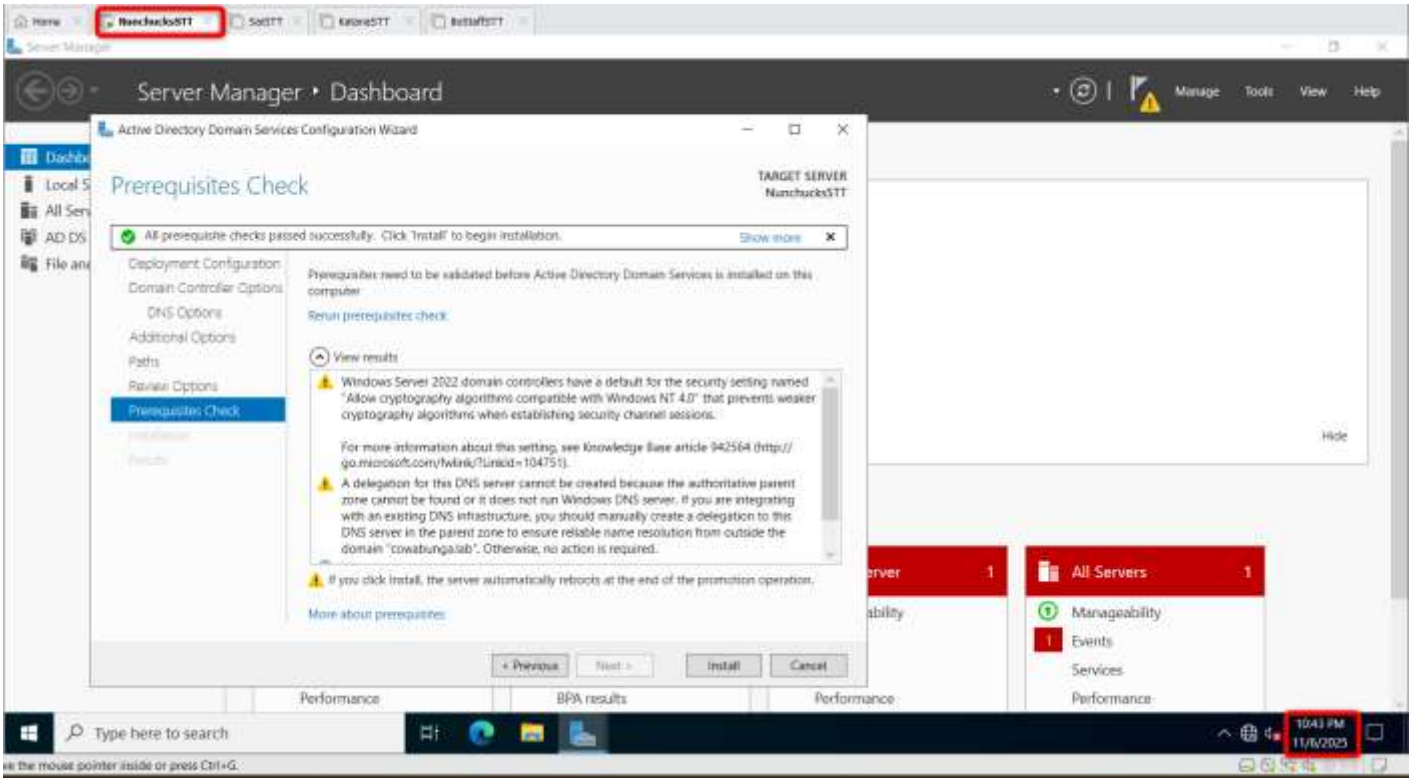
4. *Configure the Domain Name*



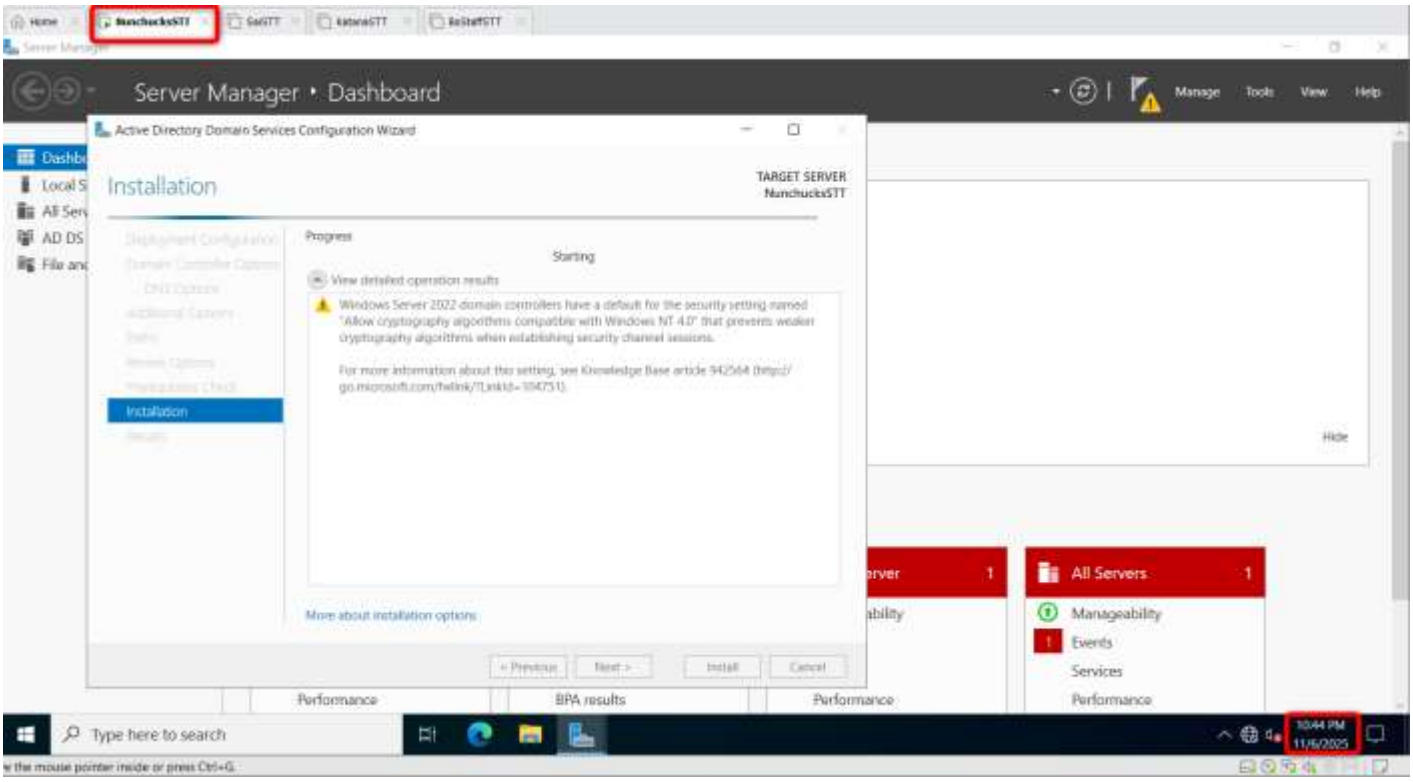
5. *Configure the Domain Name*



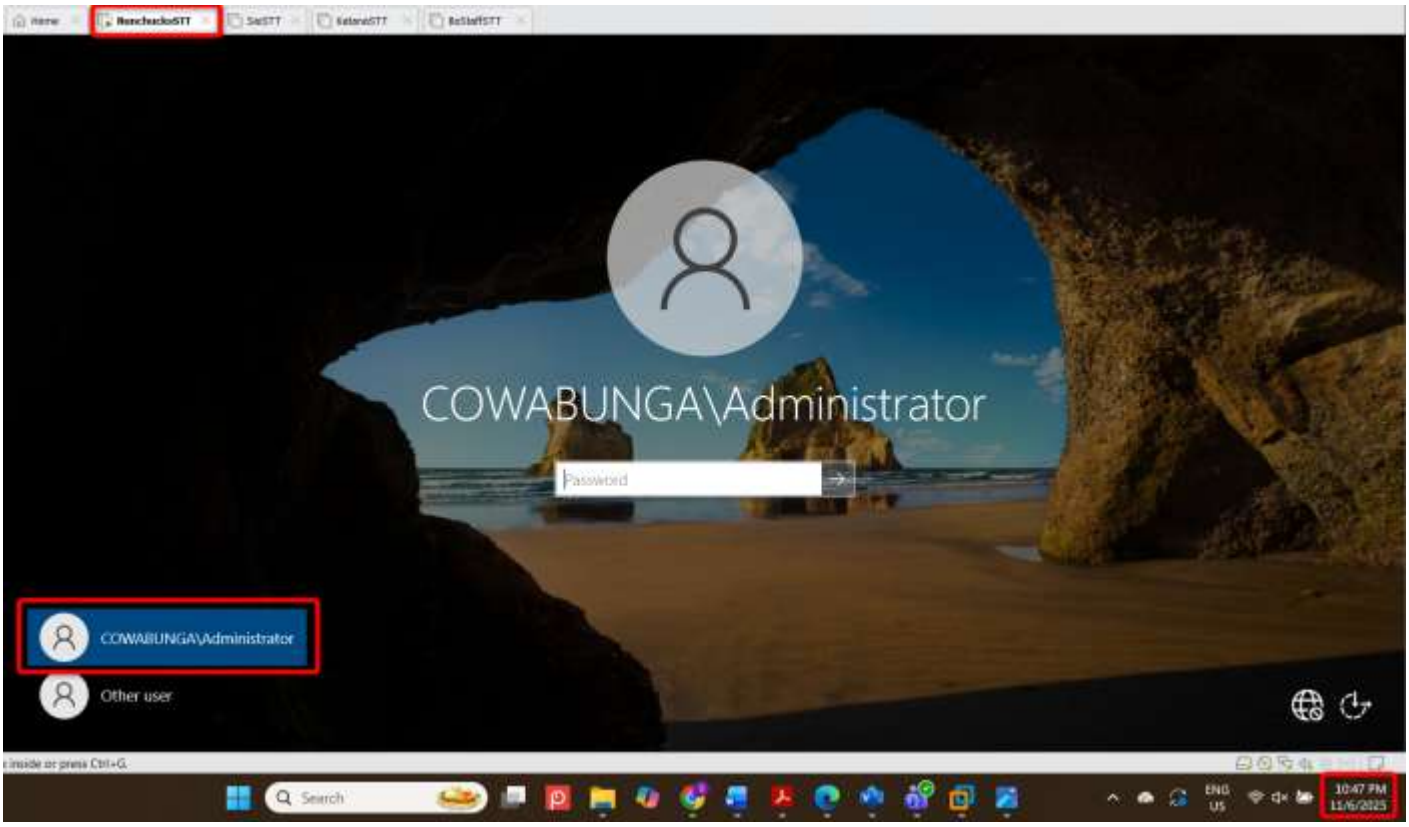
6. *Configure the Domain Name*



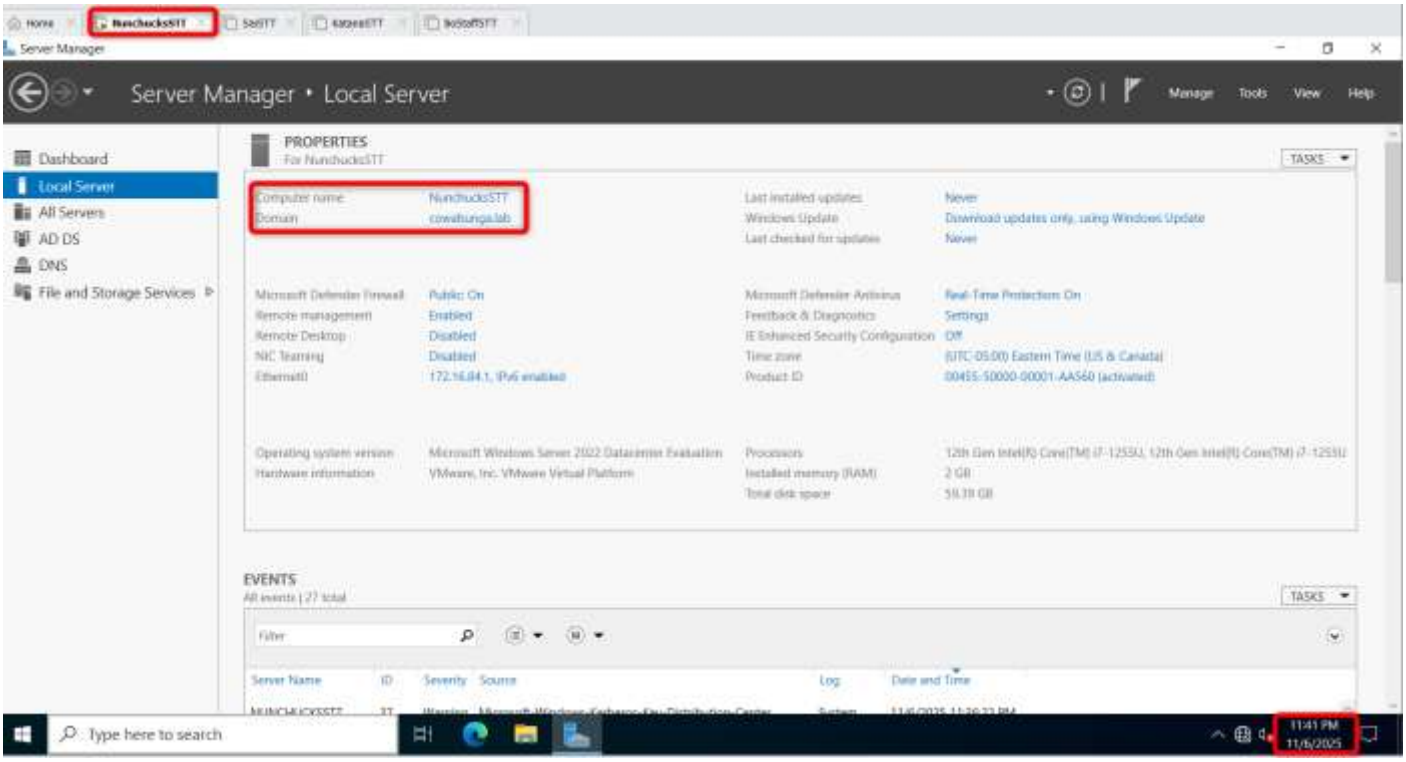
7. *Configure the Domain Name*



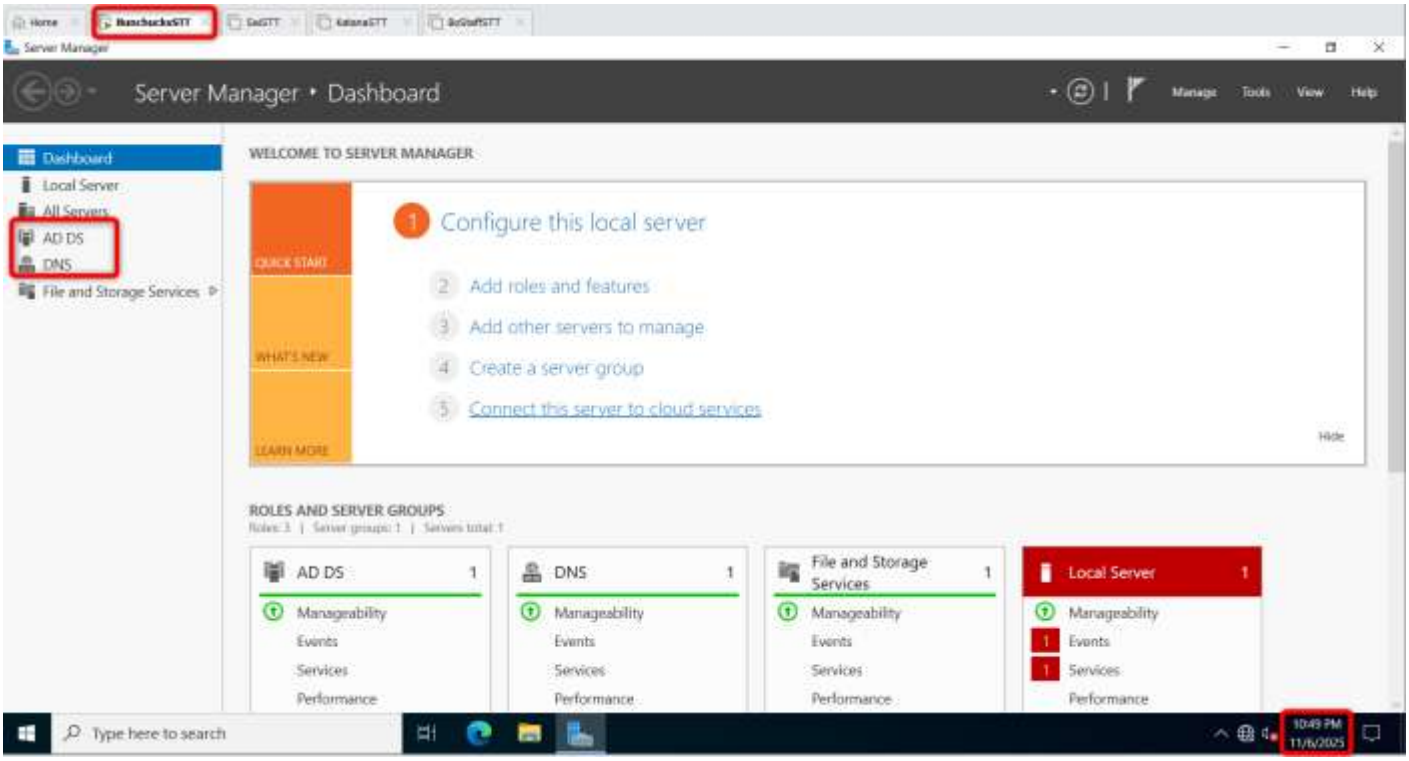
8. *Configure the Domain Name*



9. Configure the Domain Name



10. Configure the Domain Name



11. Configure the Domain Name

Using PowerShell for OU creation

Set up as follows:

OU Creation (via PowerShell)

Step 1: Run the following command for each OU:

New-ADOrganizationalUnit -Name “TurtlesSTT” -Description “TurtlesSTT OU” -DisplayName "TurtlesSTT" -ProtectedFromAccidentalDeletion $True -Path "DC=cowabunga,DC=lab"

Check the **Active Directory Users and Computers (ADUC)** to confirm the OUs were created successfully.

Group Creation

Step 2: Run this PowerShell command for each group:

New-ADGroup -Name "G_TurtlesSTT" -SamAccountName "G_TurtlesSTT" -GroupCategory Security -GroupScope Global -DisplayName "G_TurtlesSTT" -Path "OU=TurtlesSTT,DC=cowabunga,DC=lab" -Description "Security group for TurtlesSTT users"

User Creation via CSV Import

Step 3: Create a CSV File

1. Open **Notepad**
2. Use the following format as your first line:

dn,SamAccountName,UserPrincipalName,objectClass

3. Add one line per user like this:

"cn=Jane Smith,ou=Marketing,dc=example,dc=local",jsmith,jsmith@example.local,user

4. Save the file as **csvusers.csv** (for ex. Desktop)

Import the Users

Step 4: Import the users utilizing the CSV file you created in the last step

1. Open **Command Prompt**
2. Navigate to the Desktop:

cd Desktop

3. Run the import command:

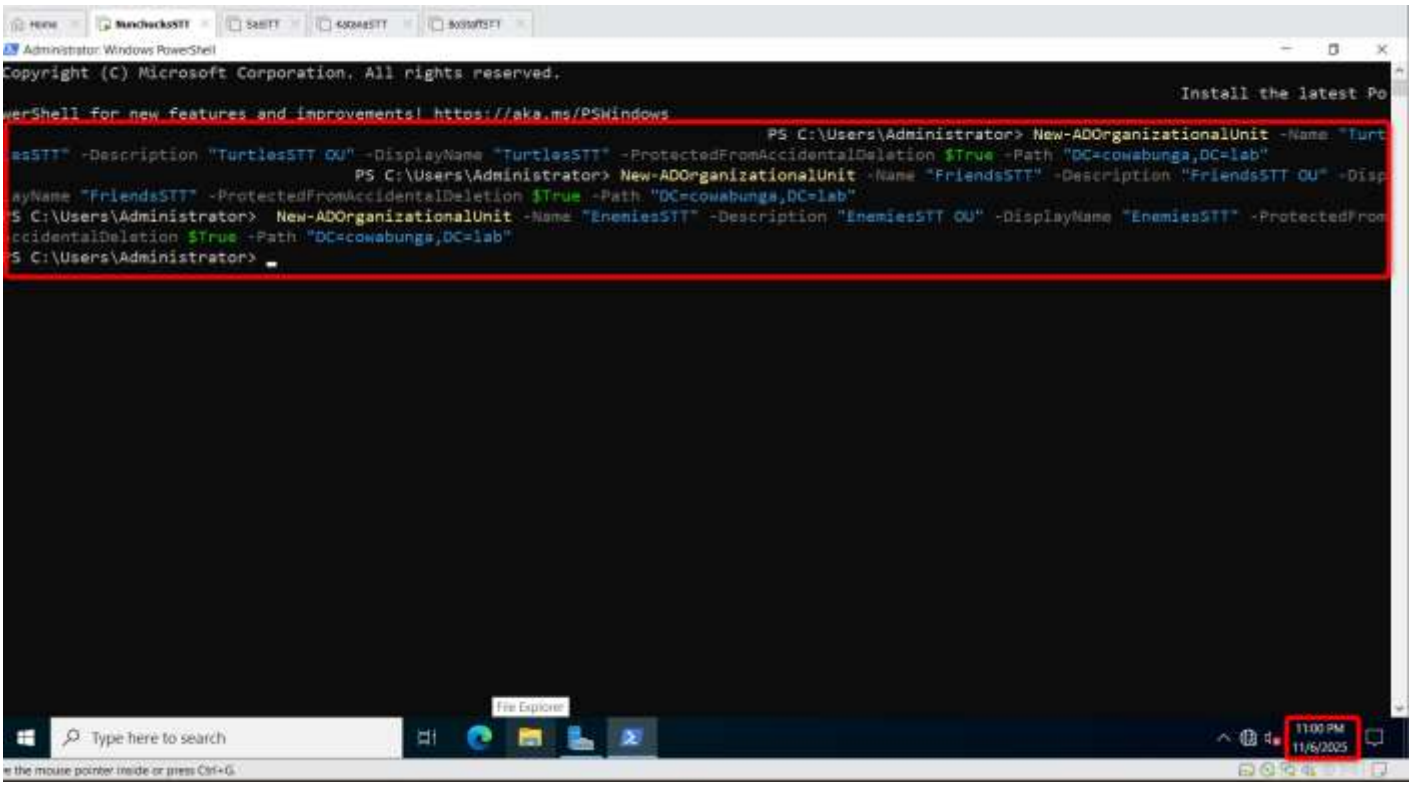
csvde -i -f csvusers.csv

You should receive a success message stating the users were imported correctly.

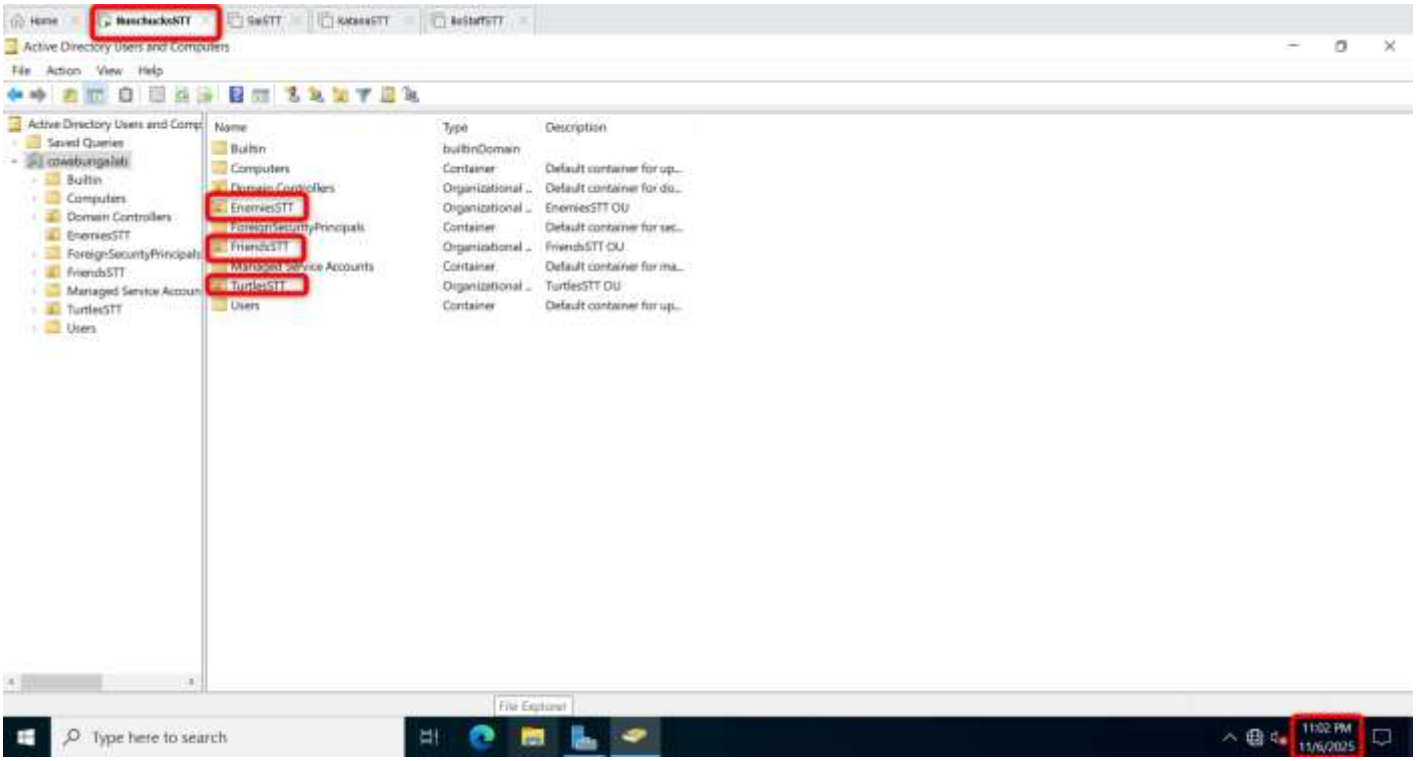
Set the Default Password and Enable the User Account

1. Run this command for for EACH OU:

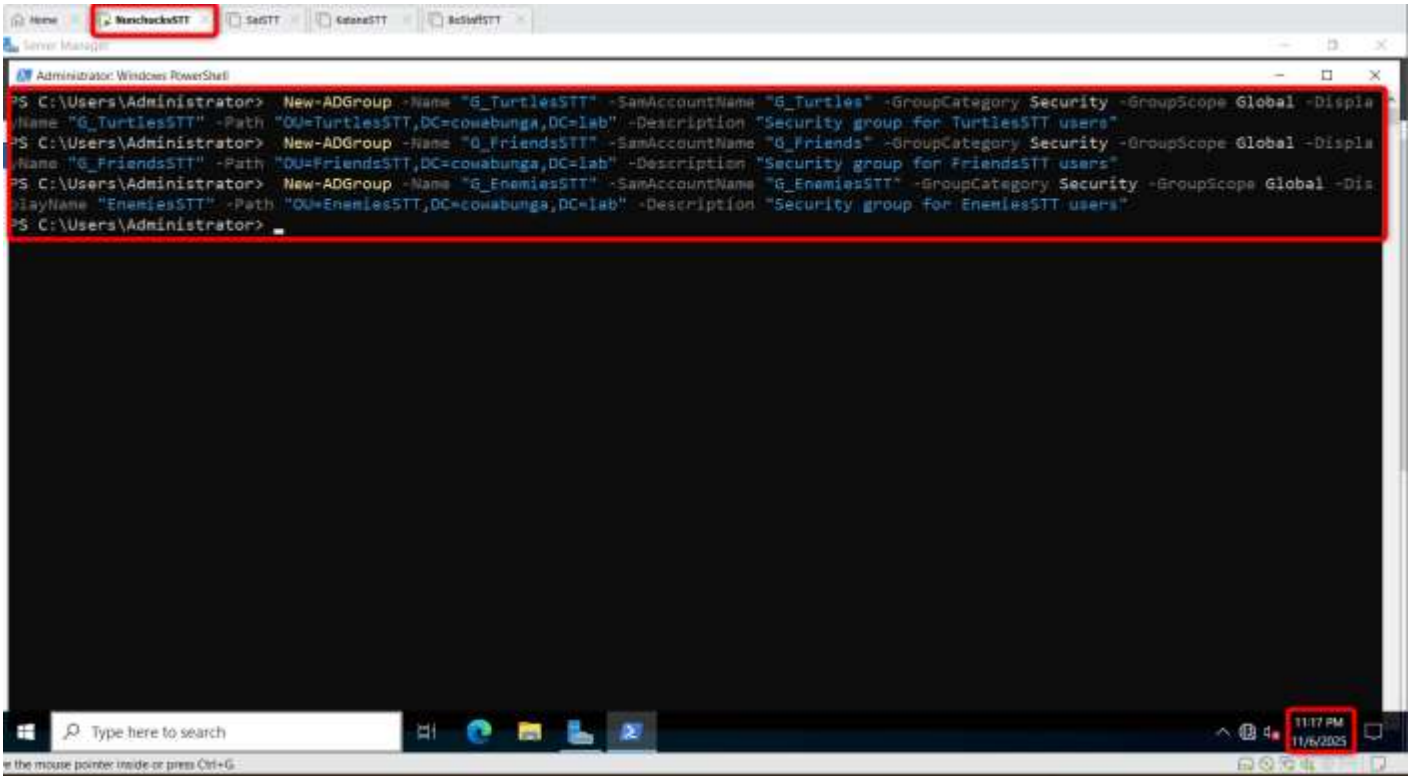
dsquery user "OU=[ou],DC=[domain],DC=[tldn]" -limit 0 | DSMOD user -pwd Password01 -disabled no -mustchpwd no -pwdneverexpires yes



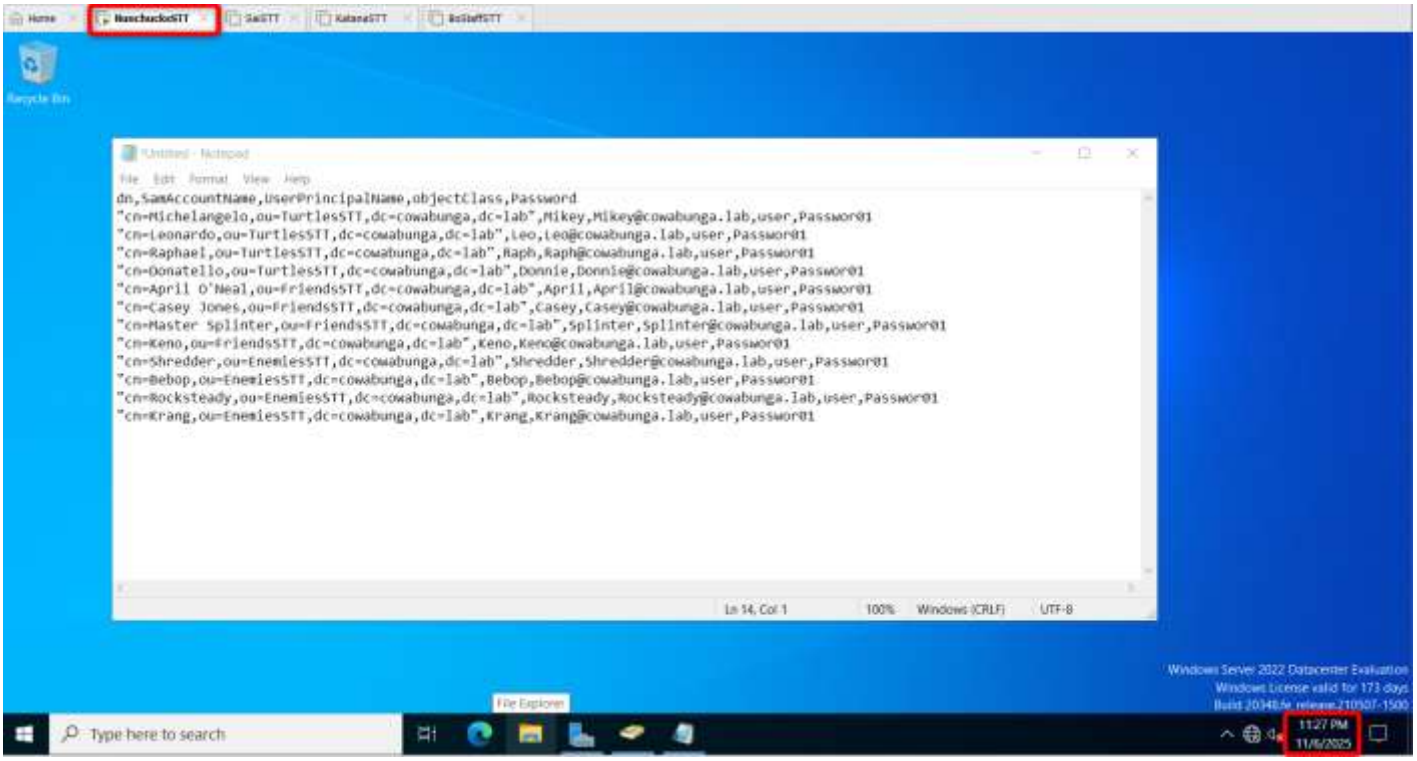
Active Directory Users and Computers (ADUC) to confirm OUs were created successfully.



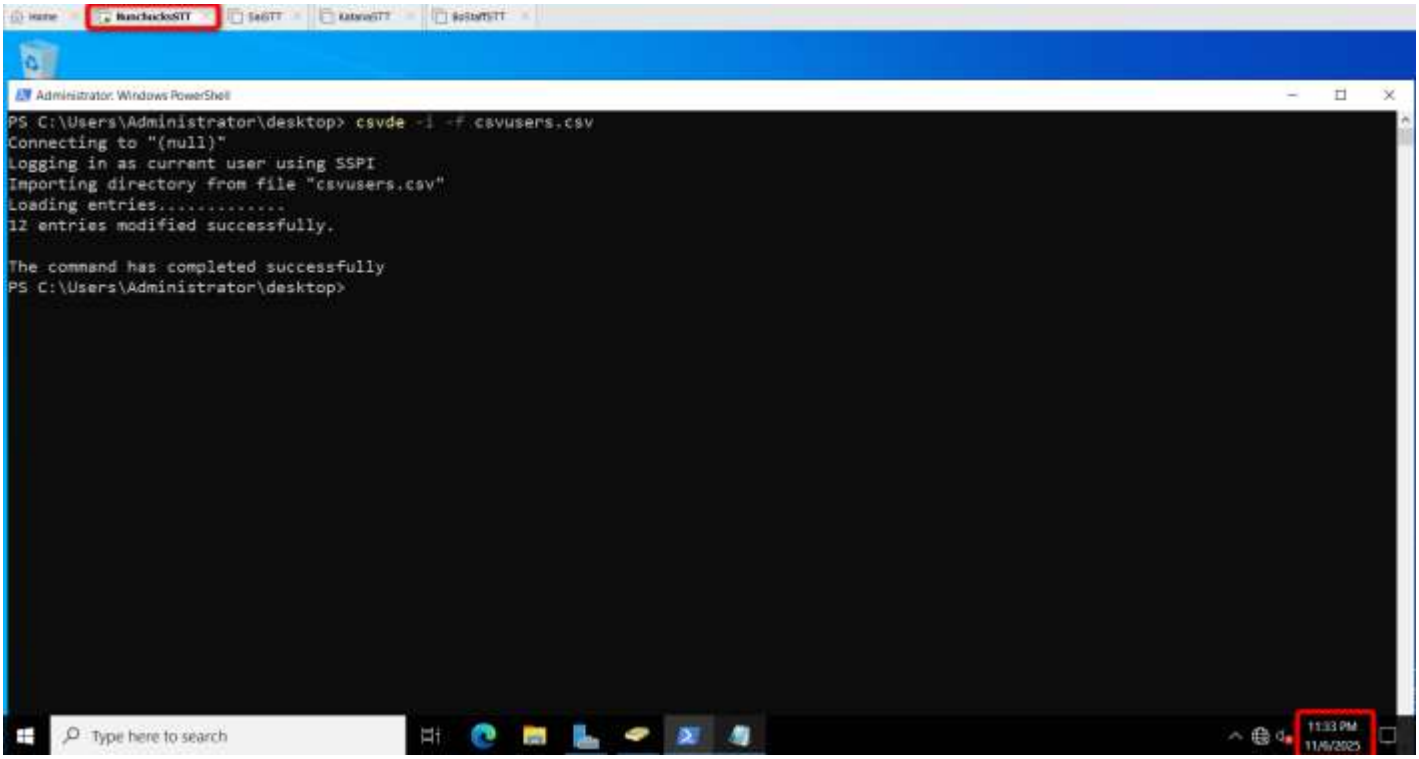
Group Creation



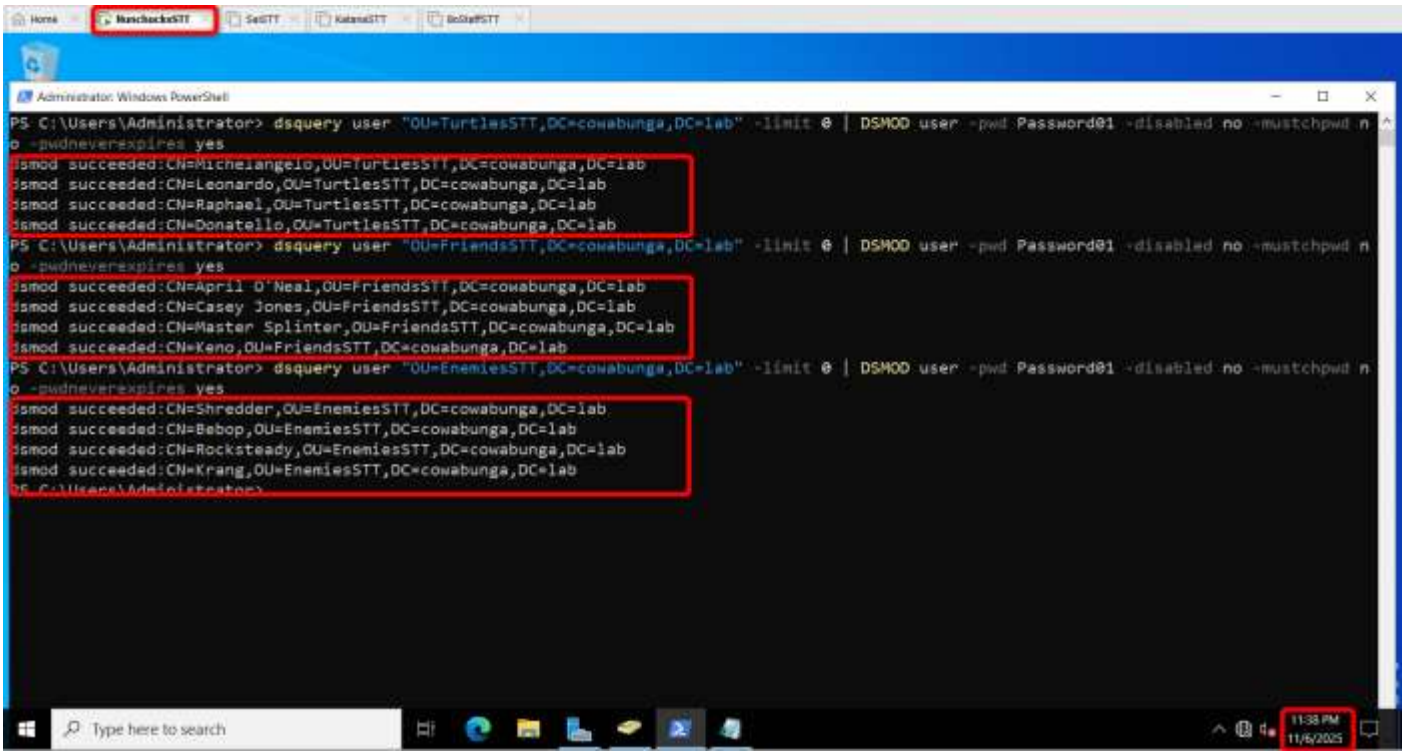
Users Creation via CSV Import



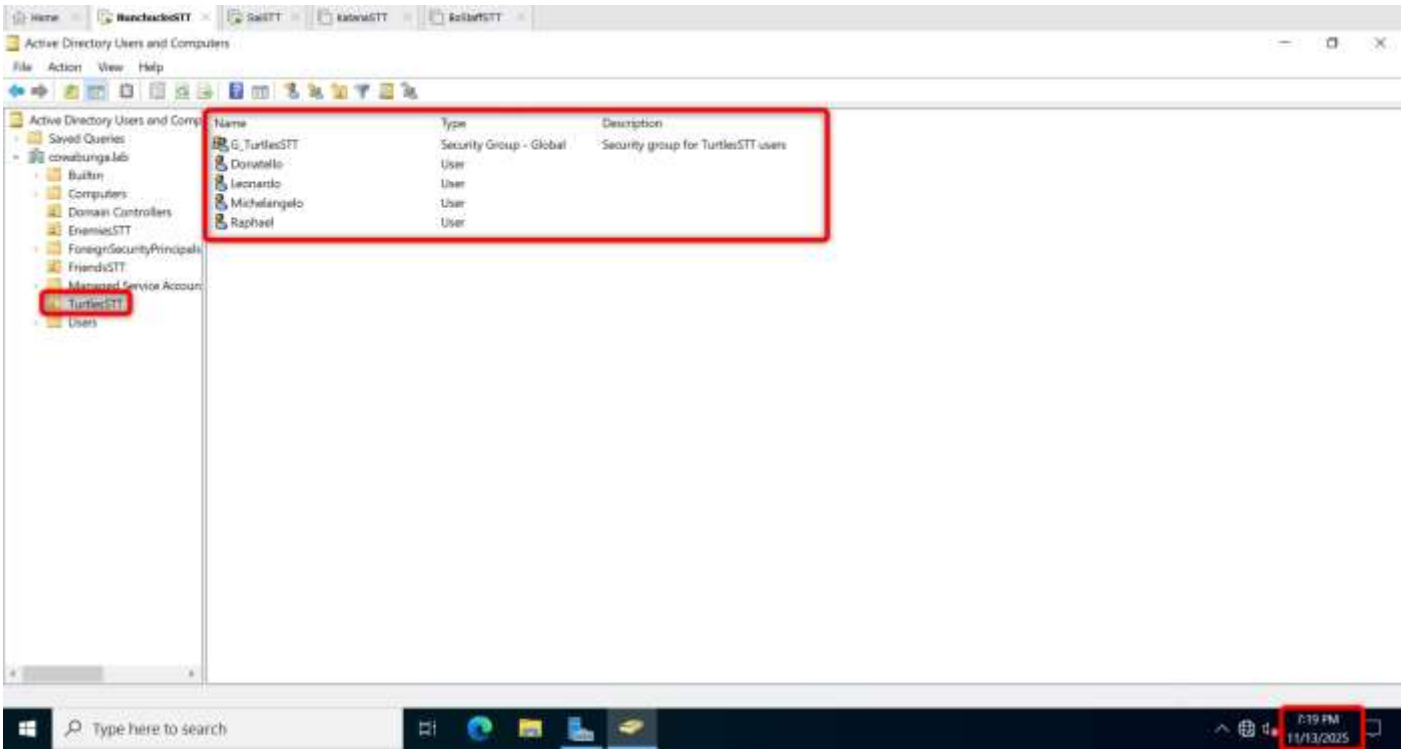
Users' creation 1



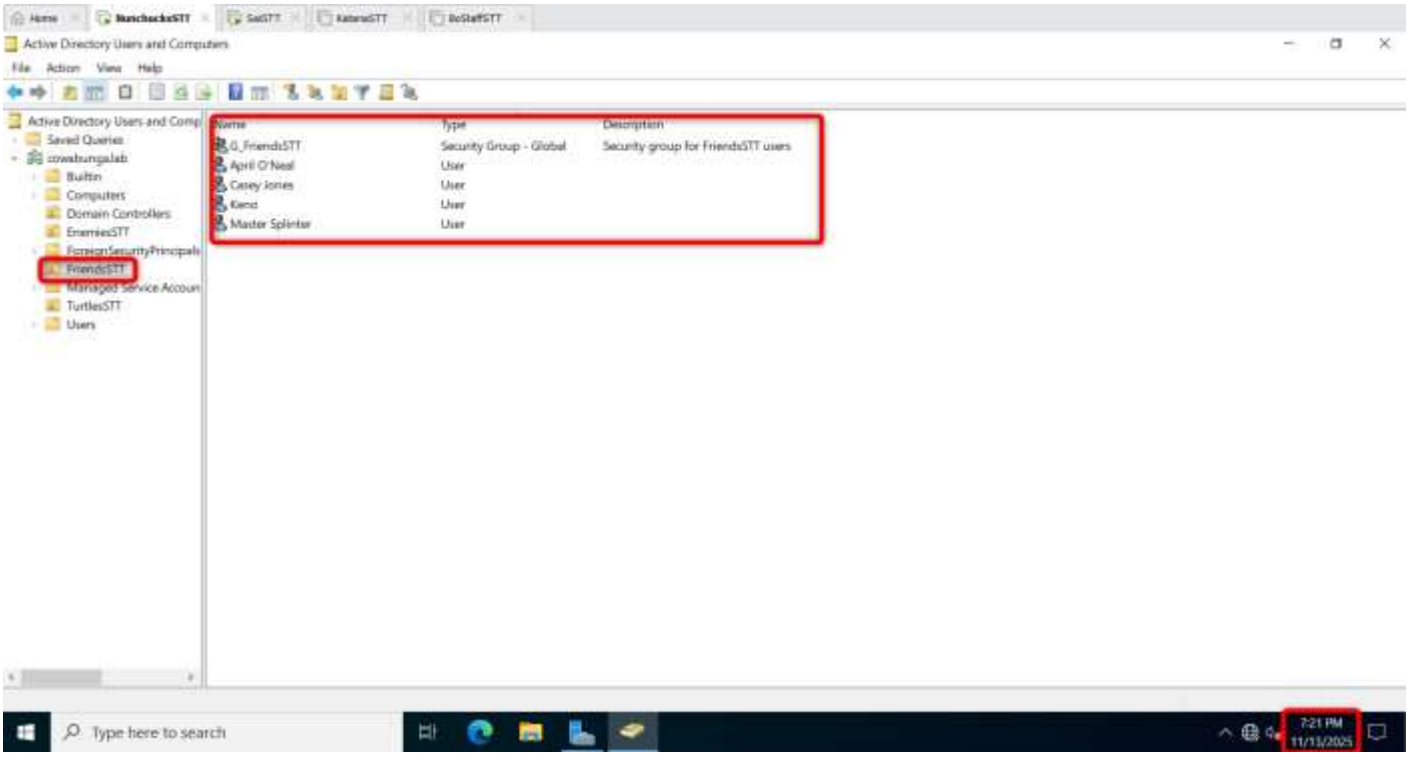
Users' creation 2



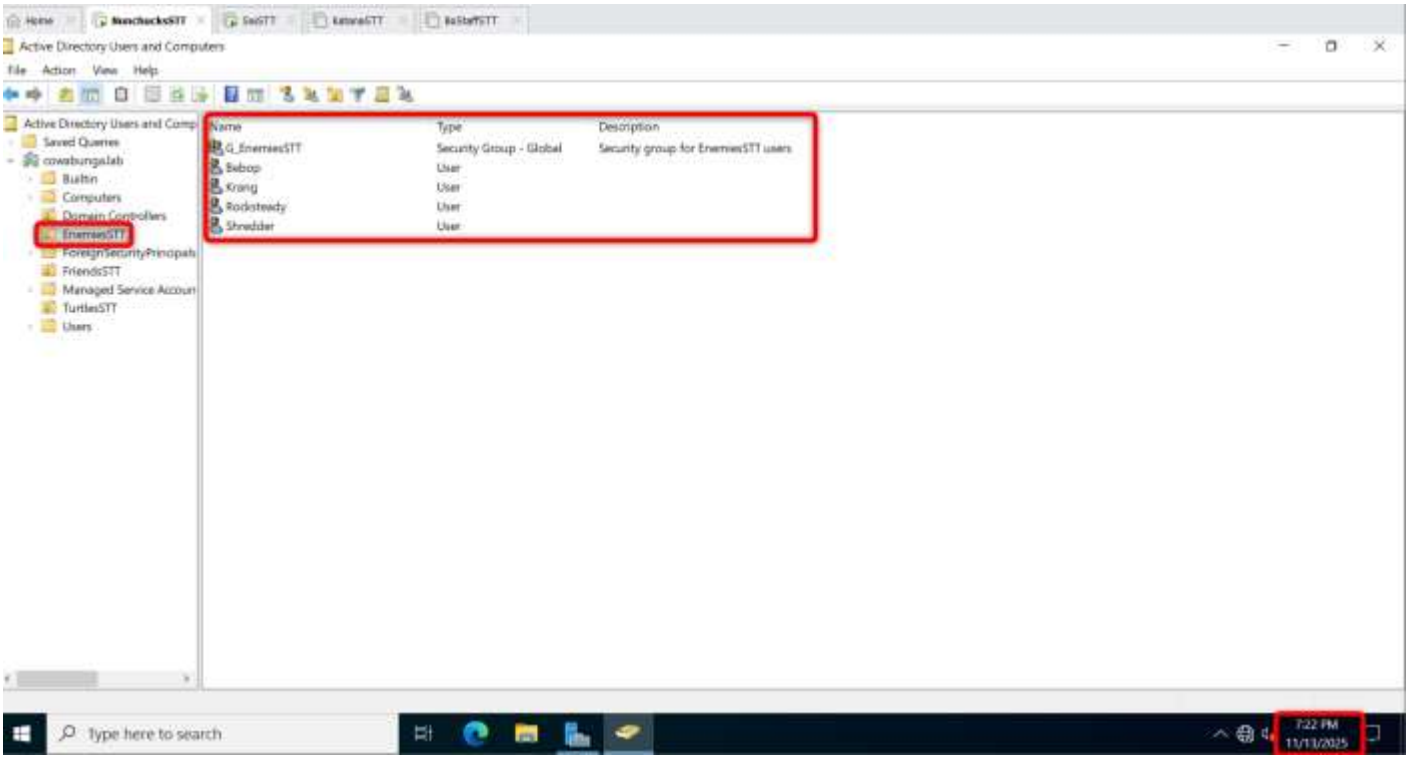
Users' creation 3



Users' creation 4



Users' creation 5



Users' creation 6

SaiSTT - Server 2

This server is running Server Core and does not have a GUI, hence the tasks below are done via the command line.

Setting up as follows:

Step 1: Install ADDS

1. Once SaiSTT is joined to the domain, you can install ADDS using the following command:

Install-WindowsFeature -name AD-Domain-Services -IncludeManagementTools

2. Restart once it is done. To check to ensure the installation was successful, run the following command:

Get-WindowsFeature -name AD*

Step 2: Configure ADDS to Promote SaiSTT to a Domain Controller

1. Run the following command to promote SaiSTT to a Domain Controller:

Install-ADDSDomainController -DomainName "cowabunga.lab" -SkipAutoConfigureDns -Credential (Get-Credential "[cowabunga\administrator")

It will take a few moments to run. Once it does, it will restart the computer automatically for you and apply some updates. Log back in once it is done restarting.

2. Go to NinchiksSTT and open up ADUC. Go to Domain Controllers and check if SaiSTT is in the list.

Step 3. Installing DHCP

1. Install DHCP First:

Install-WindowsFeature DHCP -IncludeManagementTools

2. Check your features to ensure it installed:

Get-WindowsFeature -Name DHCP*

3. Screenshot showing DHCP installed. It has an x in the box next to it in the list.

4. Authorize the server:

Add-DhcpServerInDC -DnsName saistt.cowabunga.lab -IPAddress 172.16.84.2

5. Double-check that this has worked:

```
Get-DhcpServerInDC
```

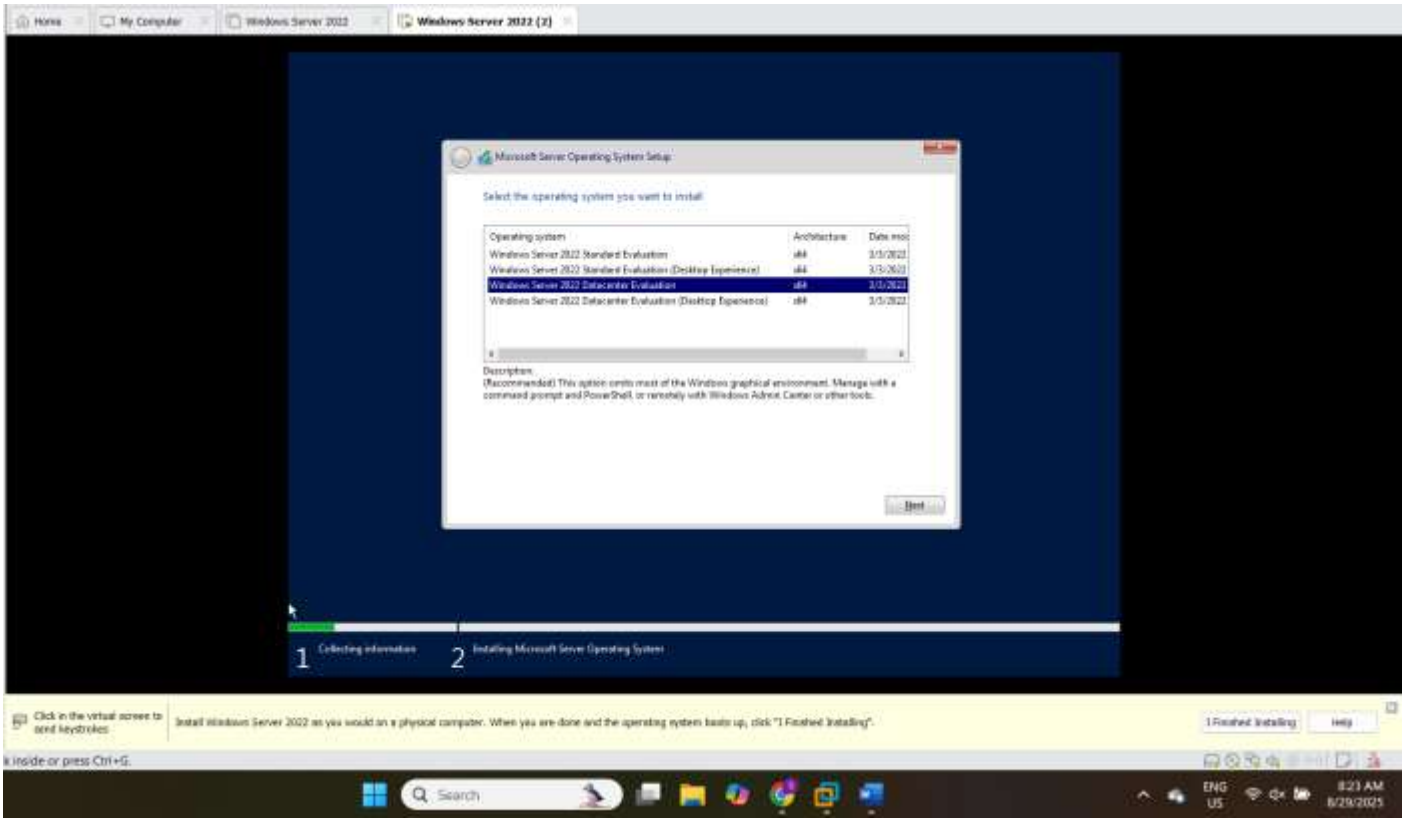
6. Add the scope:

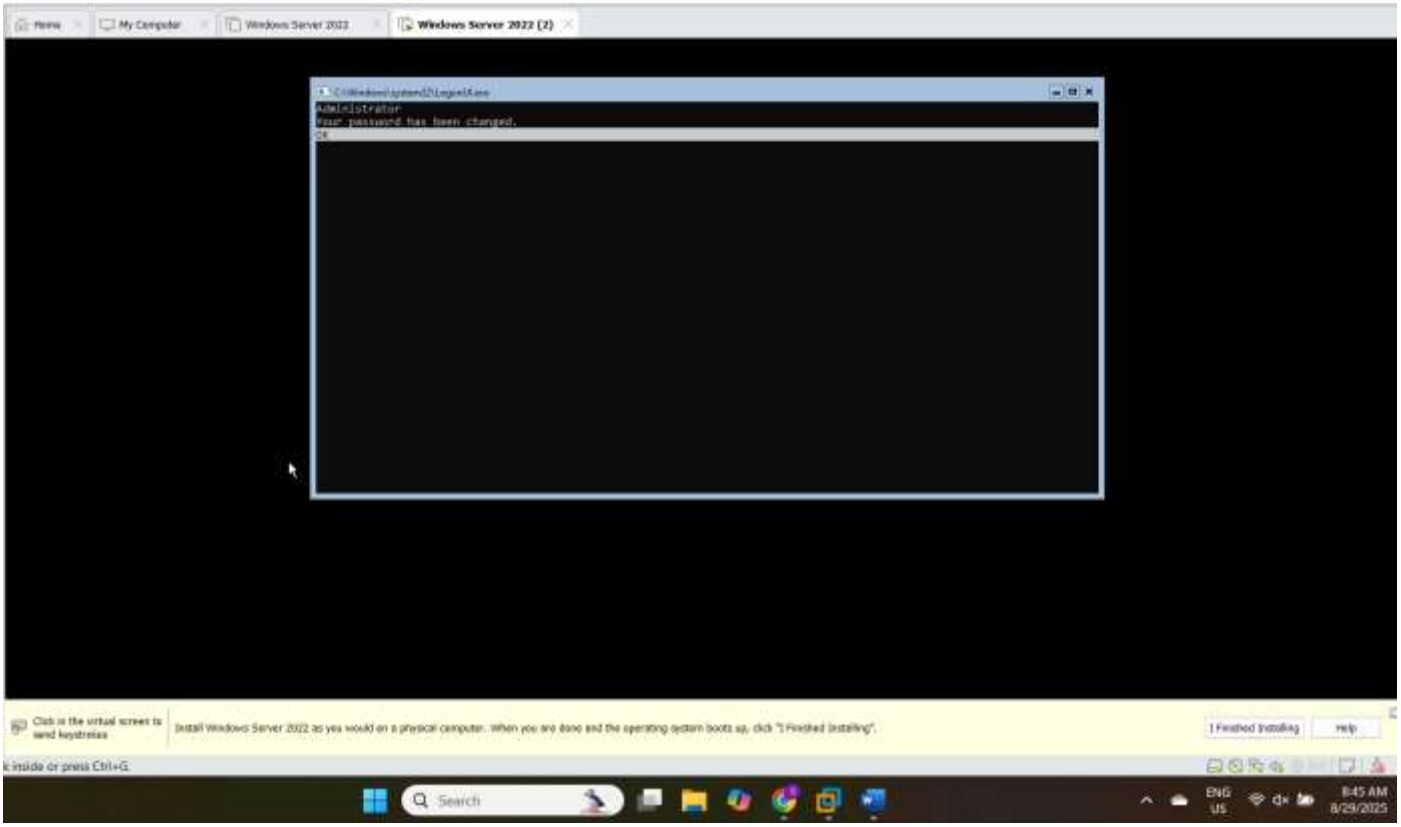
```
Add-DhcpServerv4Scope -Name "CowabungaSTT" -StartRange 172.16.84.15 -EndRange
172.16.84.31 -SubnetMask 255.255.255.0
```

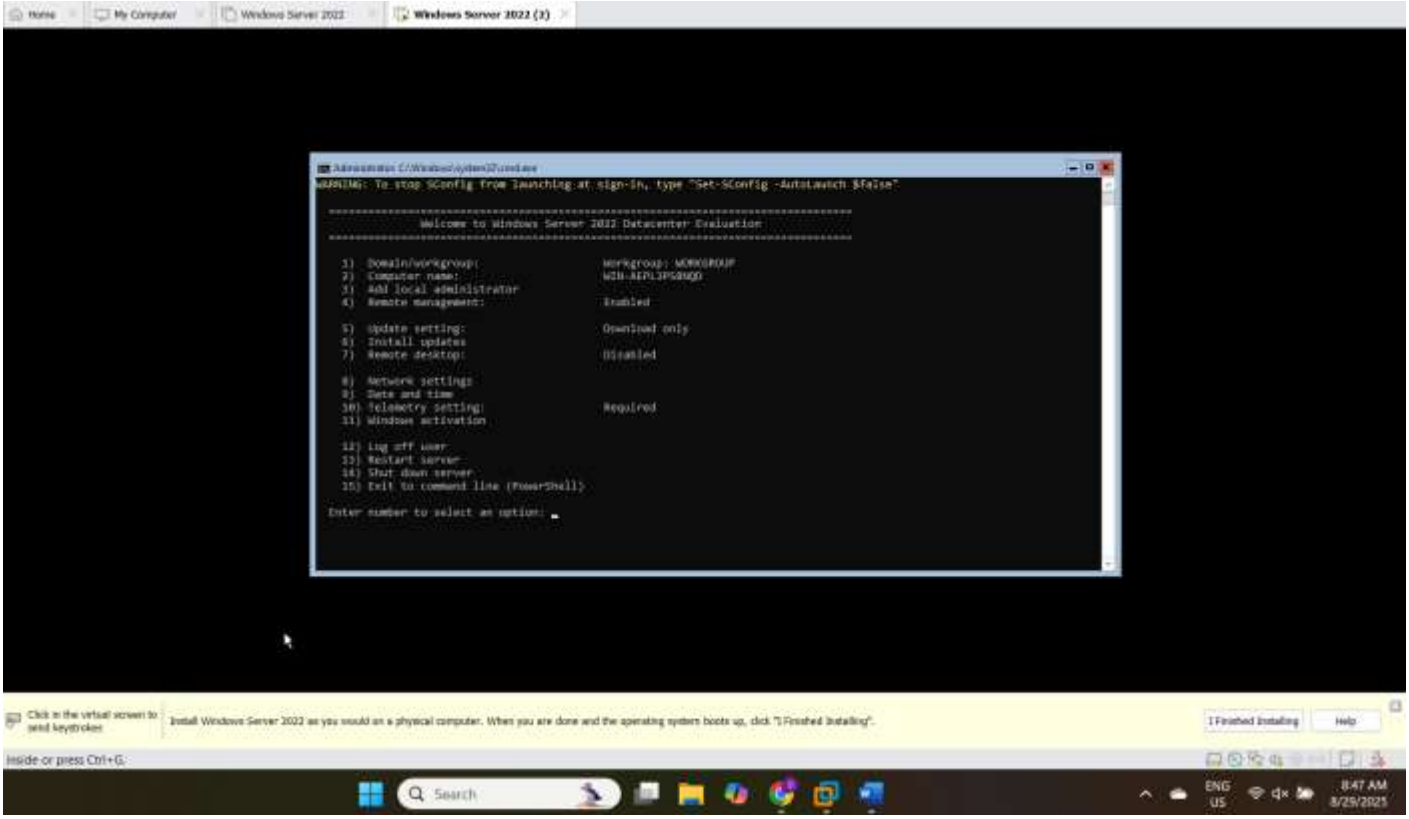
7. Check the range by using the following command:

```
Get-DhcpServerv4Scope
```

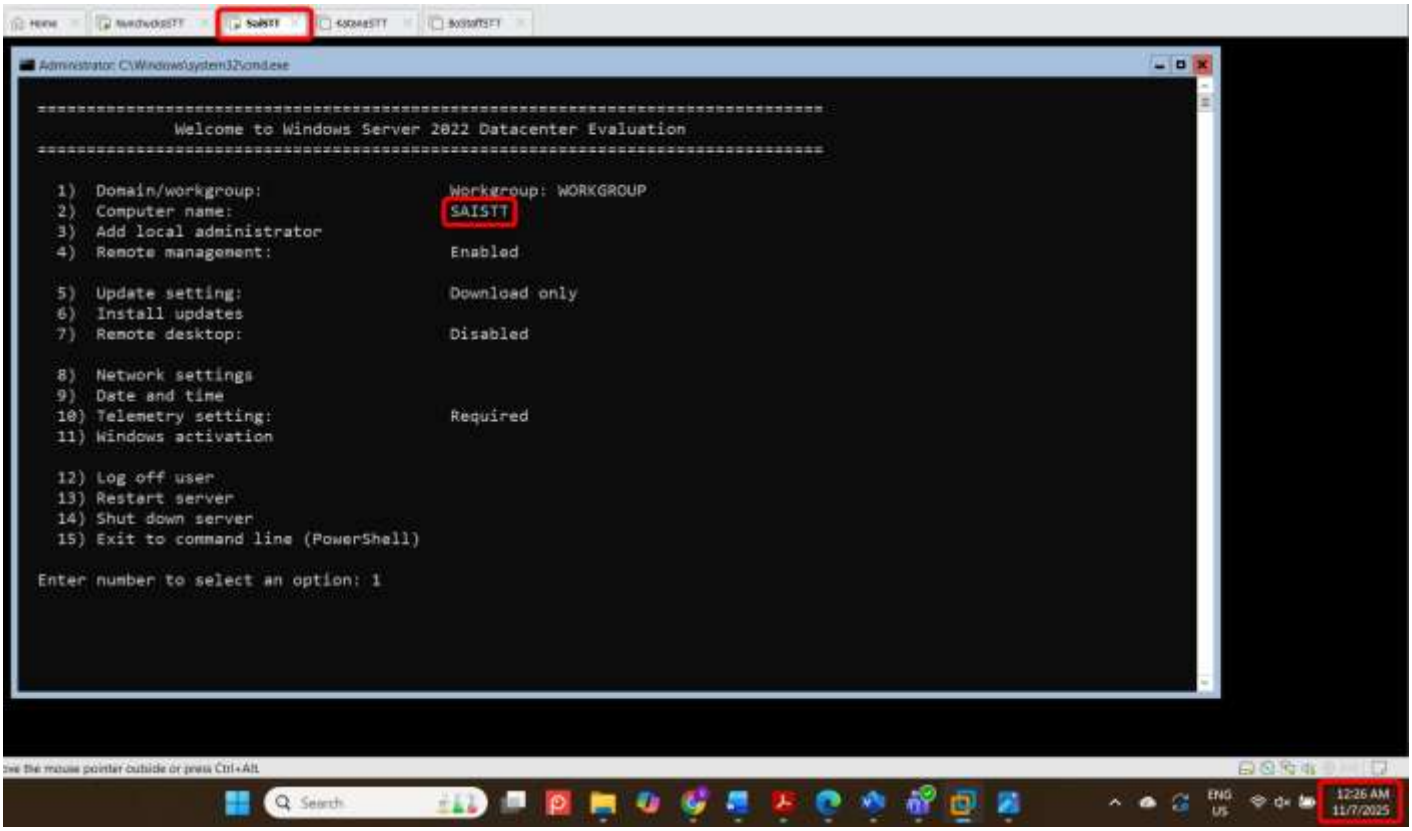
Setting up the server.



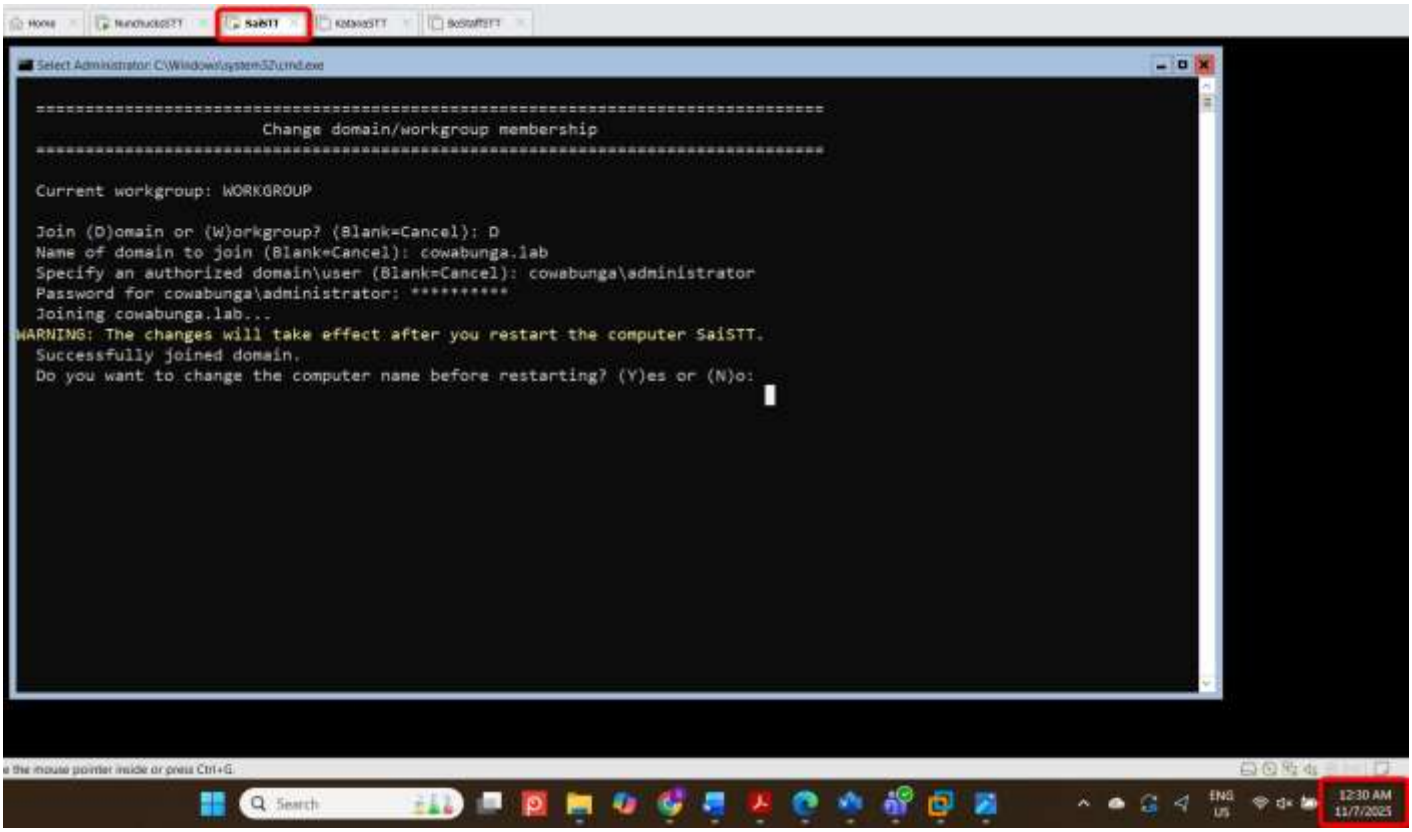



Home
My Computer
Windows Server 2022
Windows Server 2022 (1)
Administrator: C:\Windows\system32\cmd.exe
WARNING: To stop SConfig from launching at sign-in, type "Set-SConfig -AutoLaunch $False"
Welcome to Windows Server 2022 Datacenter Evaluation
1) Domain/workgroup: Workgroup: WORKGROUP
2) Computer name: WIN-AEPL13PS8UQ
3) Add local administrator
4) Remote management: Enabled
5) Update setting: Download only
6) Install updates
7) Remote desktop: Disabled
8) Network settings
9) Date and time
10) Telemetry setting: Required
11) Windows activation
12) Log off user
13) Restart server
14) Shut down server
15) Exit to command line (PowerShell)
Enter number to select an option:
Click in the virtual screen to send keystrokes
Install Windows Server 2022 as you would on a physical computer. When you are done and the operating system boots up, click "I Finished Installing".
I Finished Installing
Help
Inside or press Ctrl+G.
Search
ENG
US
8:47 AM
8/25/2025

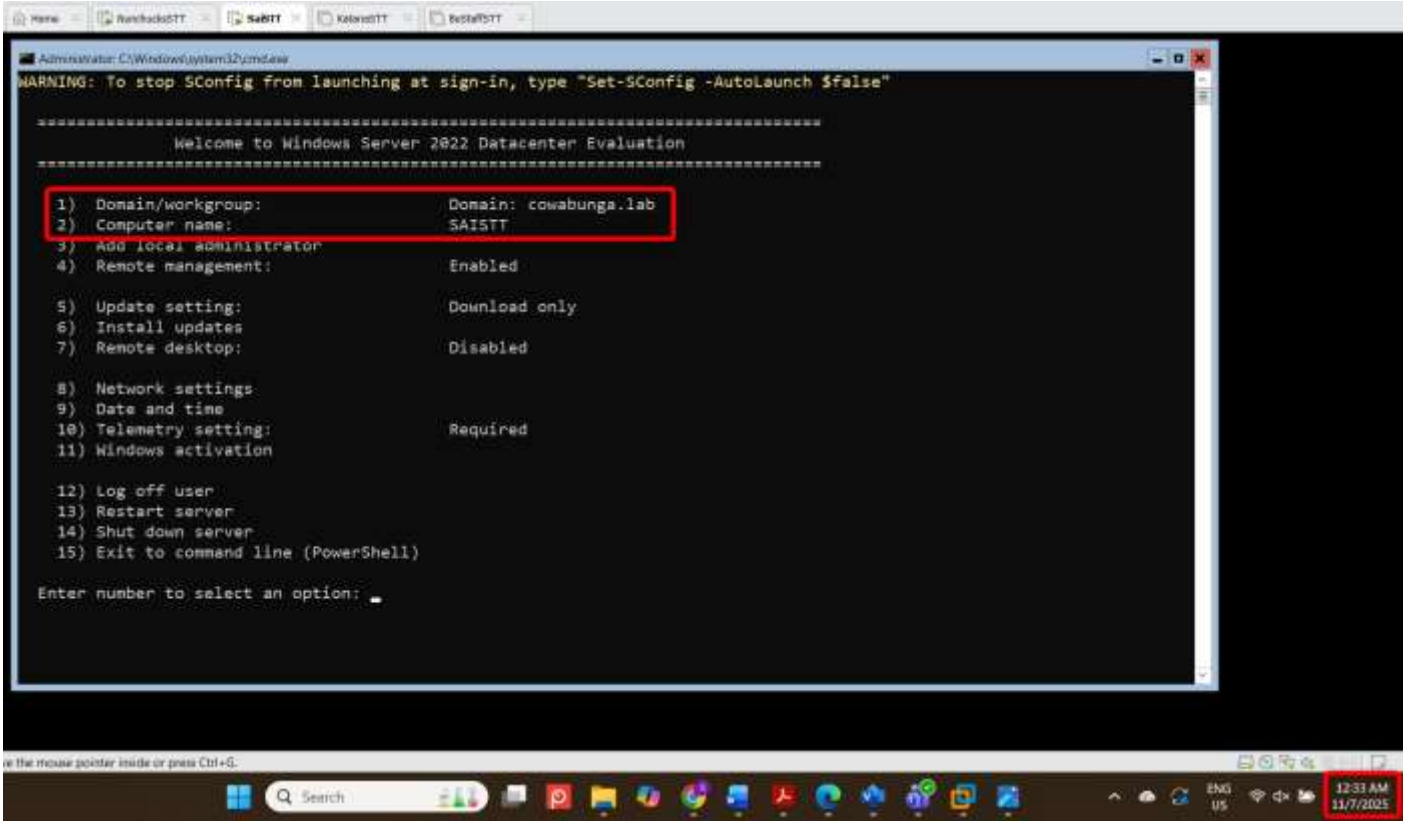
Join SaiSTT to the domain



Join SaiSTT to the Domain 1

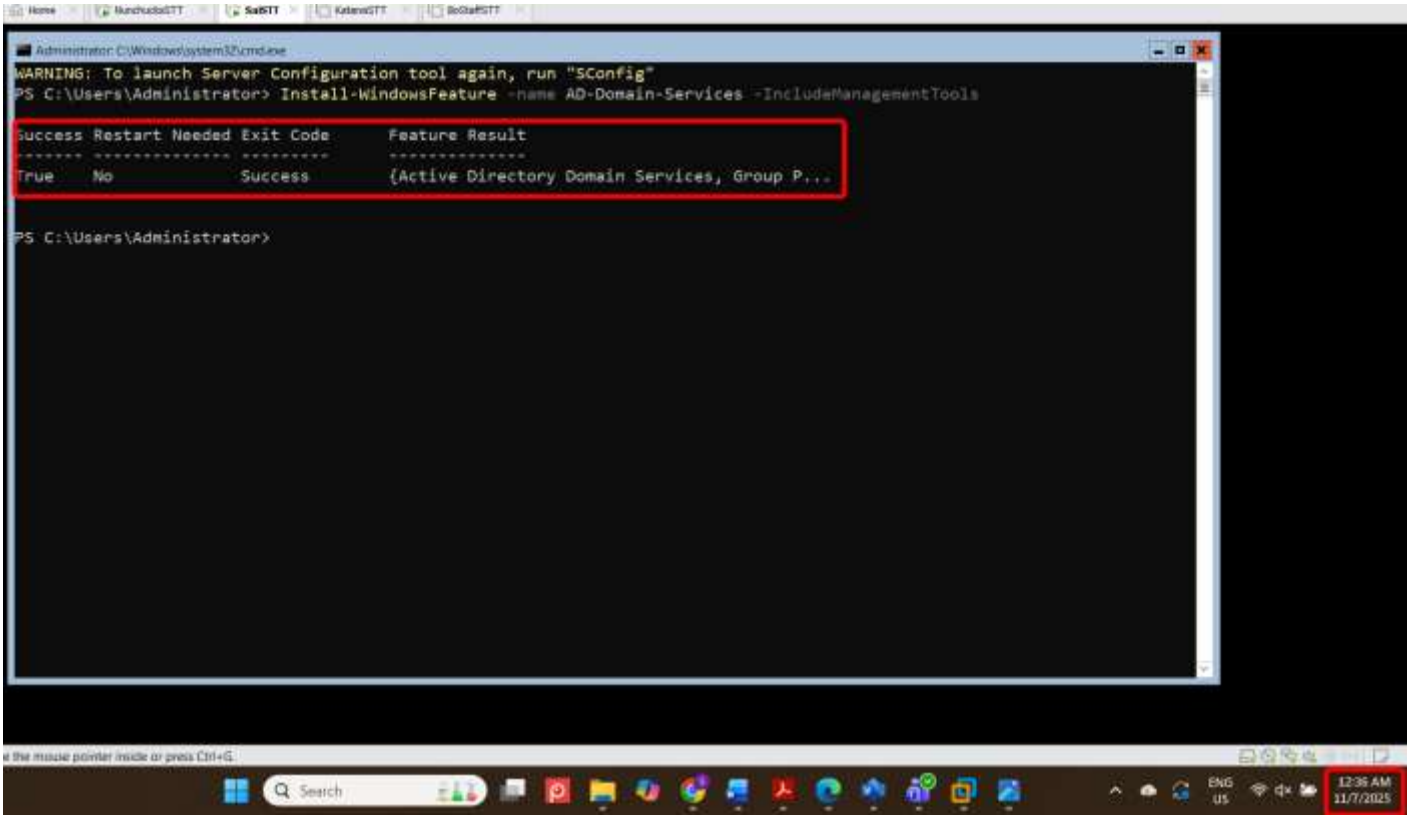


Join SaiSTT to the Domain 2

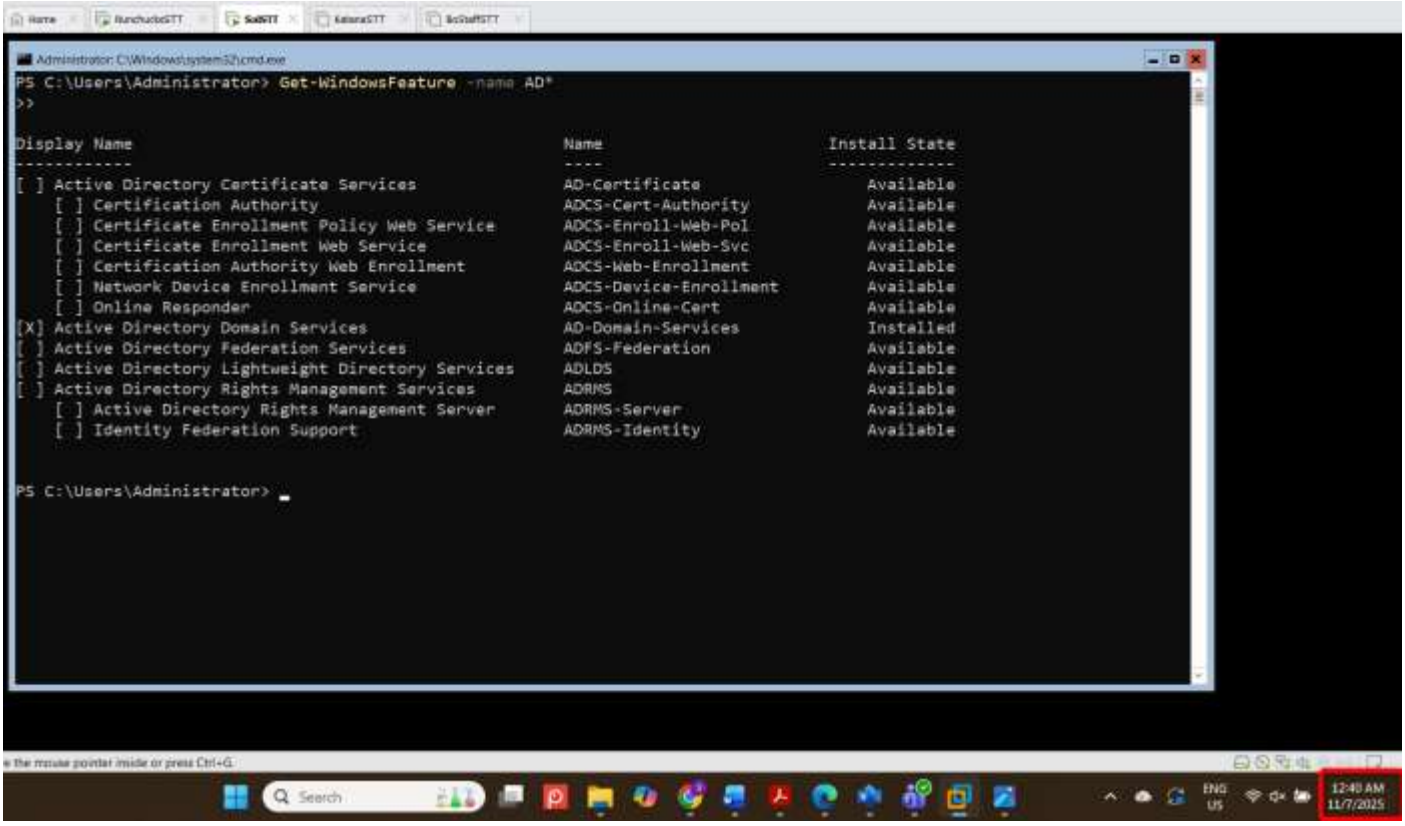


Join SaiSTT to the Domain 3

Install AD DS

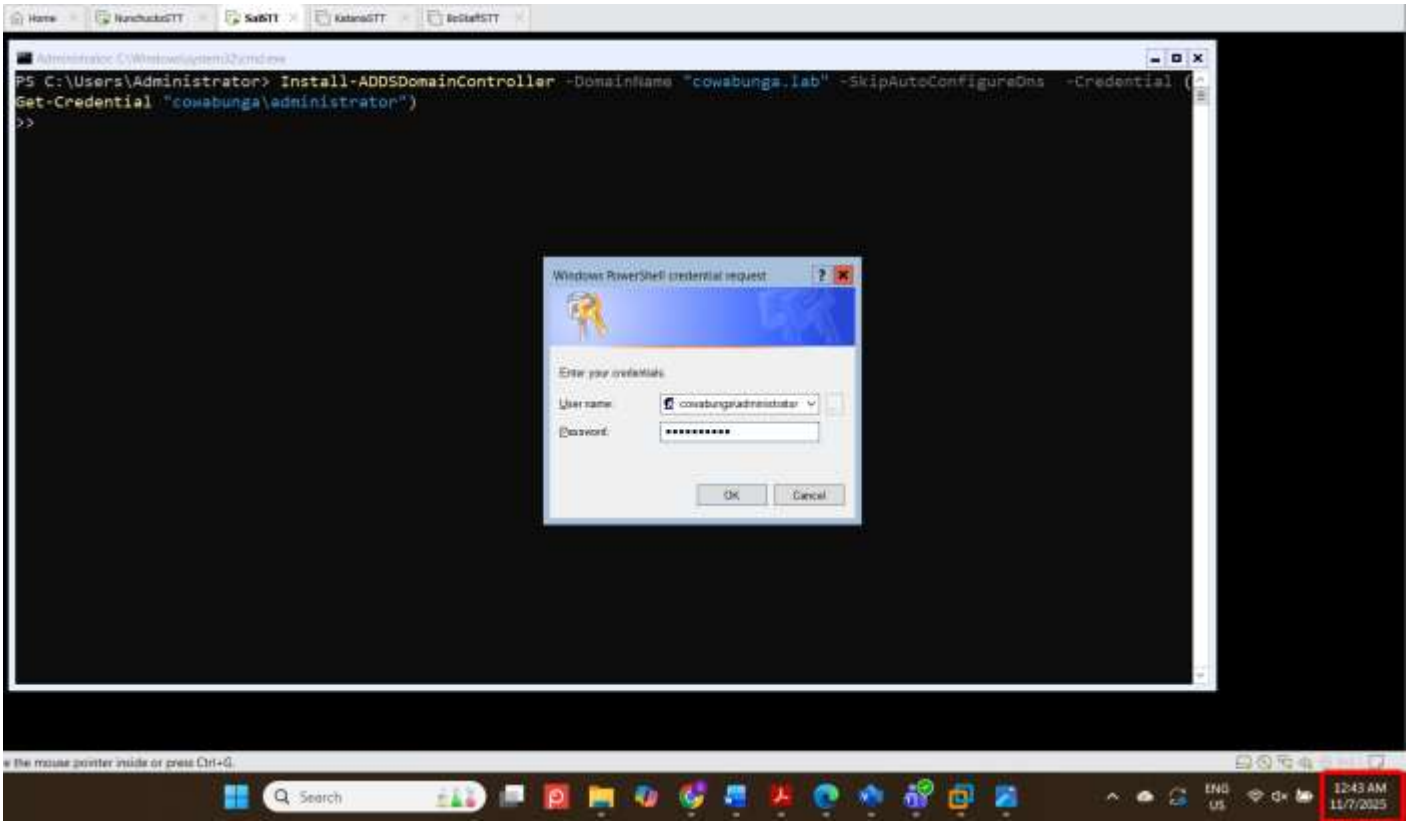


Install AD DS 1

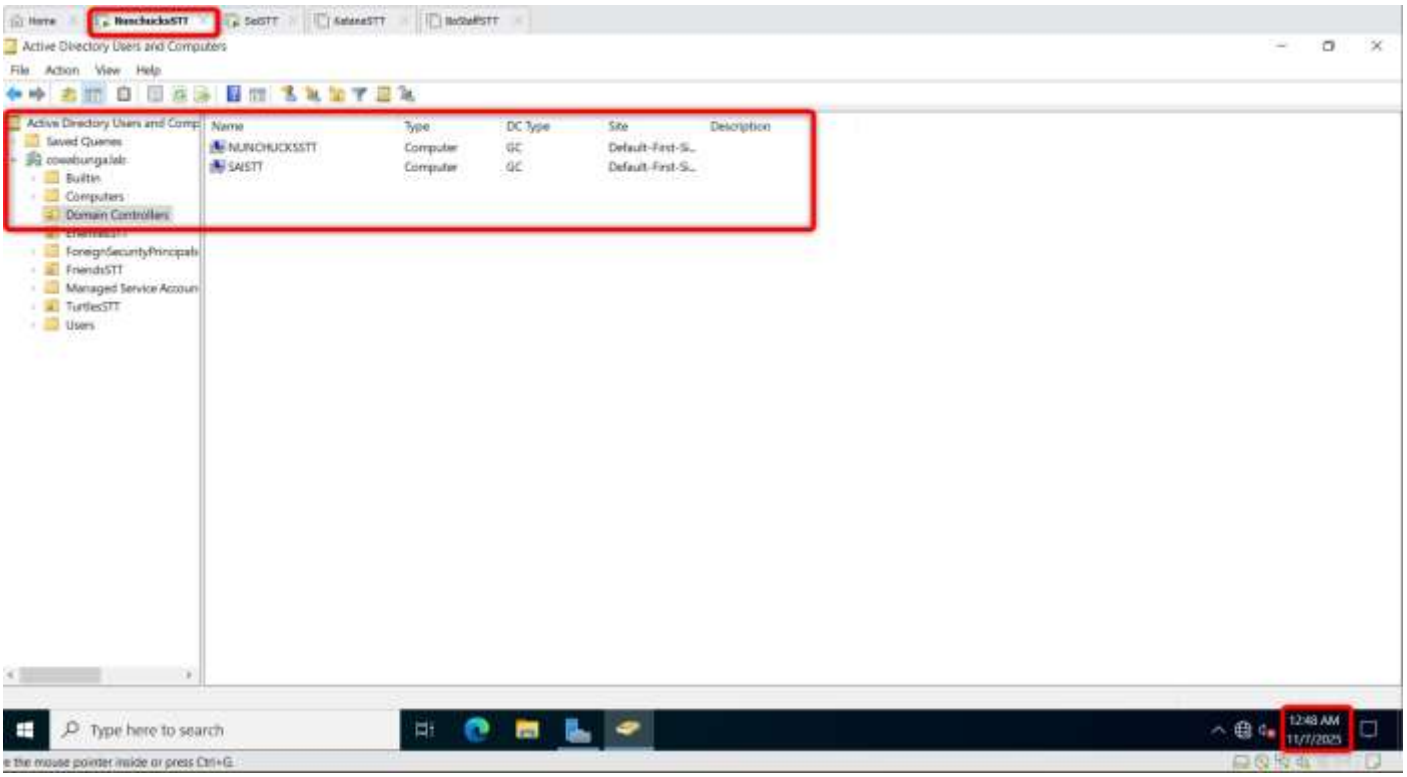


Install AD DS 2

Configure ADDS to Promote SaiSTT to a Domain Controller

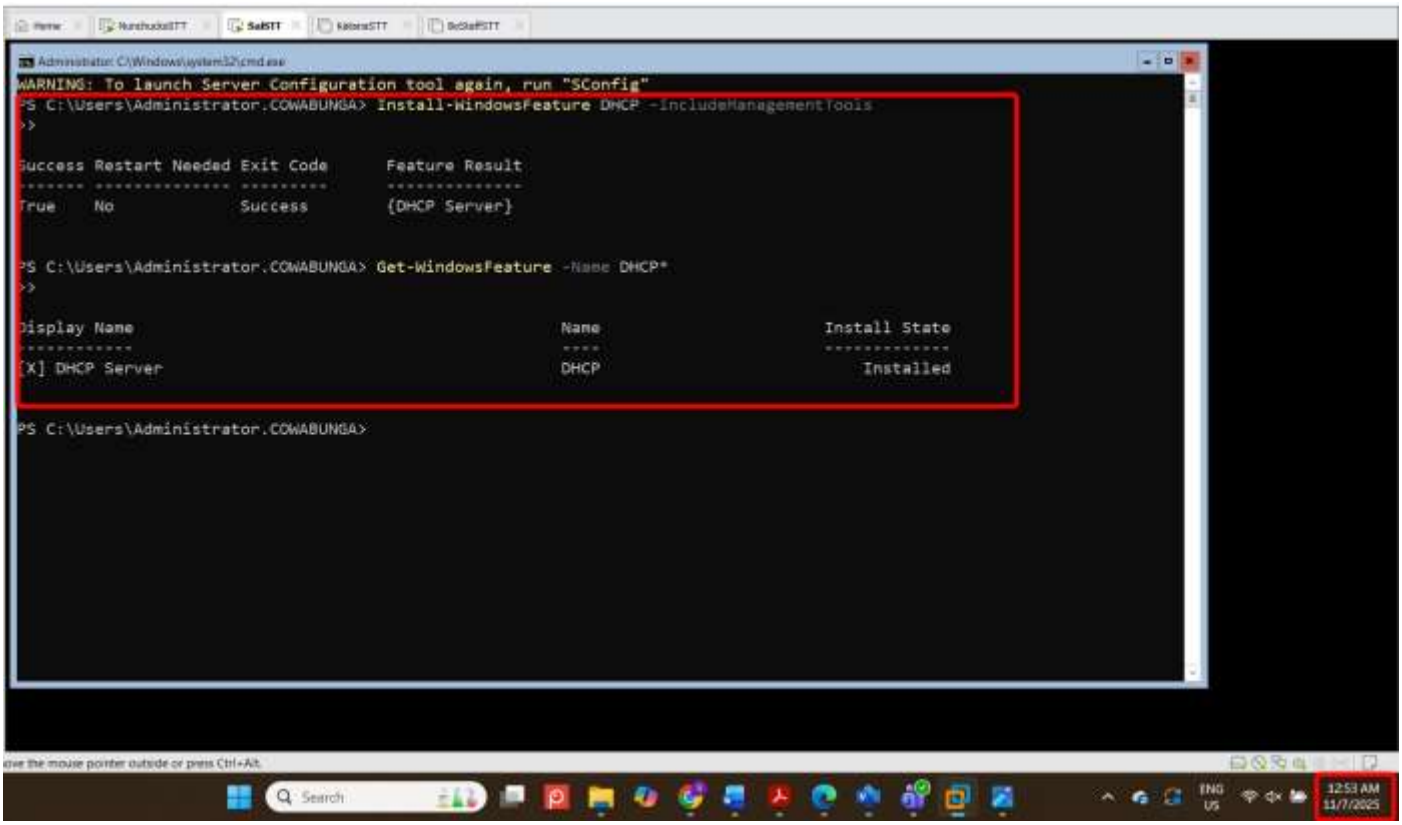


Promote SaiSTT to a Domain Controller 1

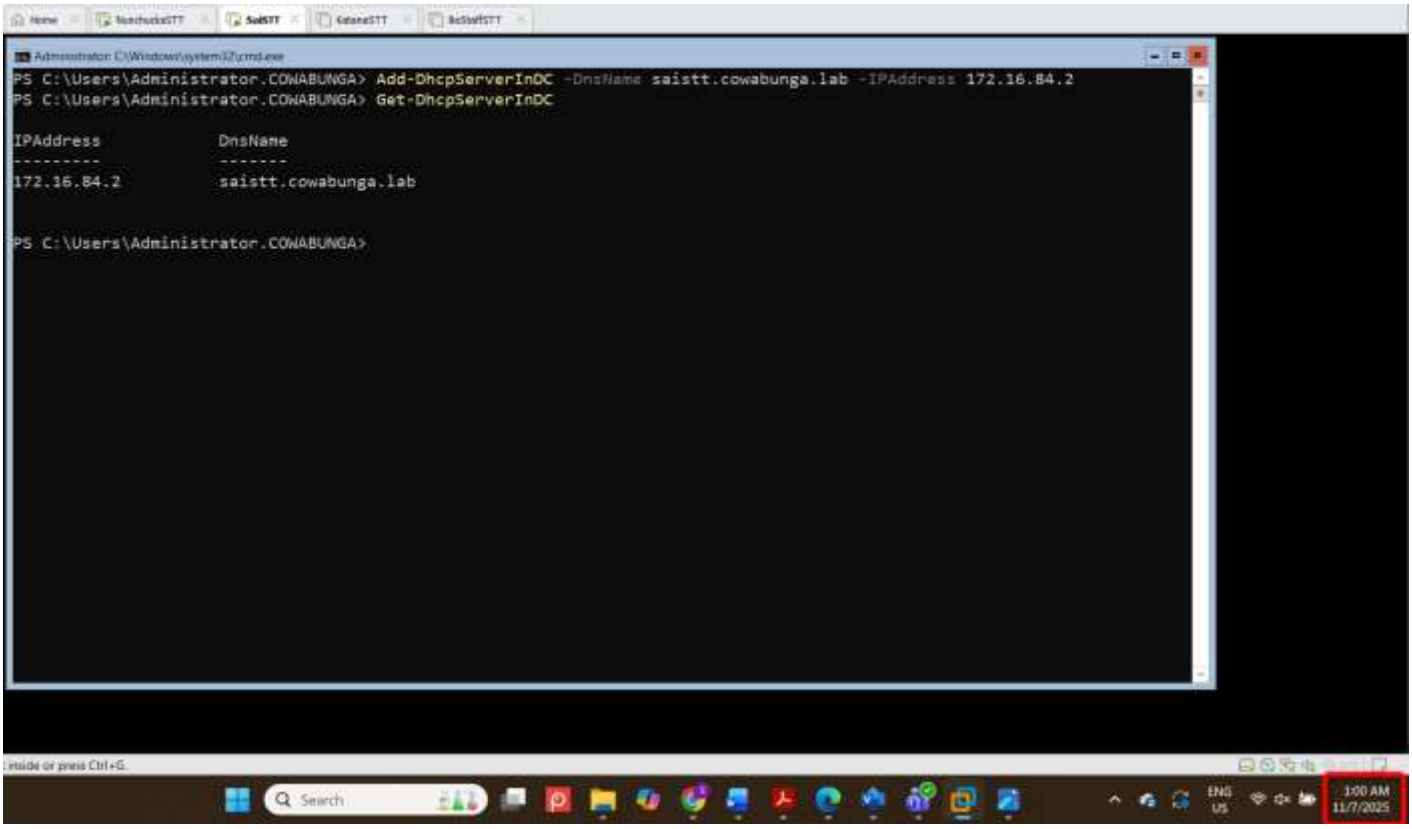


Promote SaiSTT to a Domain Controller 2

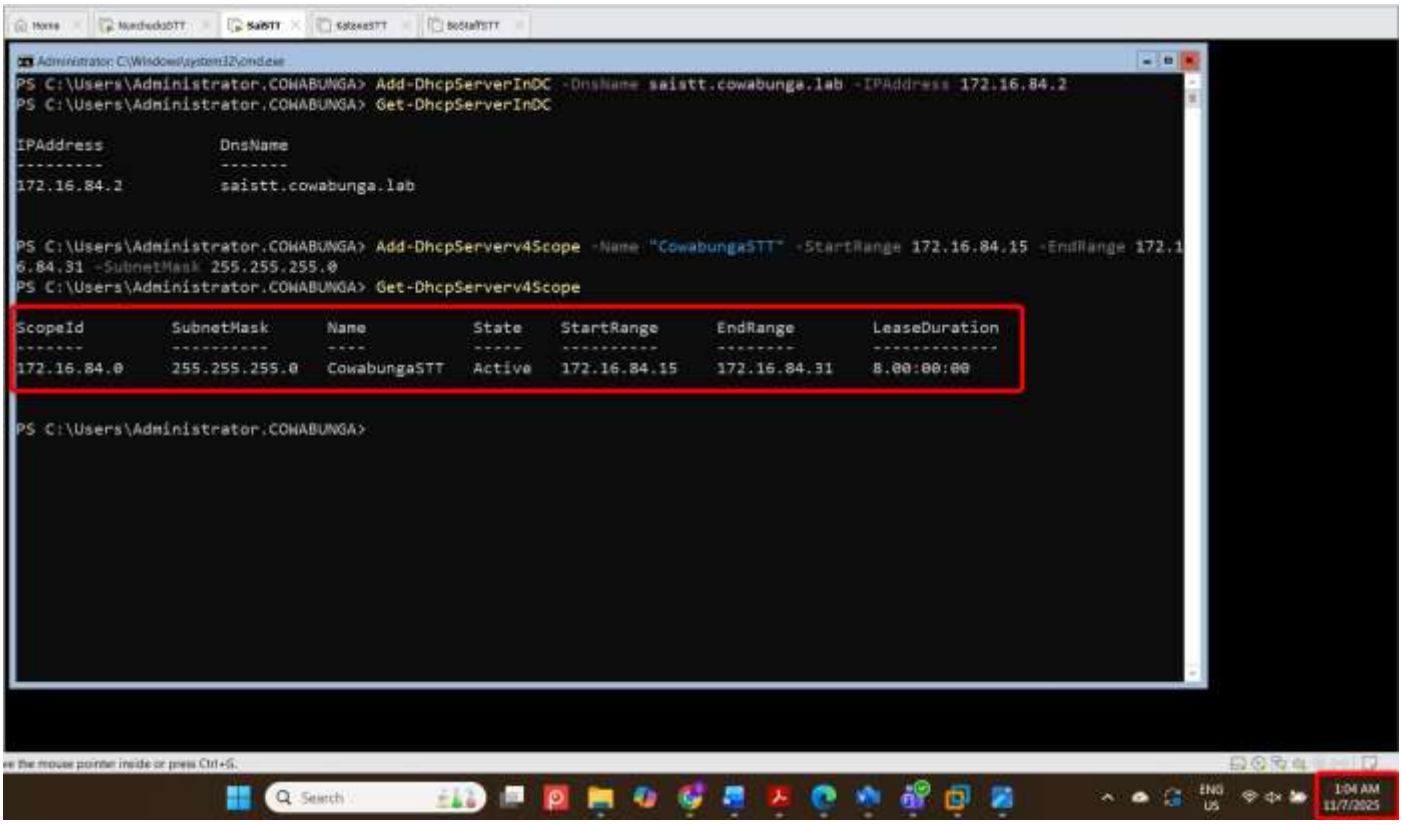
Installing DHCP



Installing DHCP 1



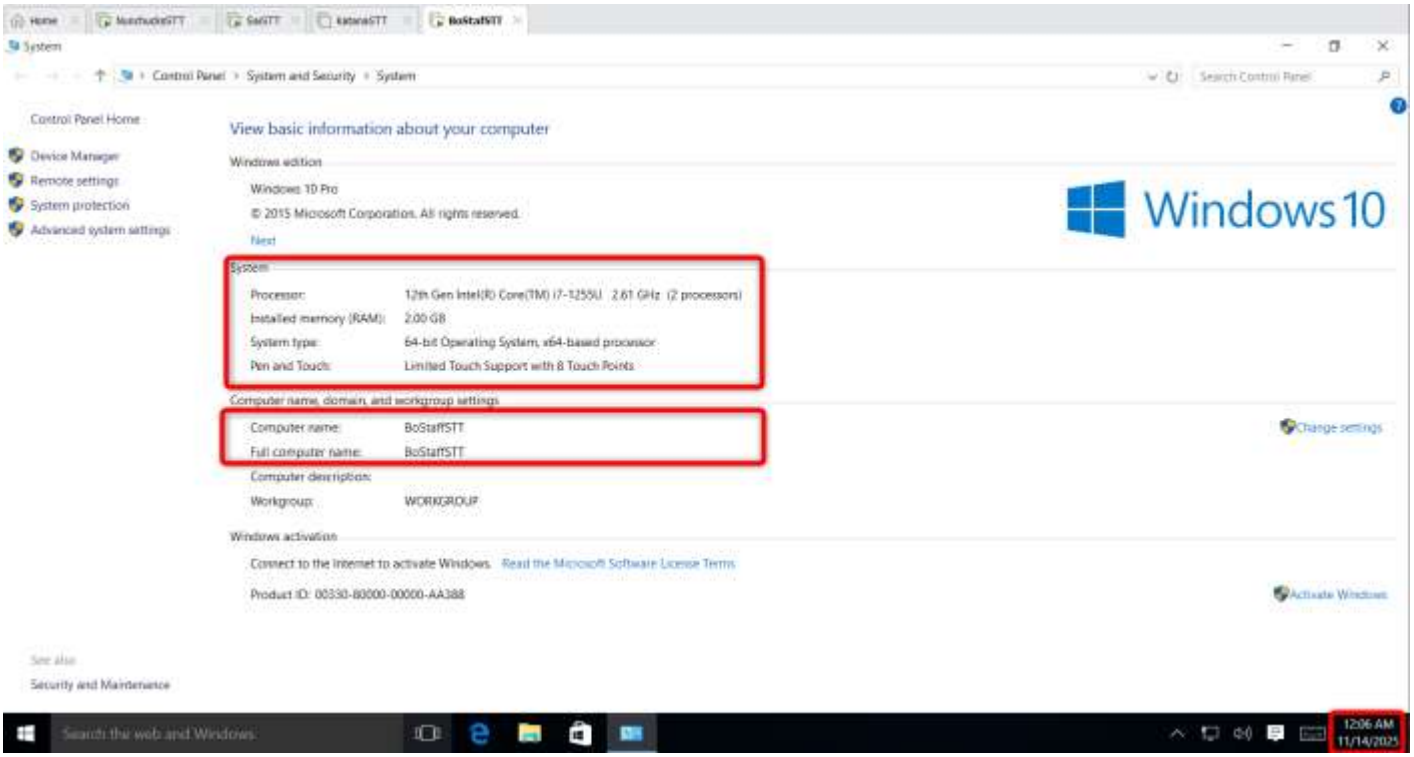
Installing DHCP 2



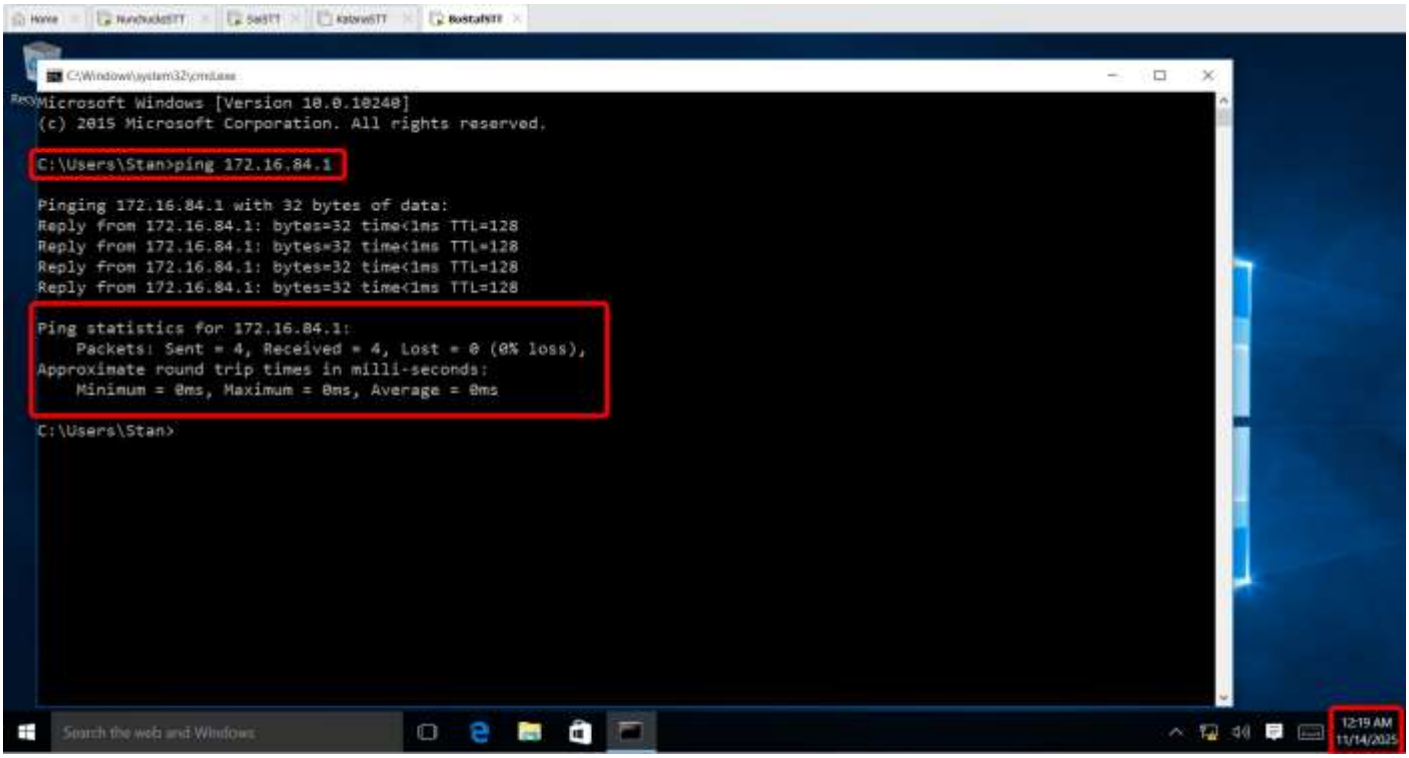
Installing DHCP 3

Client Computer Build

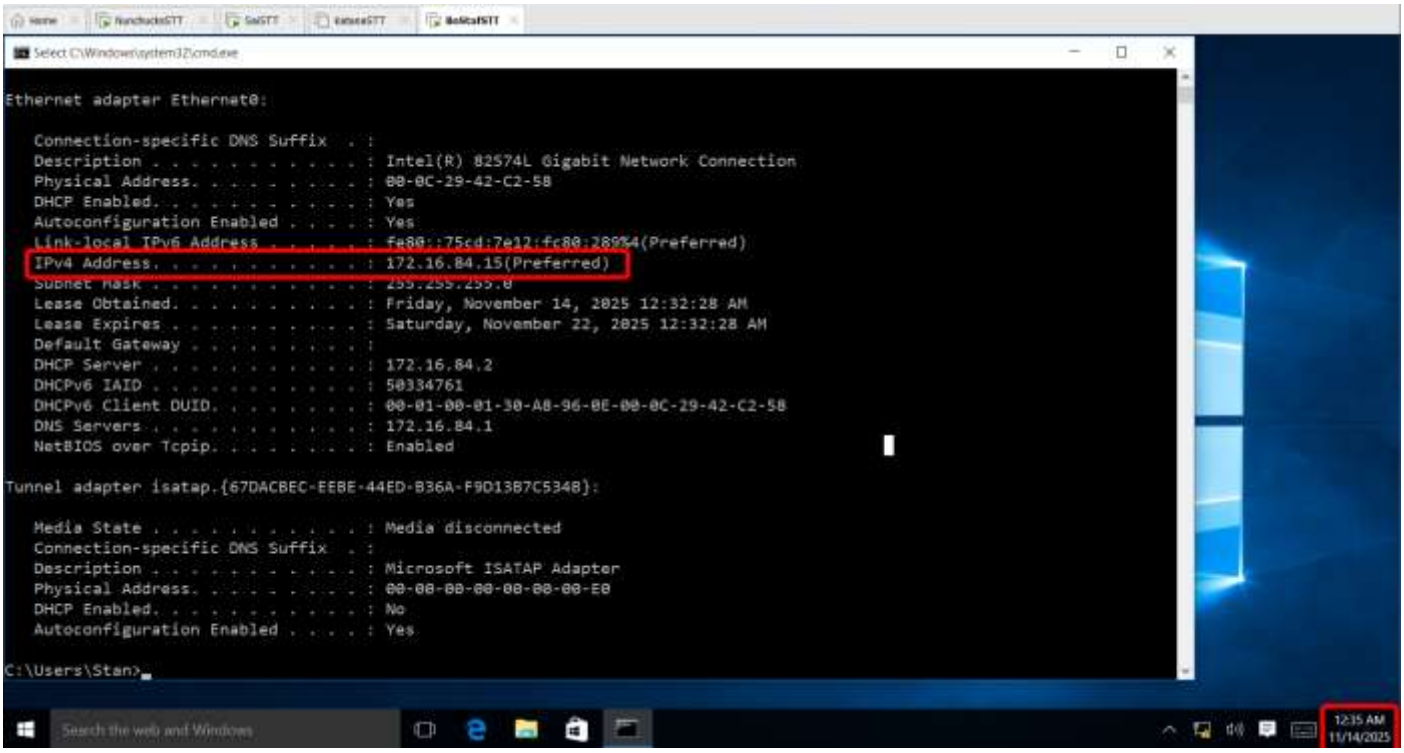
Windows 10 - System > About screen



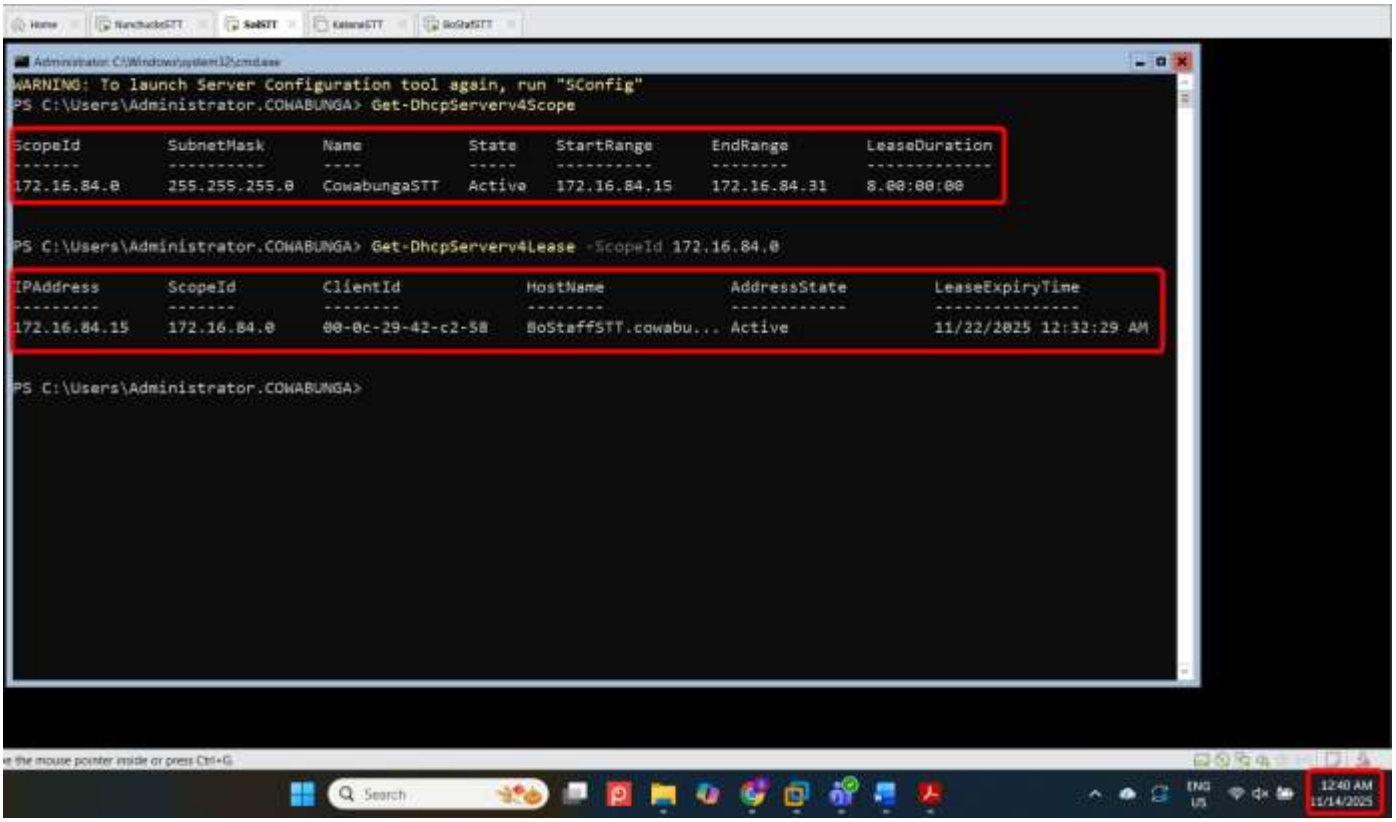
Screenshot of the ping (showing connectivity)



Screenshot from the client showing a dhcp address (Ipconfig /all)



Screenshot from the DHCP command on SaiSTT (Server 2) showing the client in the address leases

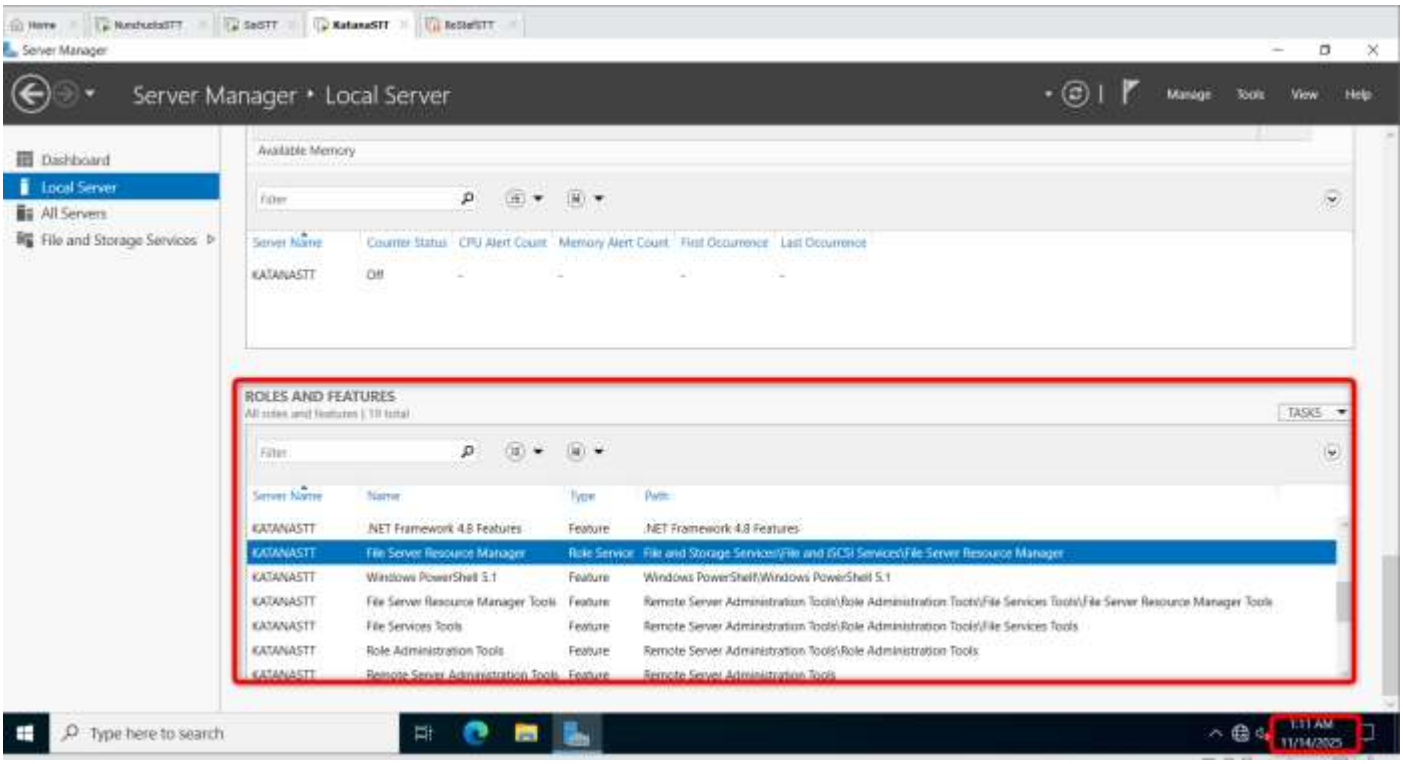


KatanaSTT - Server 3

Setting up the server as follows:

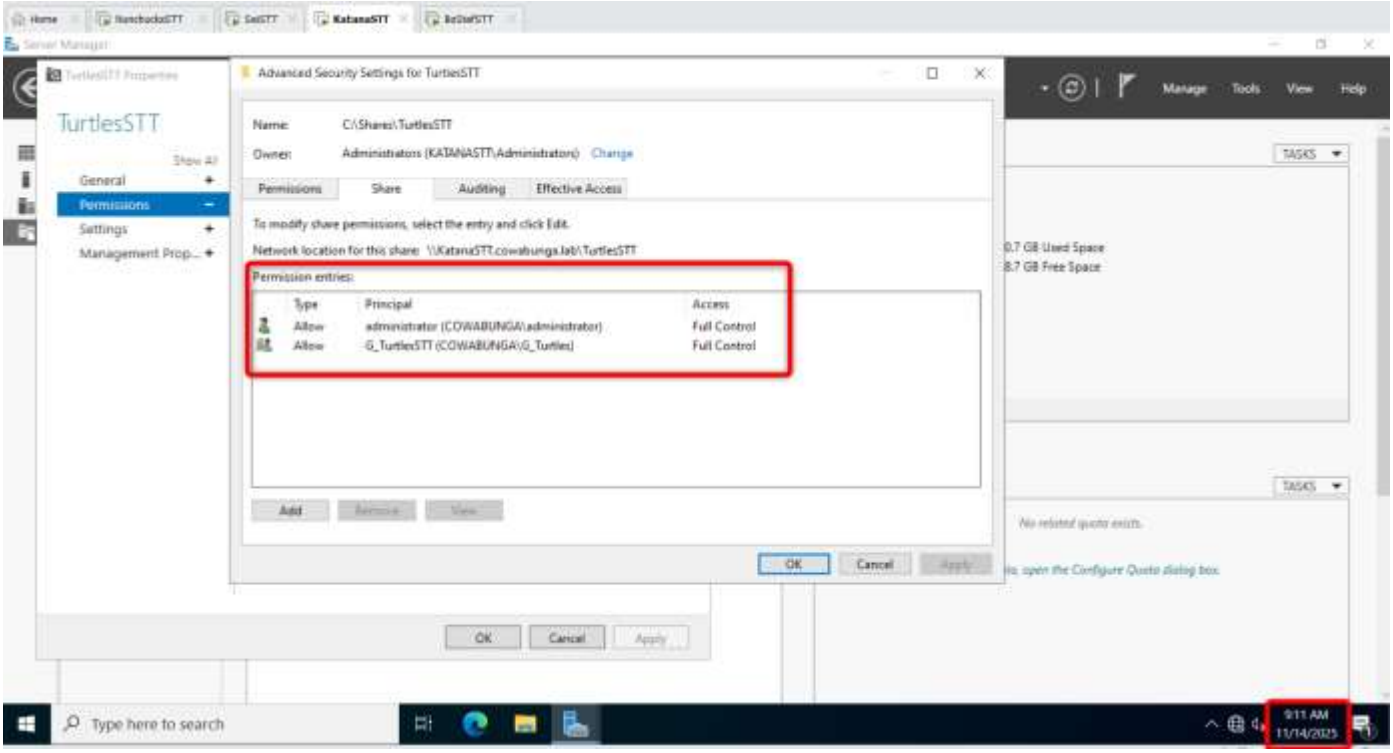
The same steps as NunchucksSTT Server 1 with GUI (ref. link Server 1 configuration)

FSRM is installed on KatanaSTT

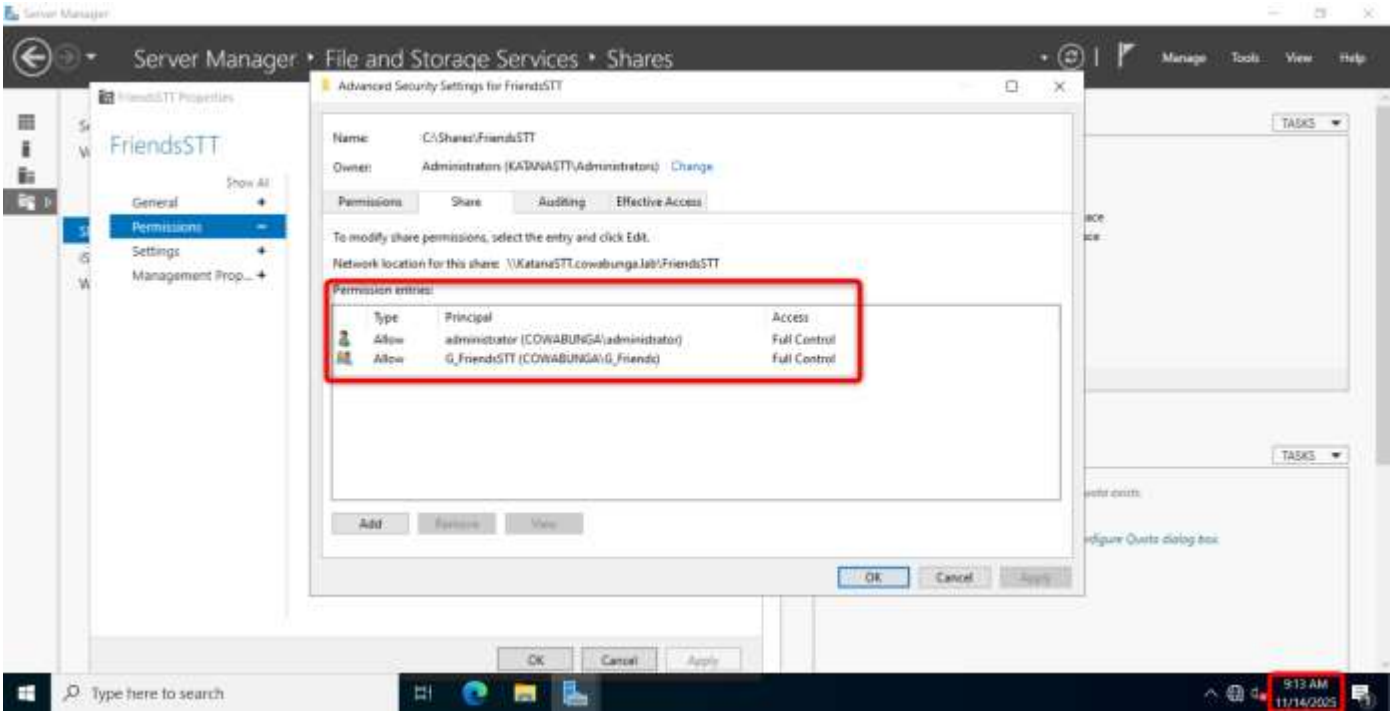


List of folders on the file server, with permissions info

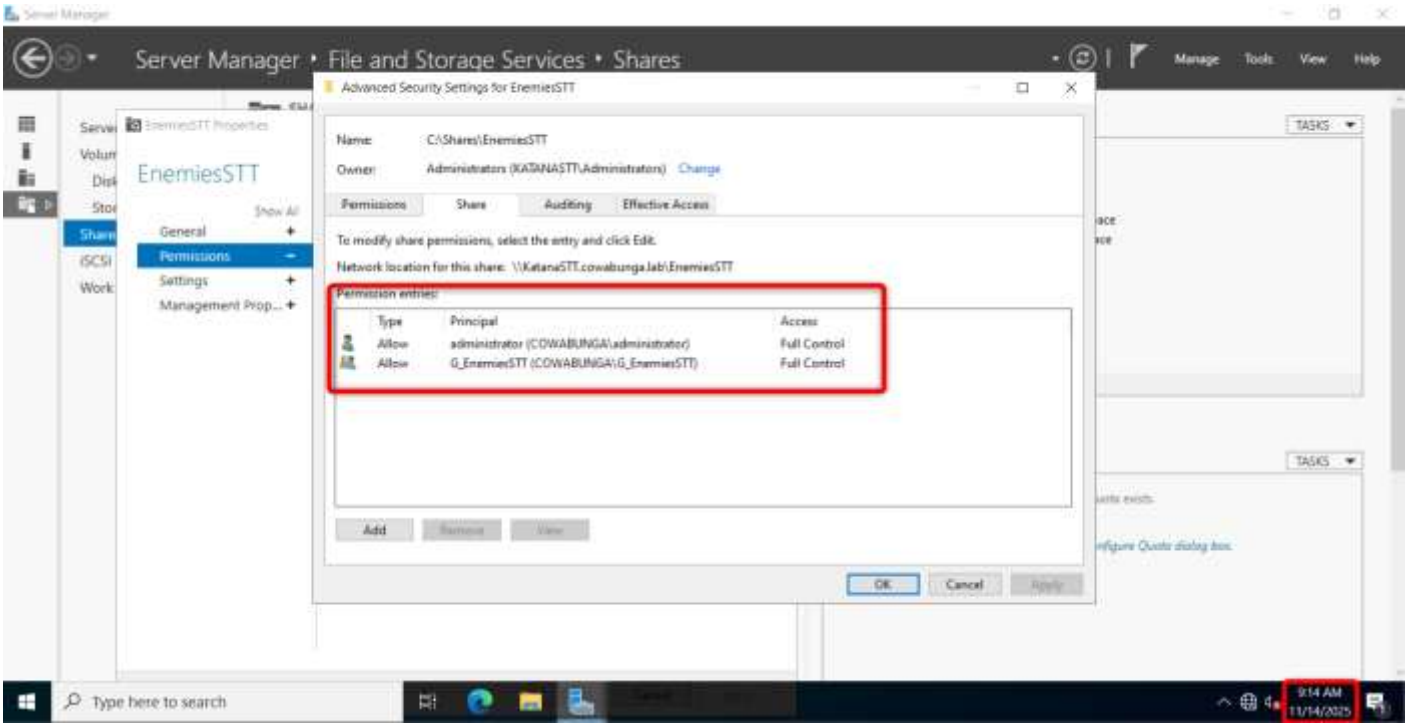
TurtlesSTT OU



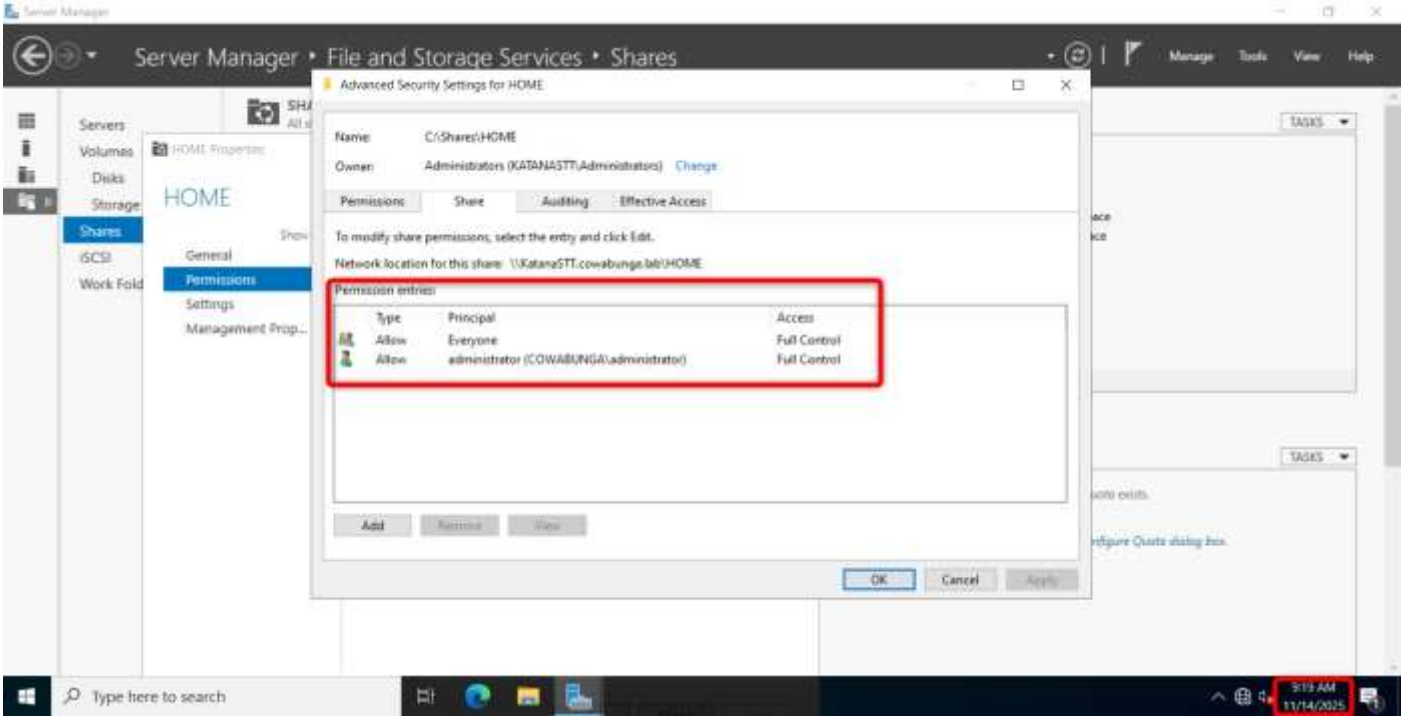
FriendsSTT OU



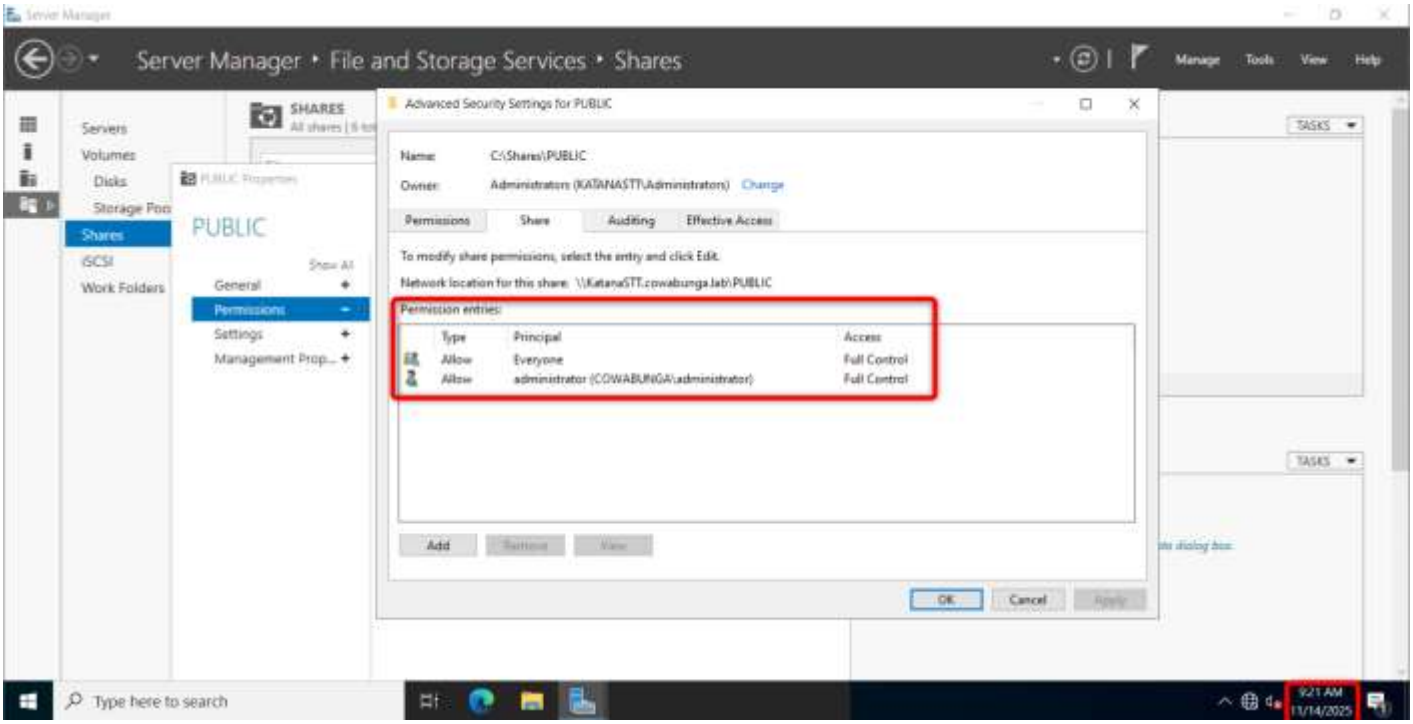
EnemiesSTT OU



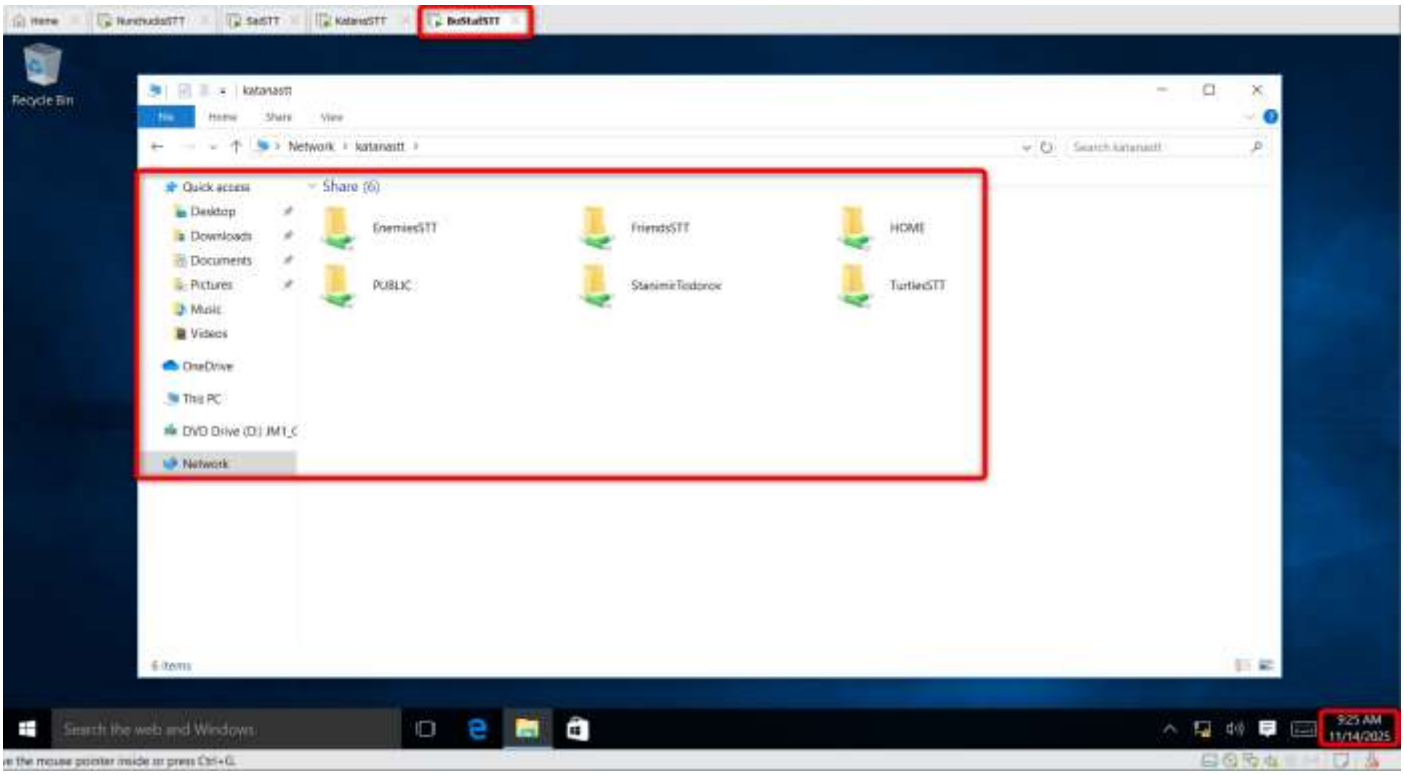
Home



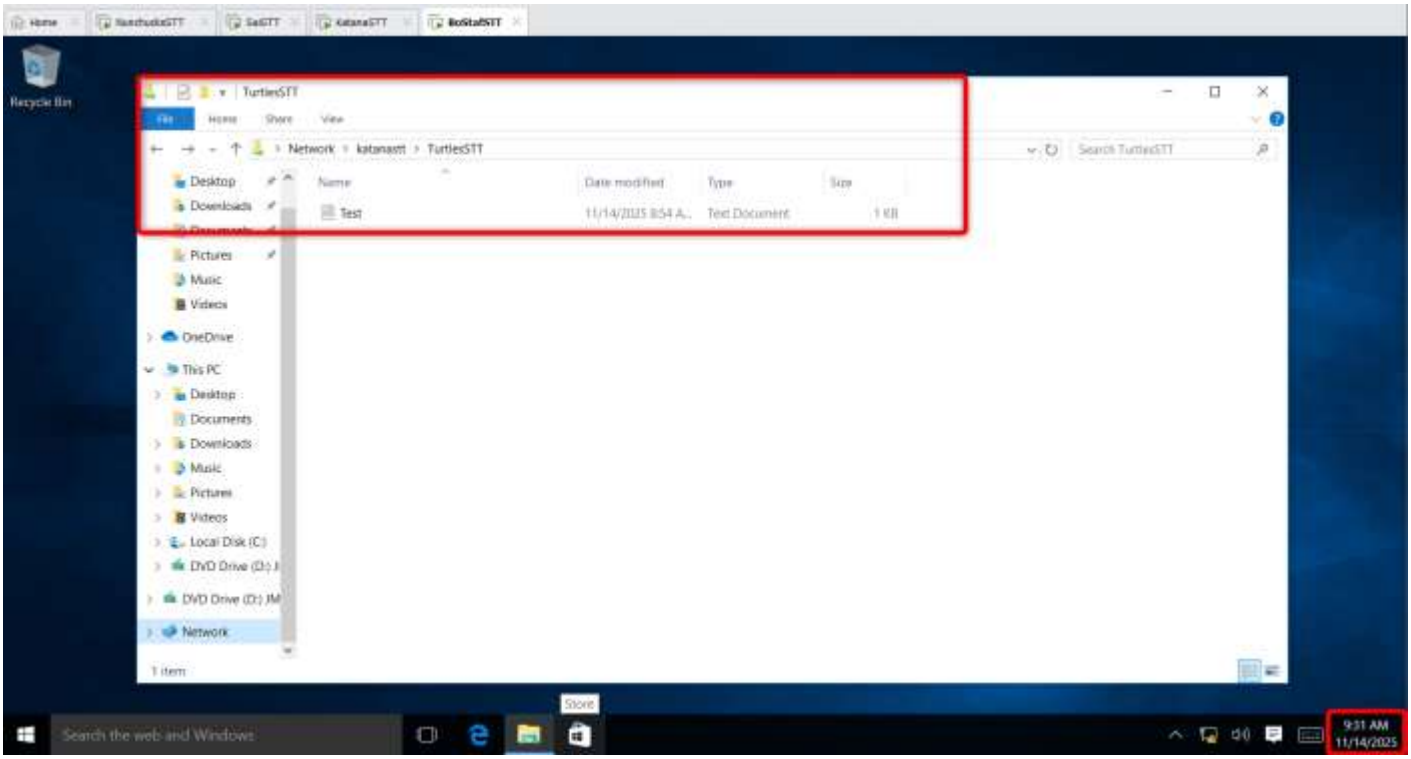
Public

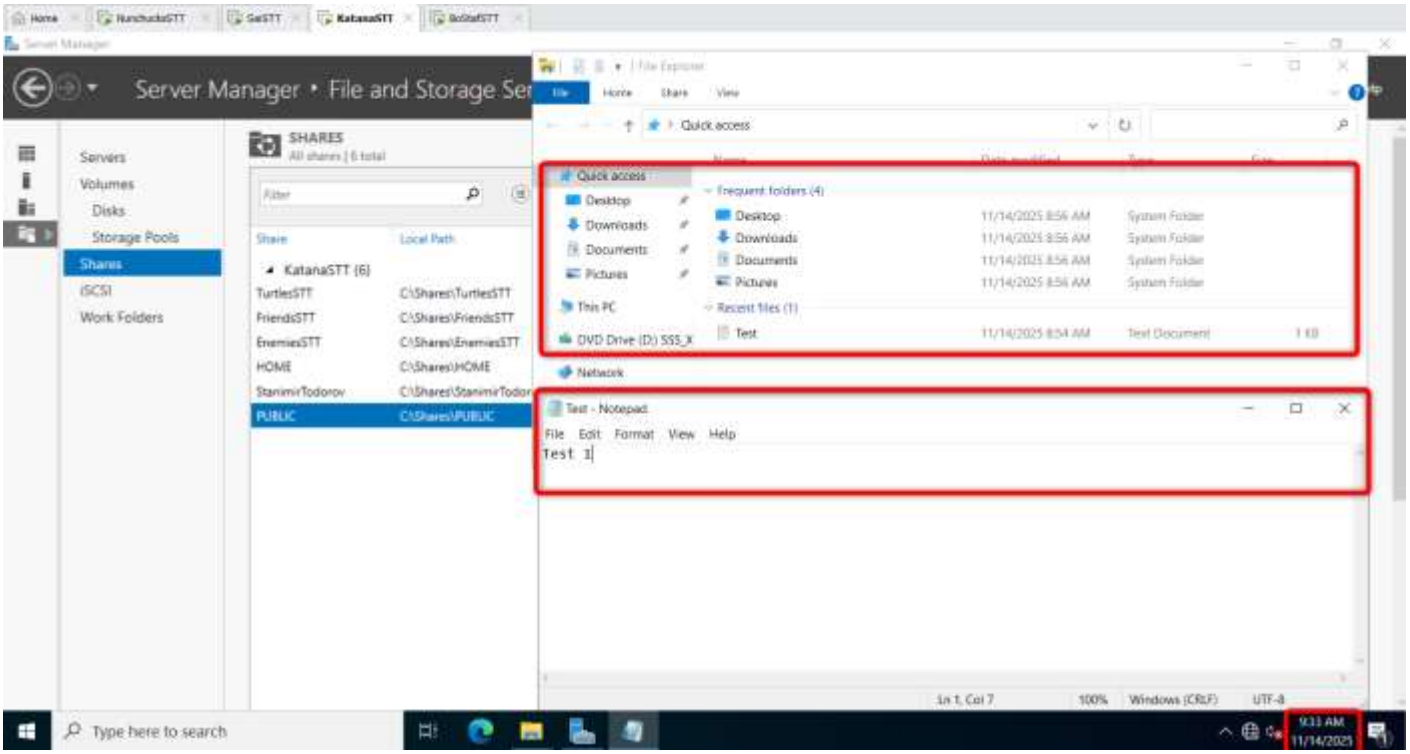


The list of folders on the client machine

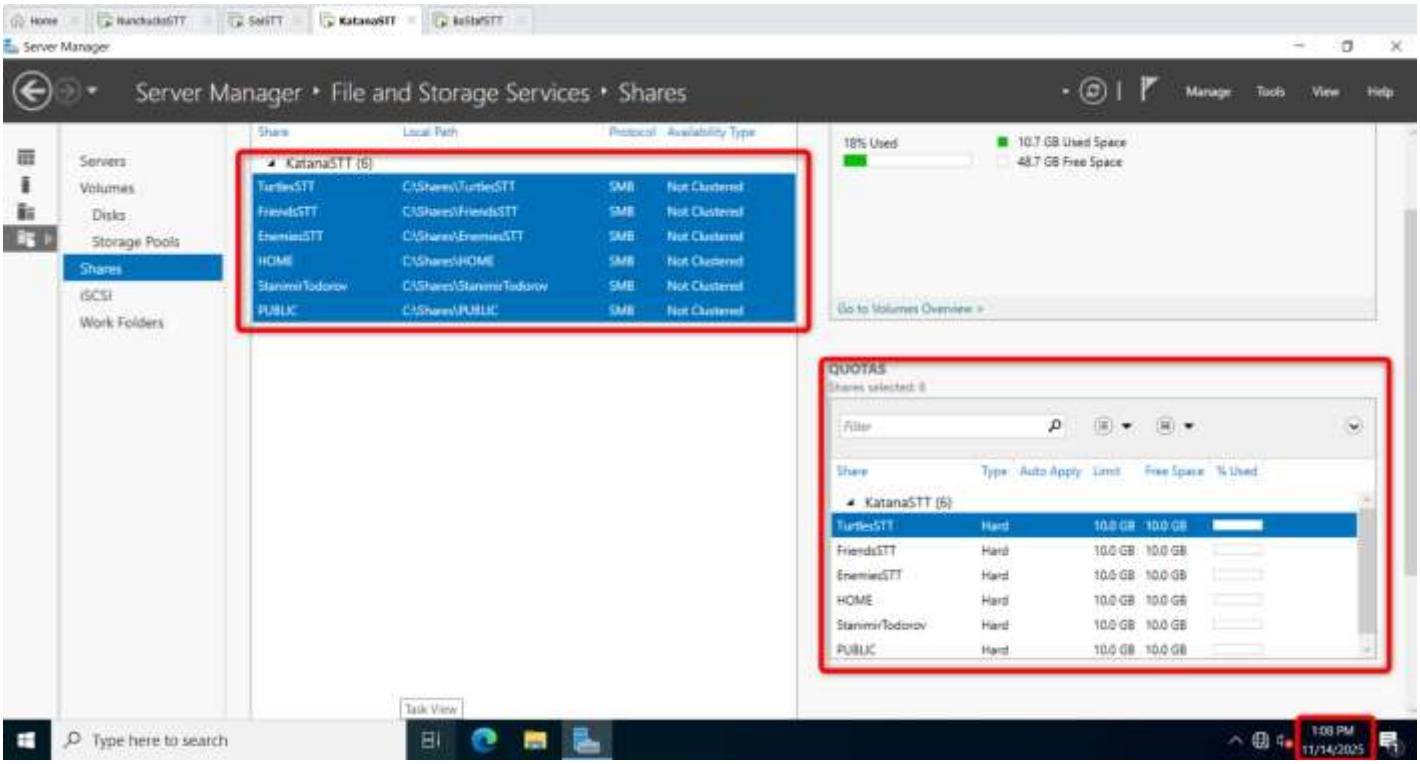


The document created on the client machine



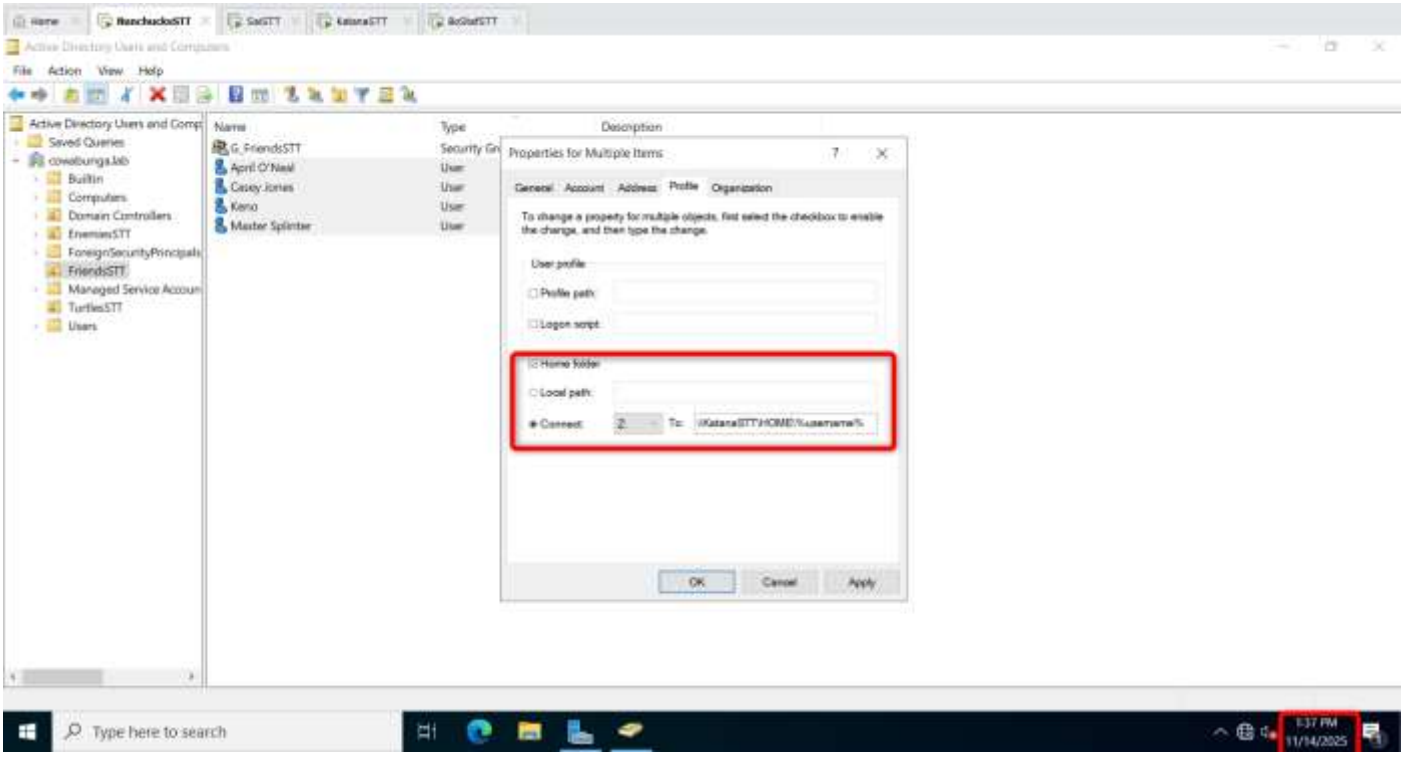


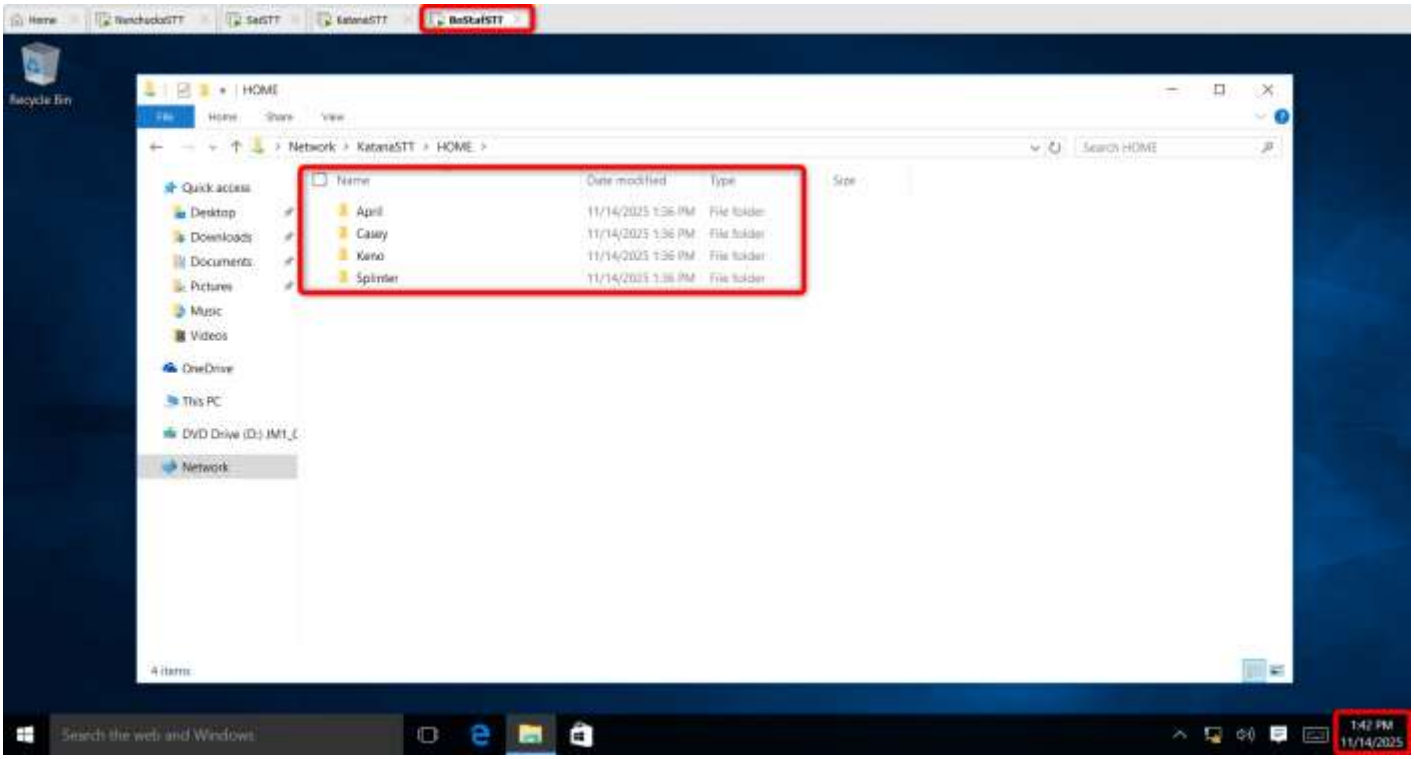
Set Storage Quotas



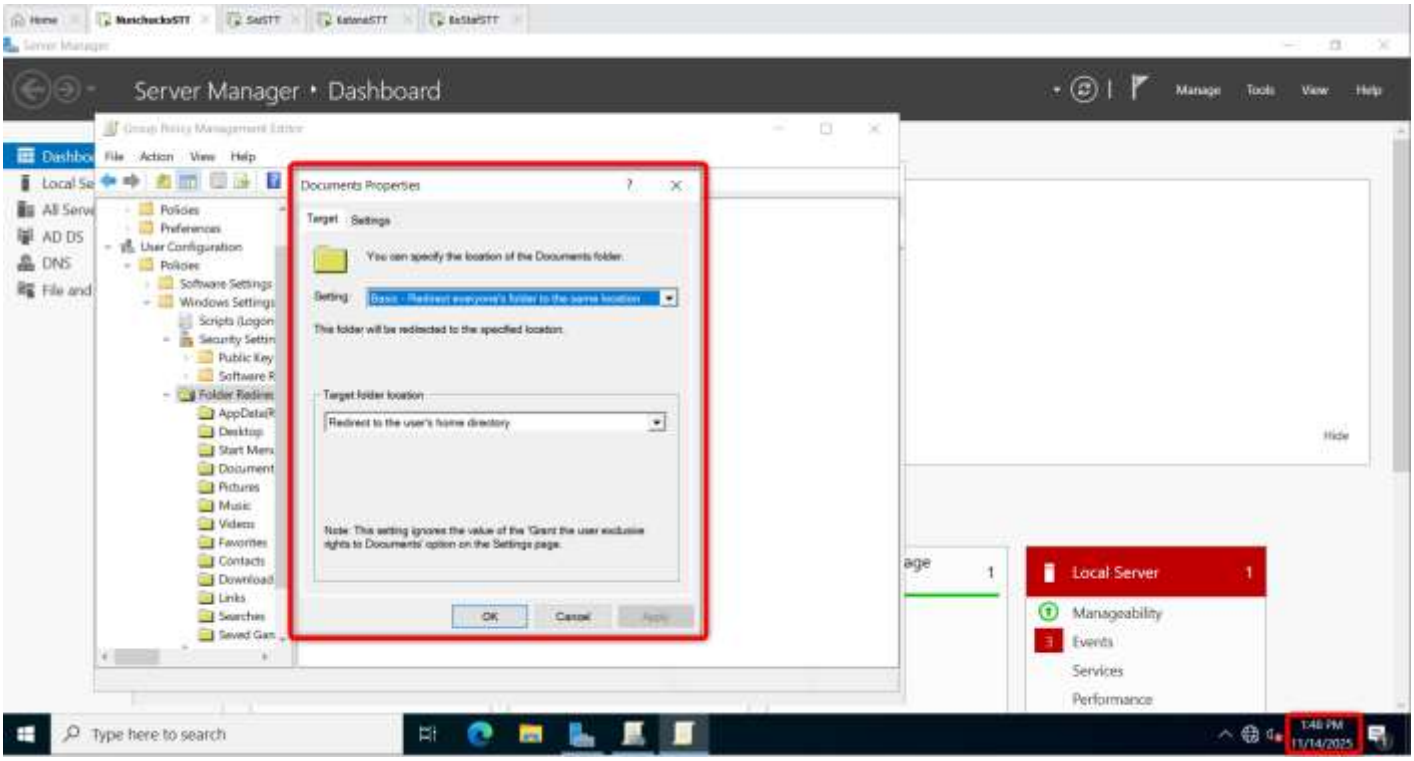
Redirect the Home Folder

Profile settings showing the drive letter and the path for the shared folder



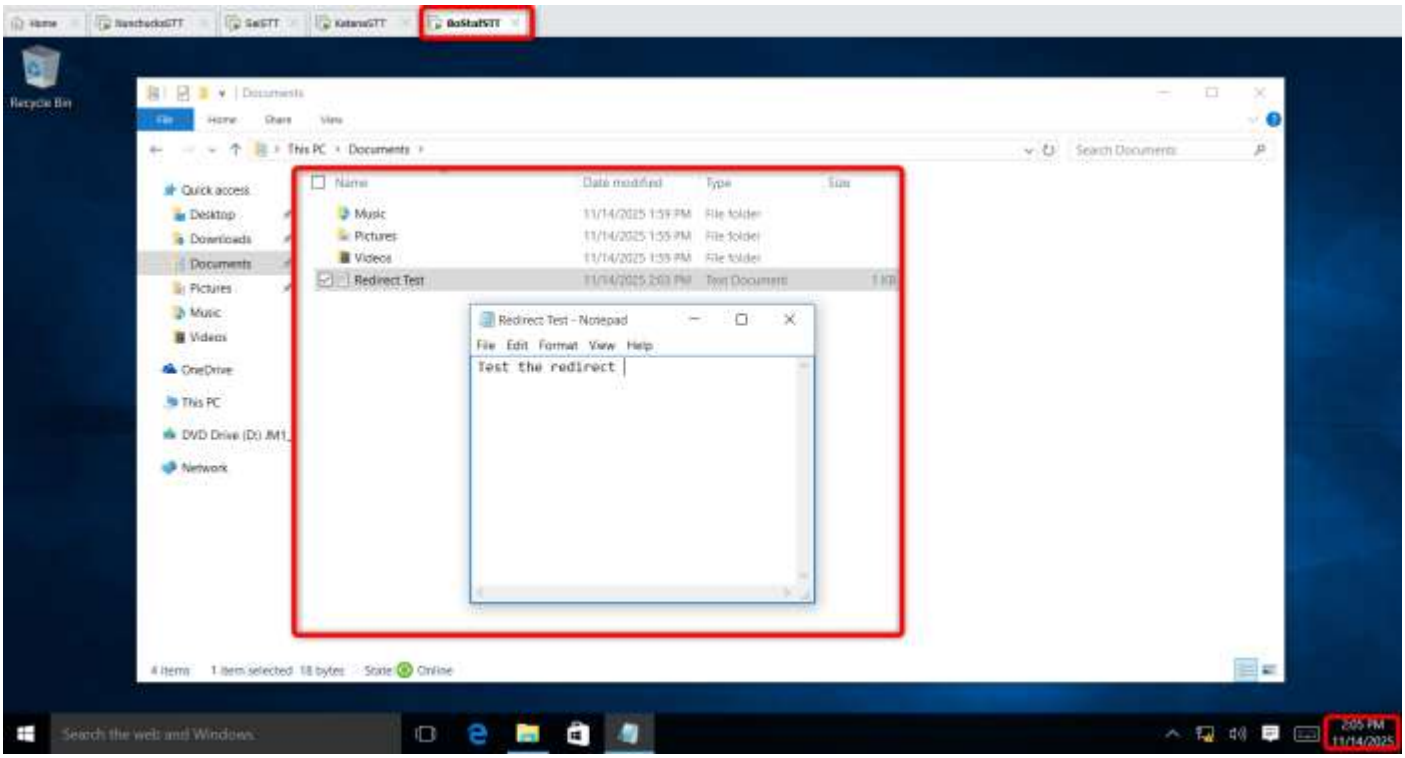


GPO Setting

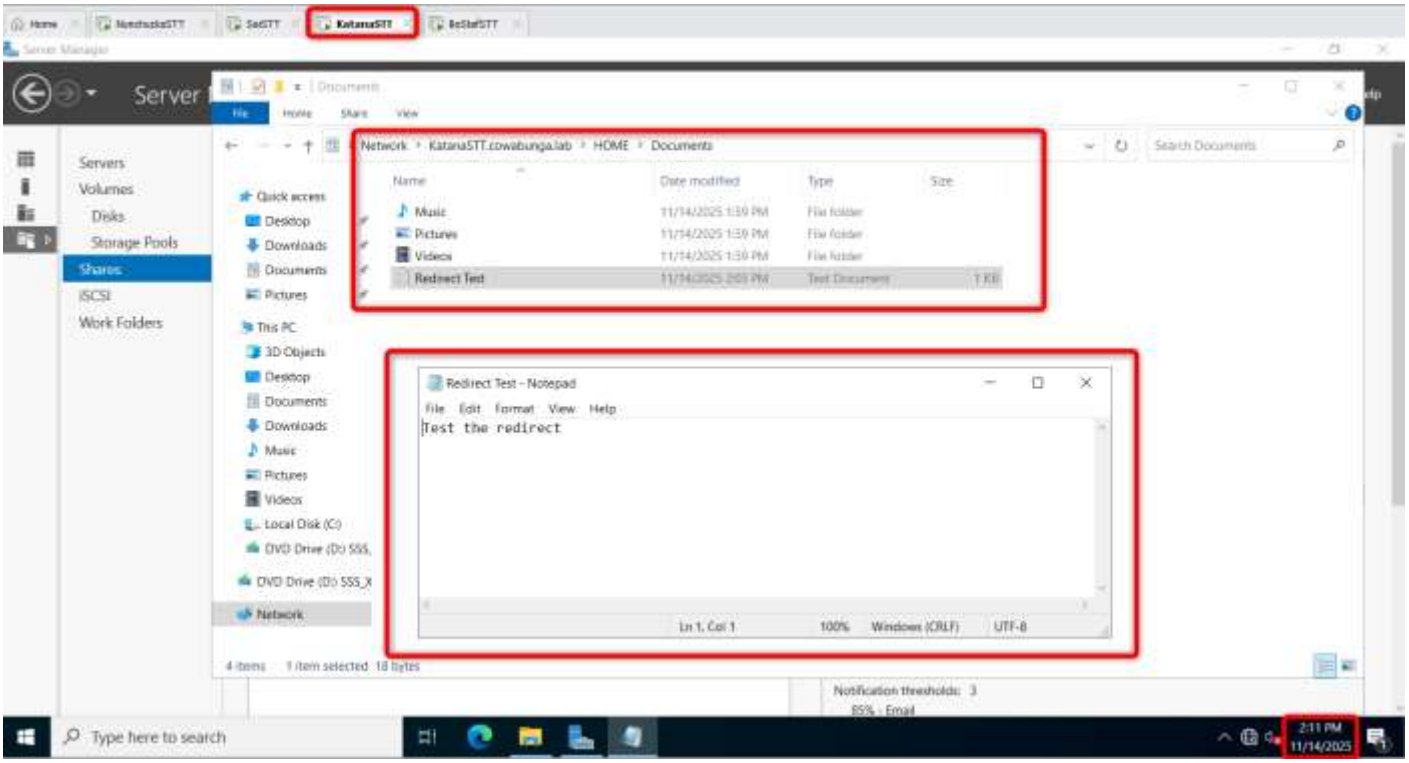

Server Manager • Dashboard
Manage
Tools
View
Help
Group Policy Management Editor
File
Action
View
Help
Policies
Preferences
User Configuration
Policies
Software Settings
Windows Settings
Scripts (Logon
Security Settin
Public Key
Software R
Folder Redirec
AppData(R
Desktop
Start Menu
Document
Pictures
Music
Videos
Favorites
Contacts
Download
Links
Searches
Saved Gam
Documents Properties
Target
Settings
You can specify the location of the Documents folder.
Setting:
Basic - Redirect everyone's folder to the same location
This folder will be redirected to the specified location.
Target folder location
Redirect to the user's home directory
Note: This setting ignores the value of the 'Grant the user exclusive rights to Documents' option on the Settings page.
OK
Cancel
Apply
Local Server
1
Manageability
Events
Services
Performance
Hide
Type here to search
1:46 PM
11/14/2025

Test showing one of the folders being redirected

File created on Client Machine



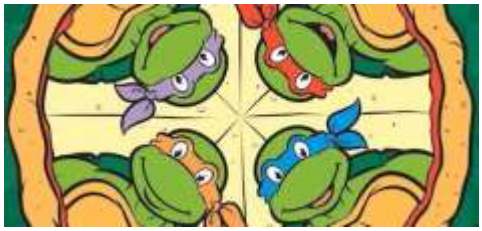
File opened on KatanaSTT server



Group Policies Build

Policies:

1. Do not allow the user to install software (turn off windows installer)
2. Disable the guest account
3. Disable forced system restarts (auto-restart)
4. Prevent users from changing the time, though allow them to view the calendar
5. Set password expiration to 203 days
6. Allow users six attempts before locking their account
7. Allow user access to keyboard and mouse but no other control panel application
8. Do not allow users to shut down the system
9. Remove the ability to use the command prompt
10. Change the default background of all computers to this picture: tmnt-pizza-cookbook-slice.png

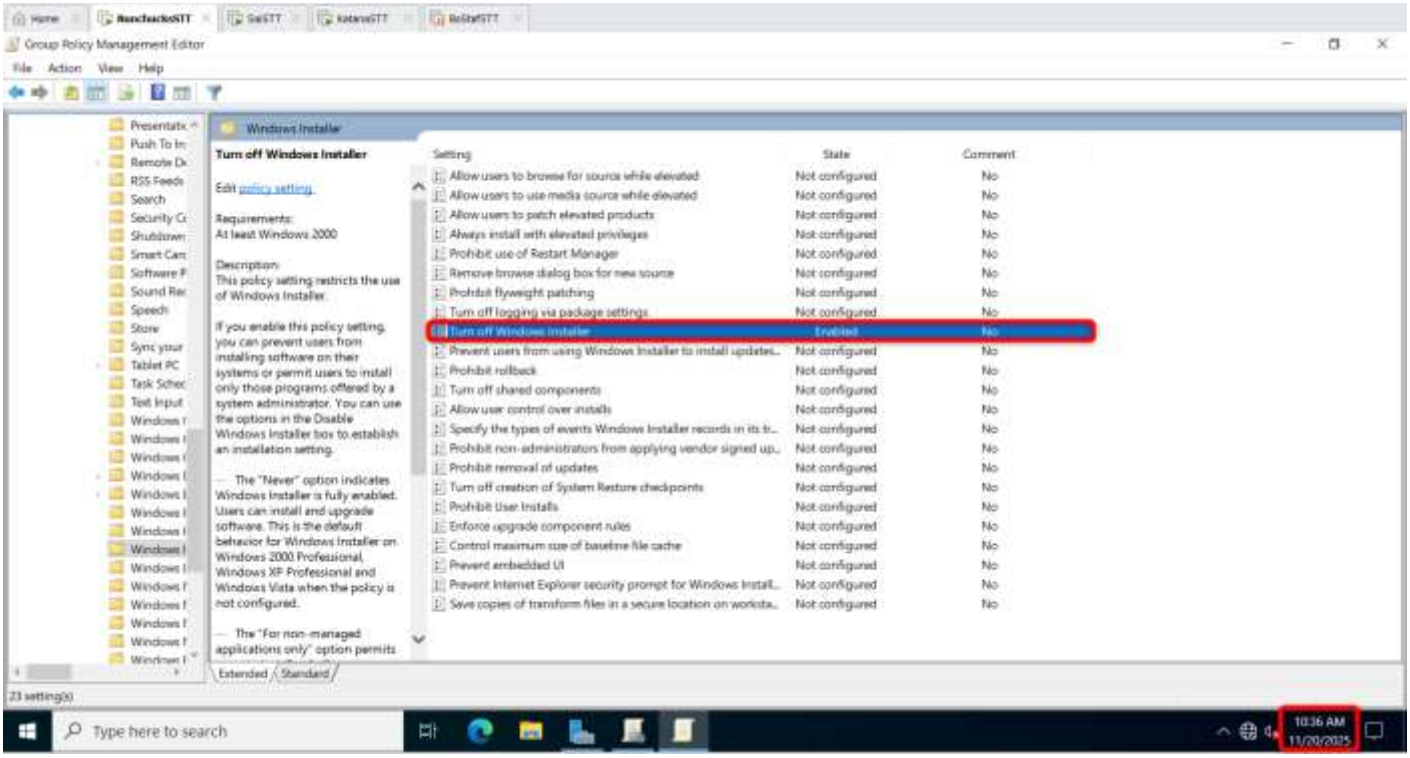


11. Disable USB devices
12. Prevent adding, dragging, dropping and closing the Taskbar´s toolbars
13. Create a startup script that deletes the contents of the "C:\Users\%username%\AppData\Local\Google\Chrome\User Data\Default\Cache" folder on shutdown
14. Set password history to 9
15. Set the default webpage to https://www.reddit.com/r/WindowsServer/Links to an external site.
16. Lock the taskbar so users cannot change its location

17. Audit object Access for Failure
18. Redirect the documents folder to the File
19. Deny log on through remote desktop services (RDS)
20. Prevent users from changing their geographic location

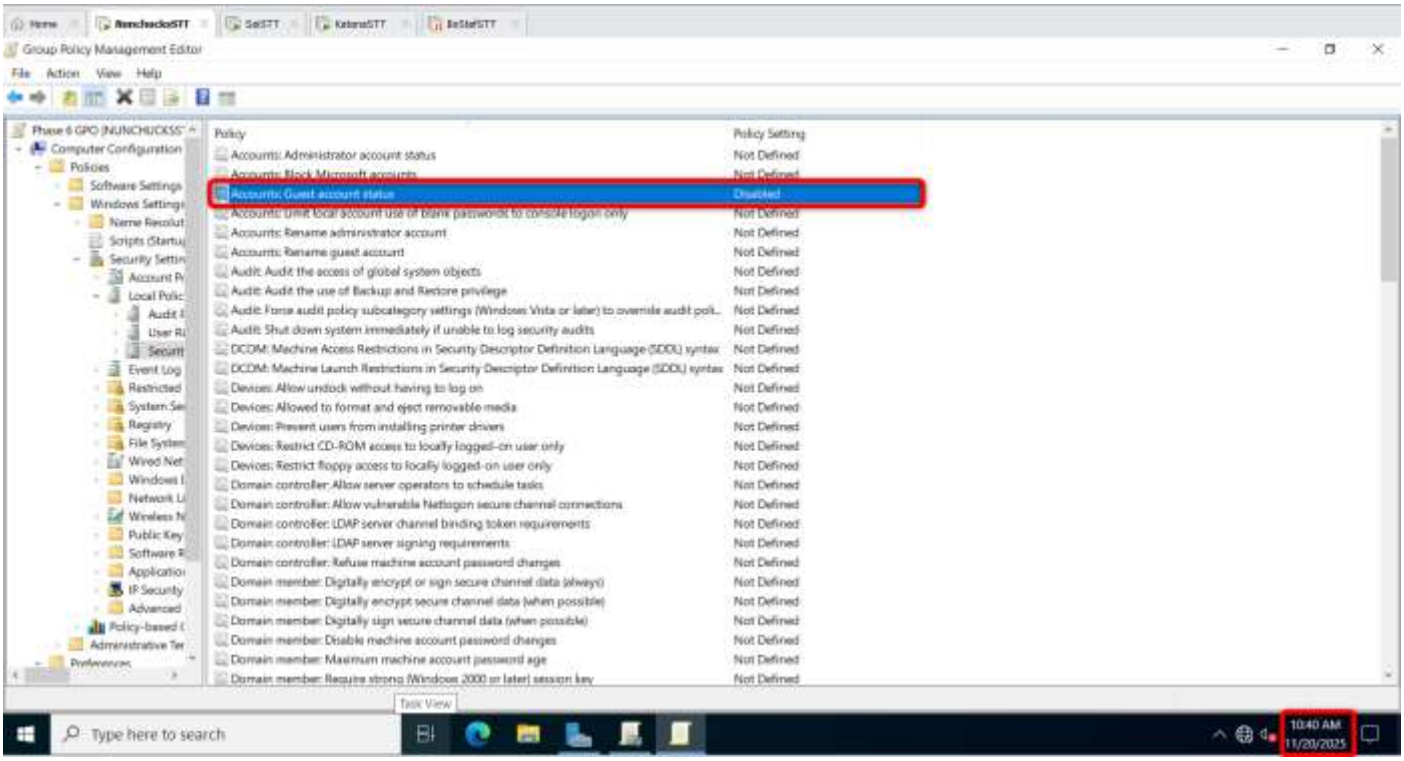
Do not allow the user to install software (turn off windows installer)

Computer Configuration → Policies → Administrative Templates → Windows Components → Windows Installer → Turn off Windows Installer → Enabled → In the dropdown Always→ Apply → OK.



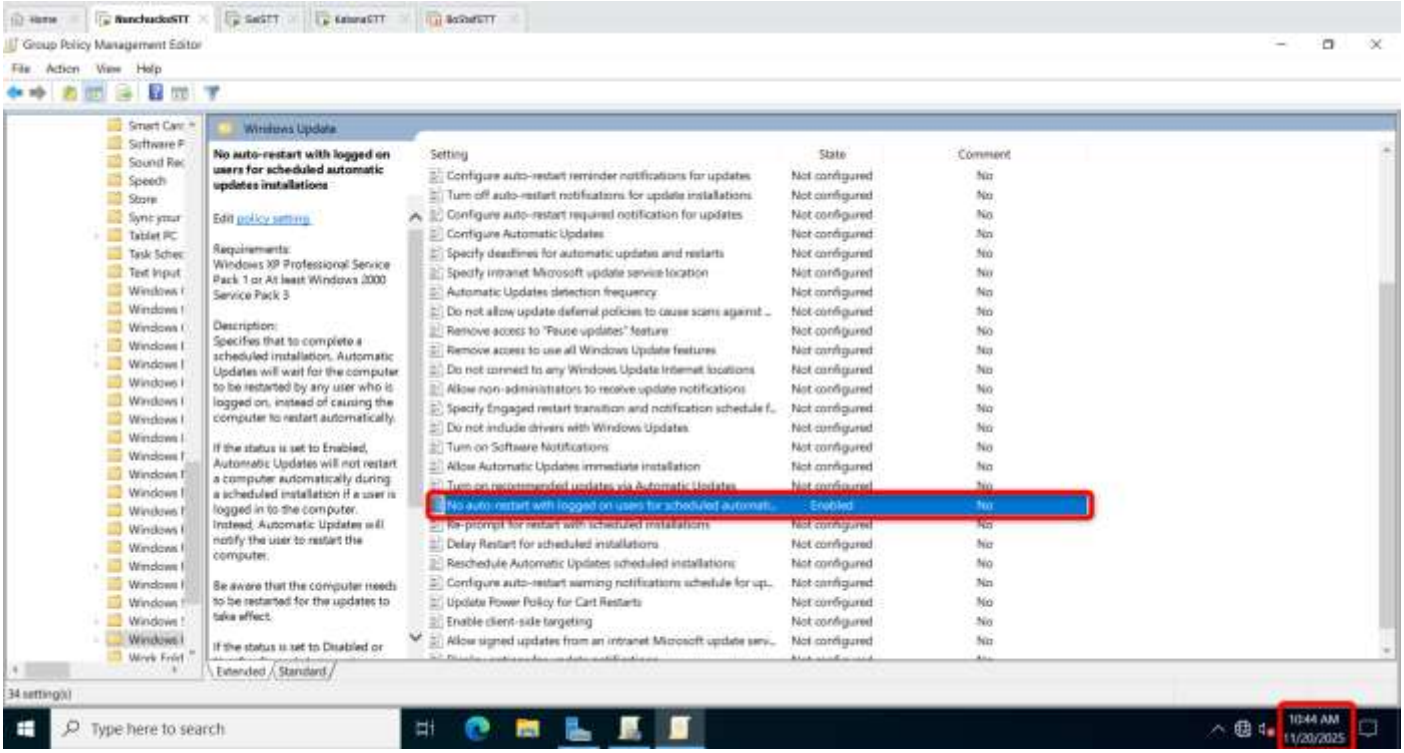
Disable the guest account

Computer Configuration → Policies → Windows Settings → Security Settings → Local Policies → Security Options



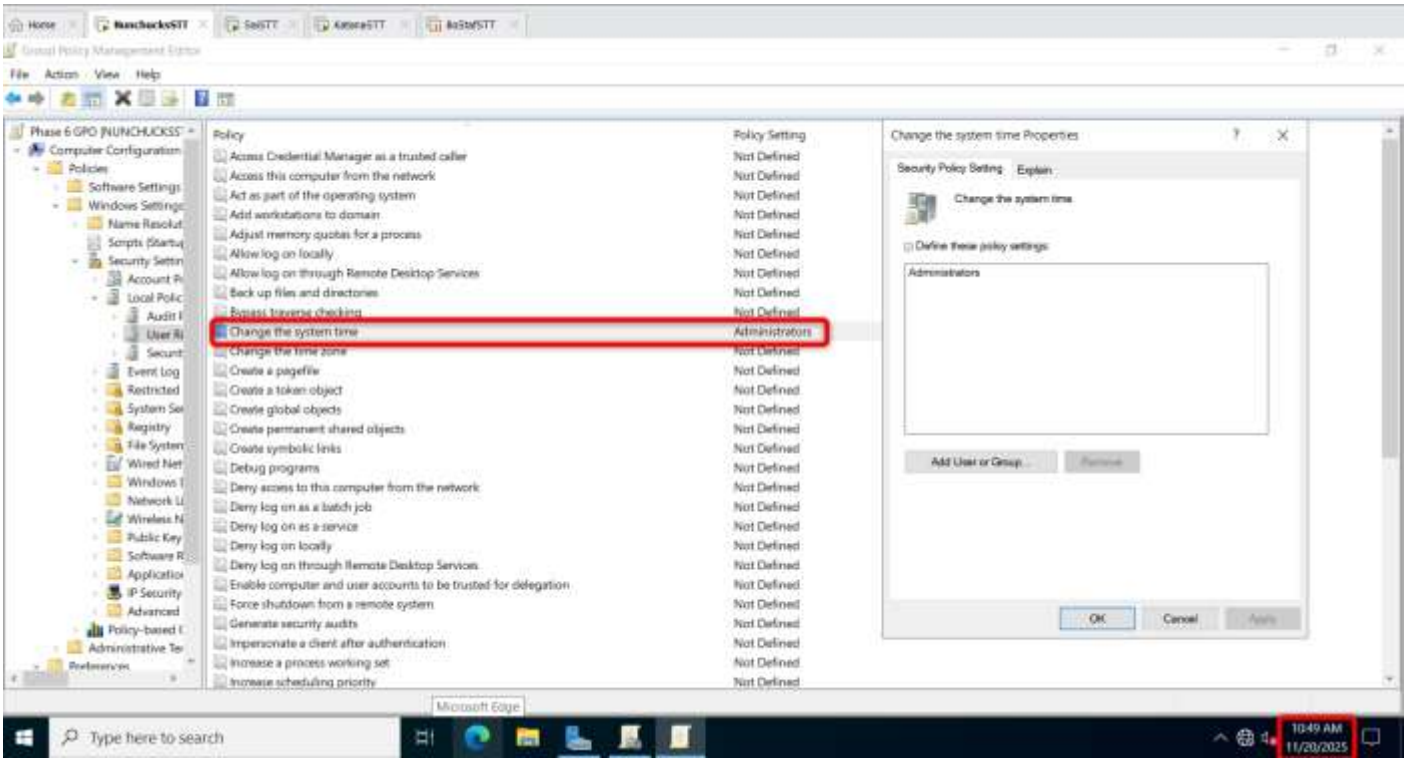
Disable forced system restarts (auto-restart)

Computer Configuration → Policies → Administrative Templates → Windows Components → Windows Update



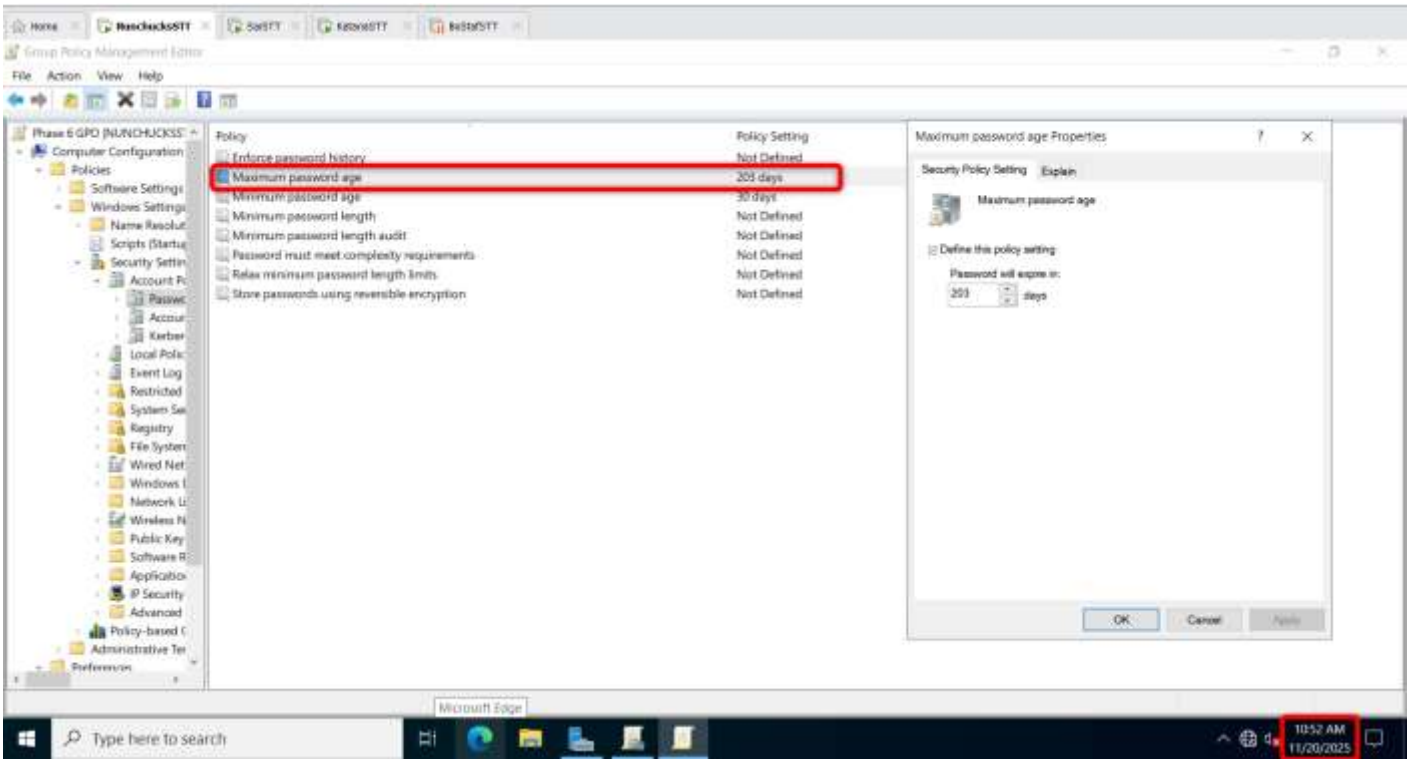
Prevent users from changing the time, though allow them to view the calendar

Computer Configuration → Policies → Windows Settings → Security Settings → Local Policies → User Rights Assignment



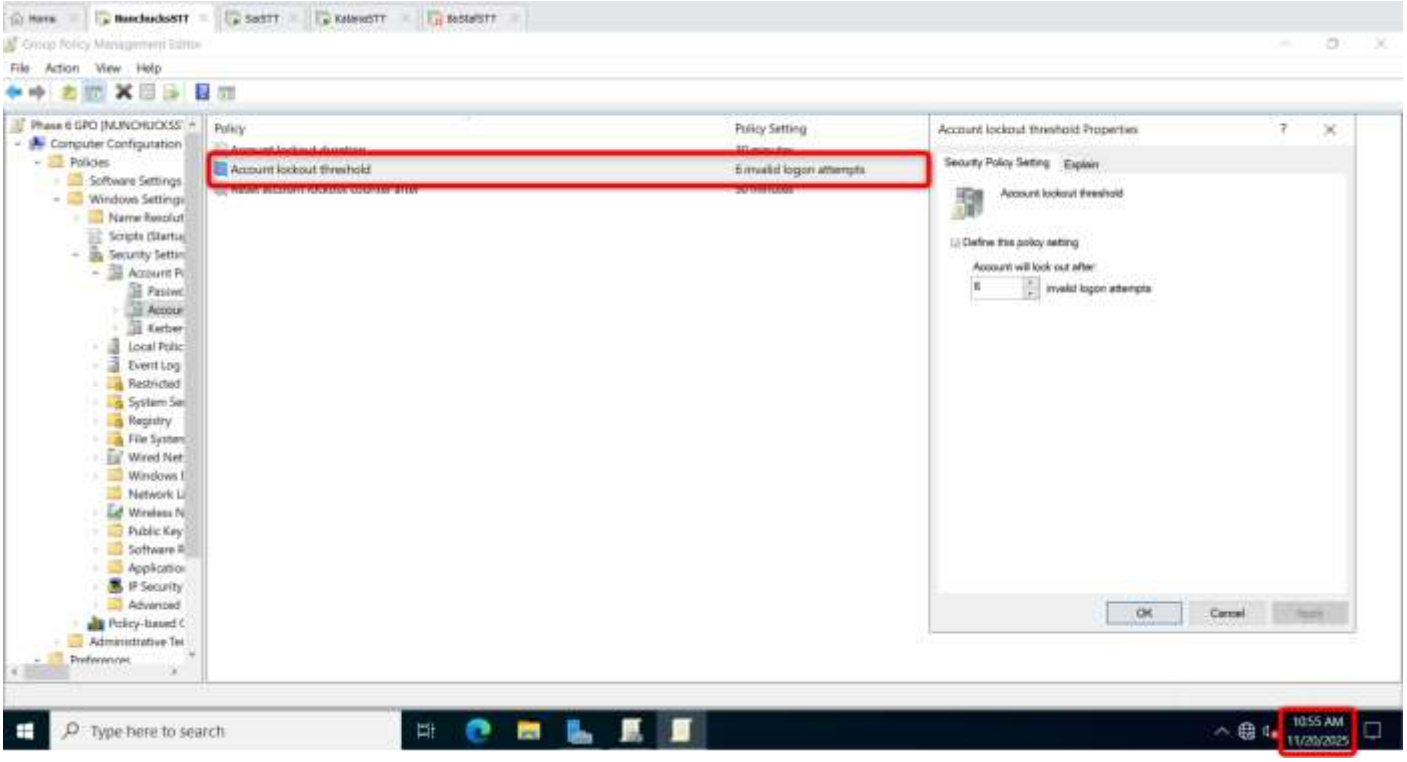
Set password expiration to 203 days

Computer Configuration → Policies → Windows Settings → Security Settings → Account Policies → Password Policy → Maximum password age



Allow users six attempts before locking their account

Computer Configuration → Policies → Windows Settings → Security Settings → Account Policies → Account lockout policy



Allow user access to keyboard and mouse but no other control panel application

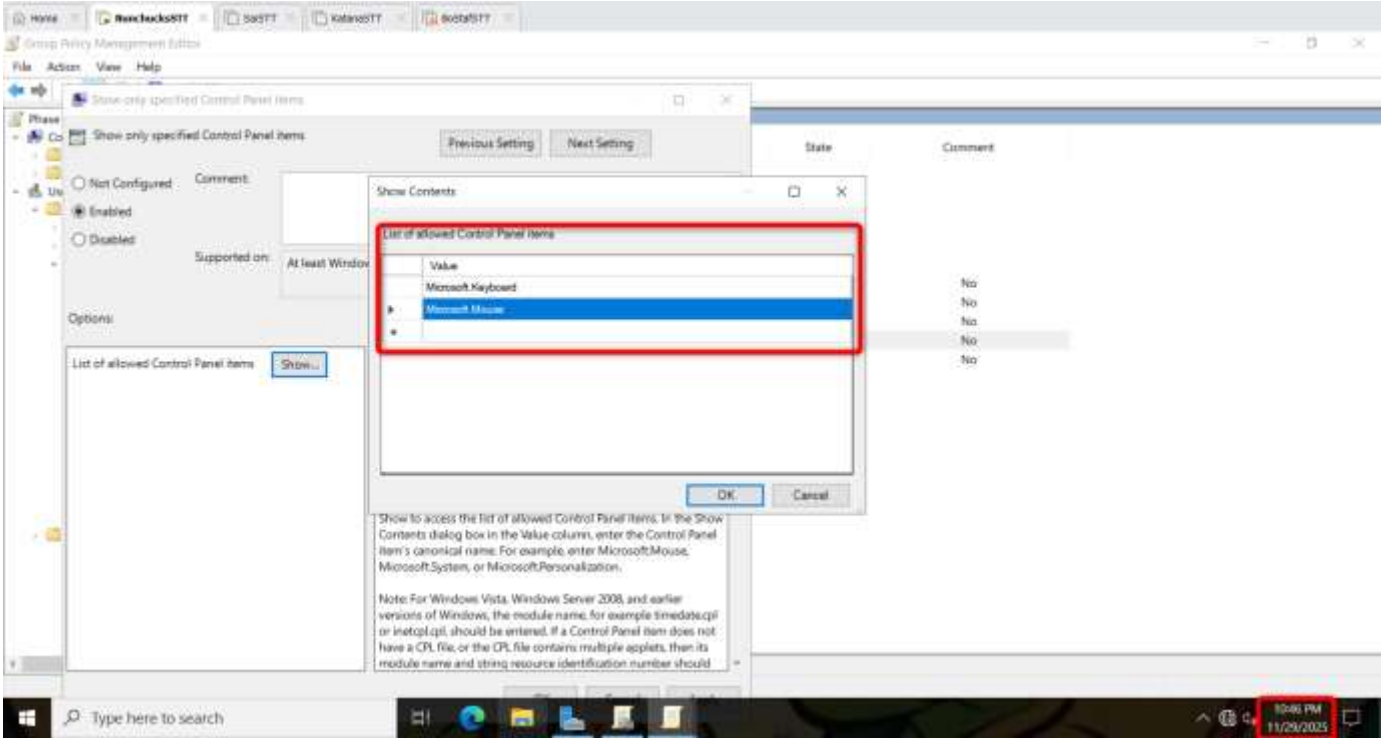
User Configuration → Policies → Administrative Templates → Control Panel

Double-click Show only specified Control Panel items.

Set the policy to Enabled → Click the Show button

Enter these names in the list:

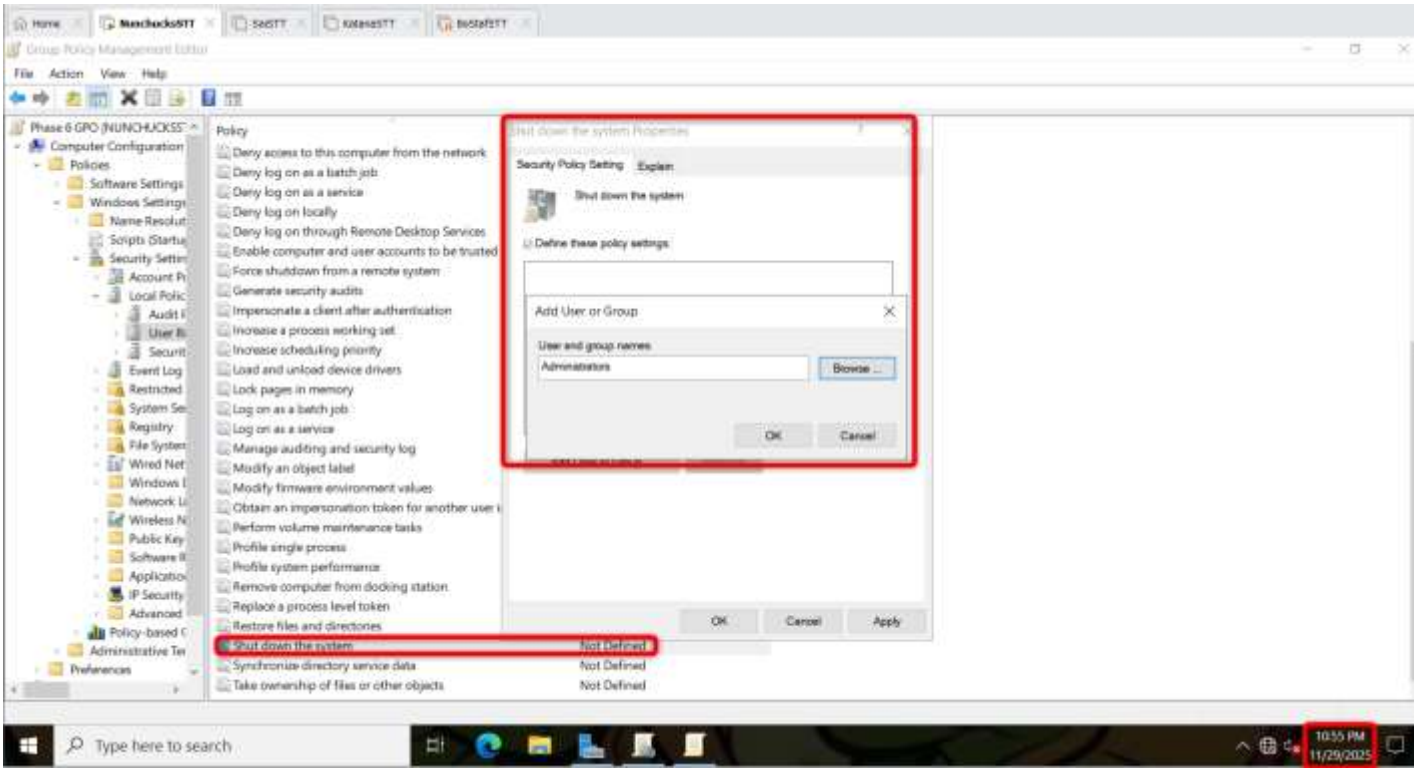
Microsoft.Keyboard and Microsoft.Mouse → OK



Do not allow users to shut down the system

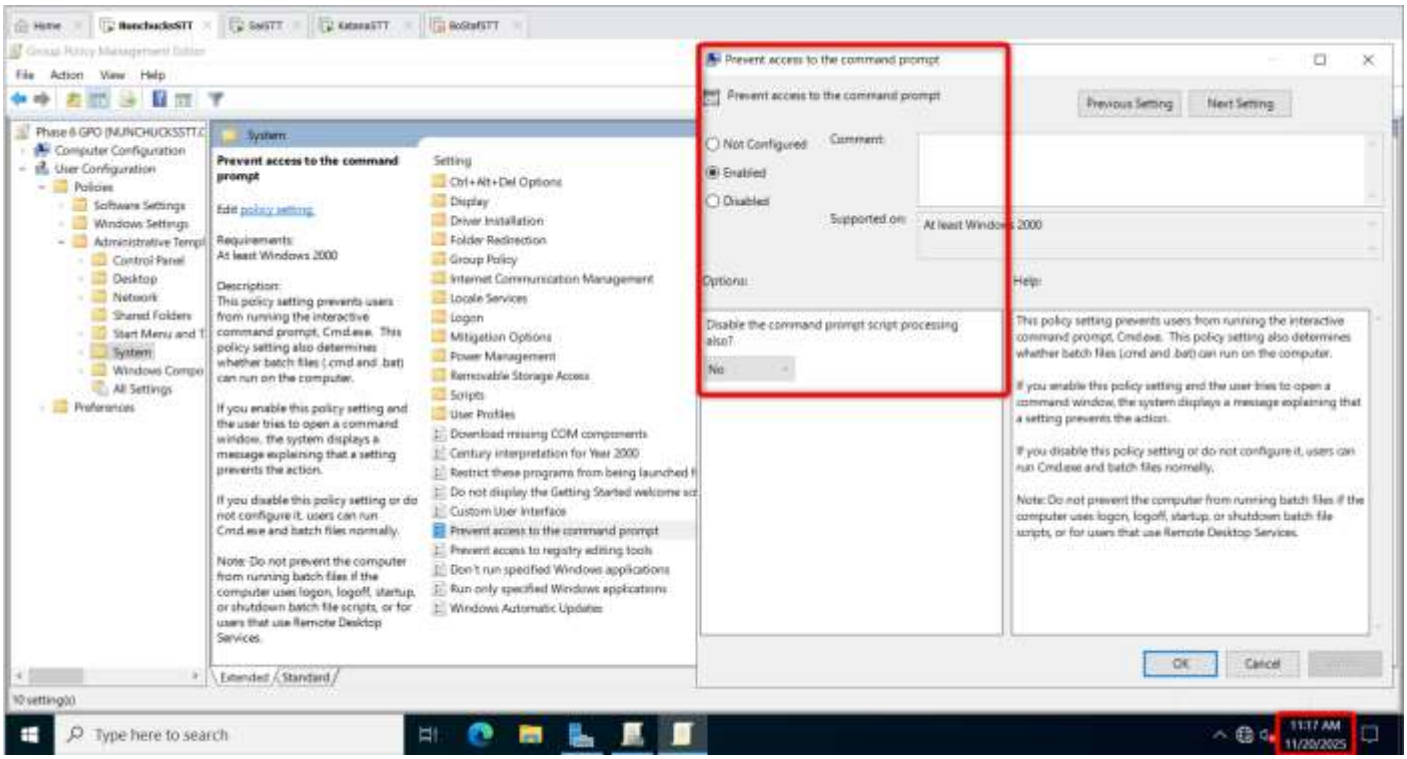
Computer Configuration → Policies -> Windows Settings → Security Settings → Local Policies → User Rights Assignment → Shut down the system (double click) → Check – Define these policy settings → Add User or Group

In this scenario only Administrators are added



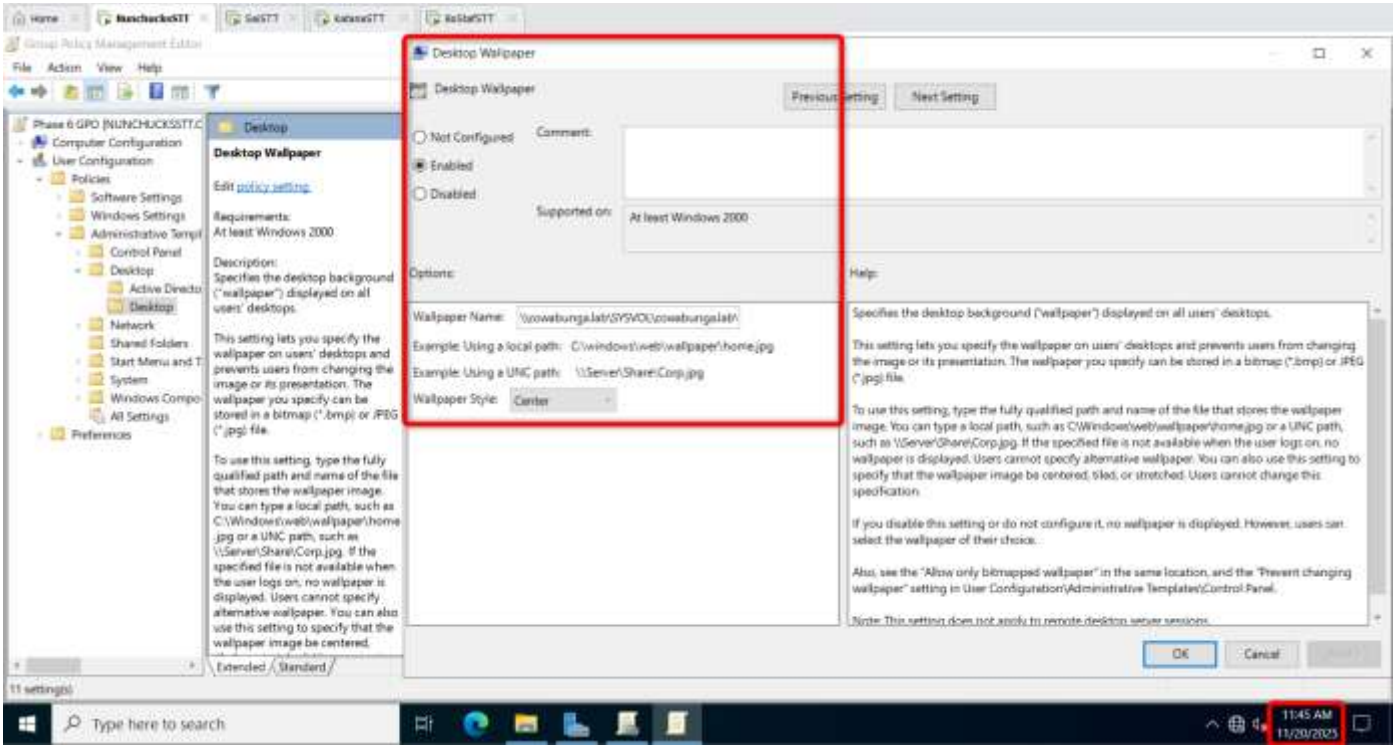
Remove the ability to use the command prompt

User Configuration → Policies → Administrative Templates → System



Change the default background of all computers to this picture: tmnt-pizza-cookbook-slice.png

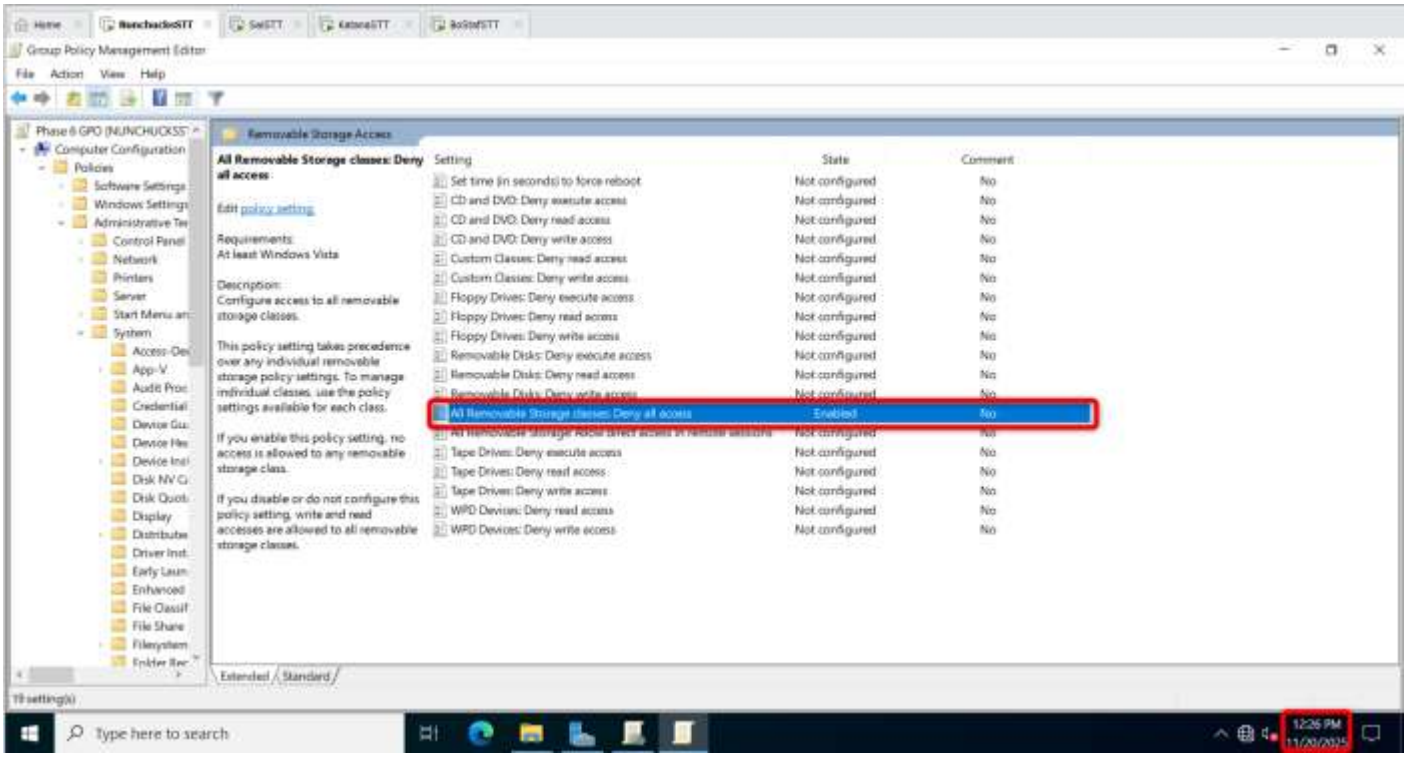
User Configuration → Policies → Administrative Templates → Desktop → Desktop → Desktop Wallpaper



Disable USB devices

Computer Configuration → Policies → Administrative Templates → System → Removable Storage Access

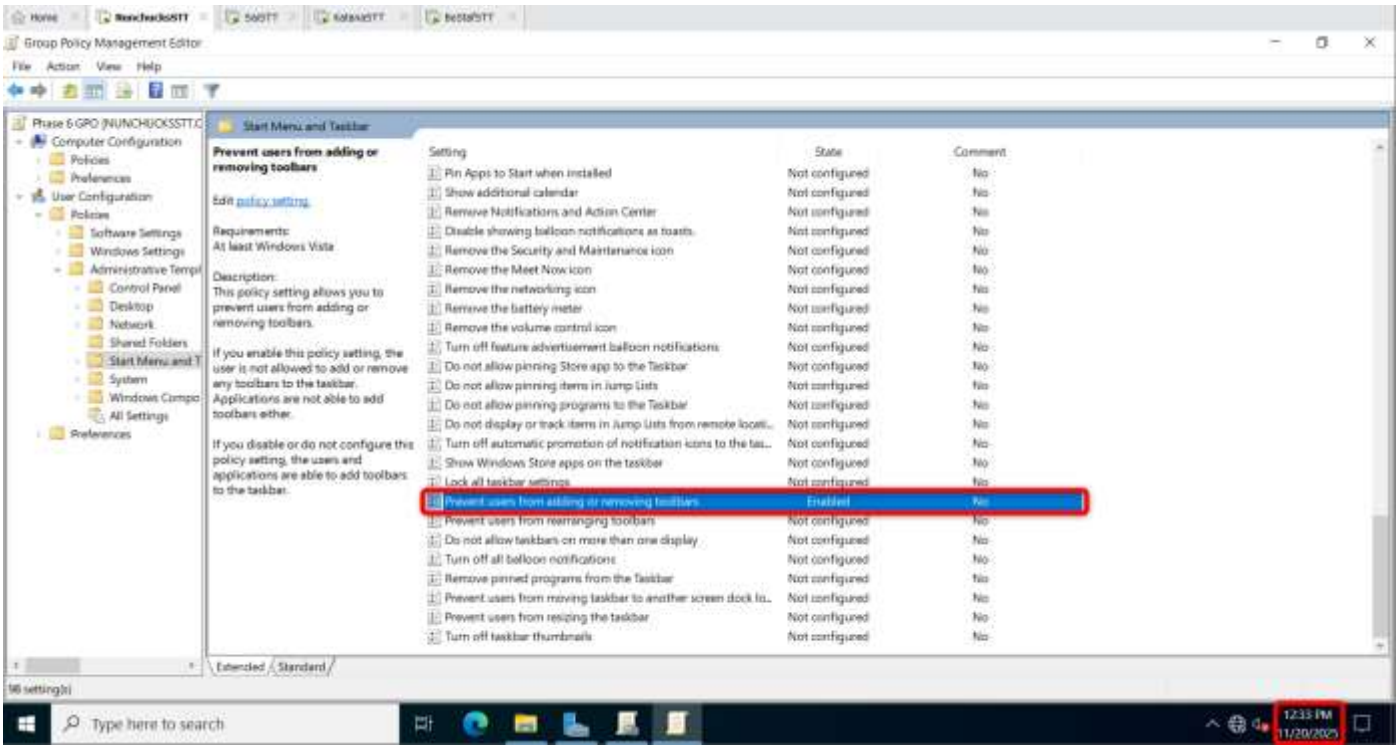
All Removable Storage classes → Deny all access → Enabled → OK



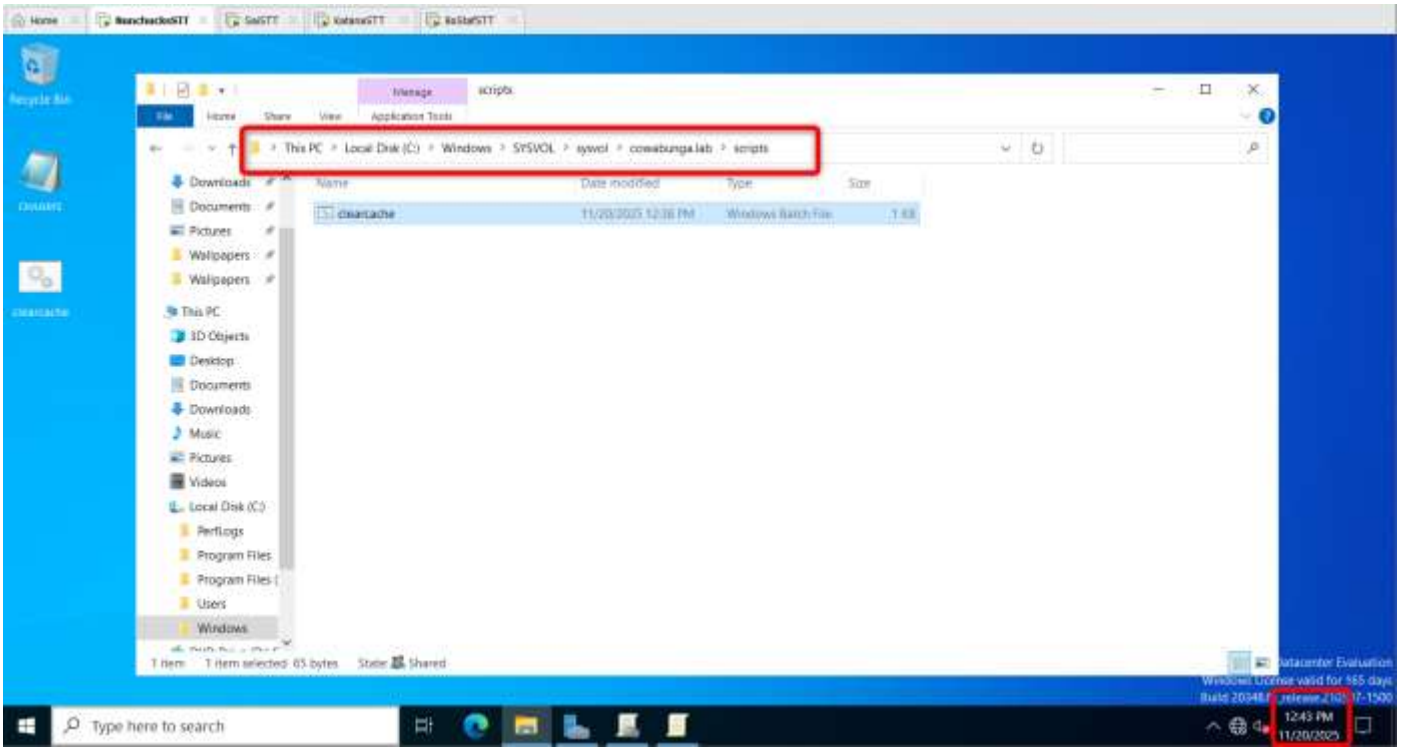
Prevent adding, dragging, dropping and closing the Taskbar´s toolbars

User Configuration → Policies → Administrative Templates → Start Menu and Taskbar

Prevent users from adding or removing toolbars → Enabled → Apply → OK

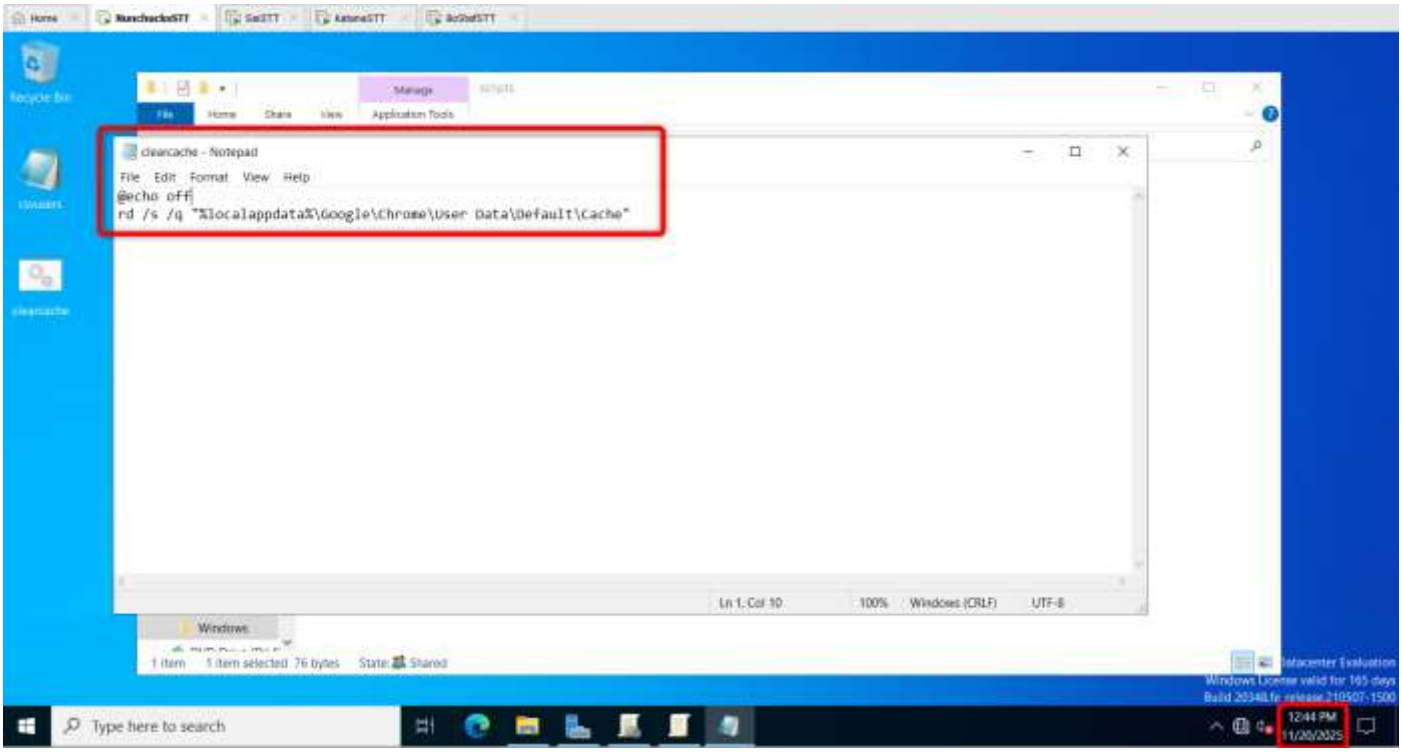


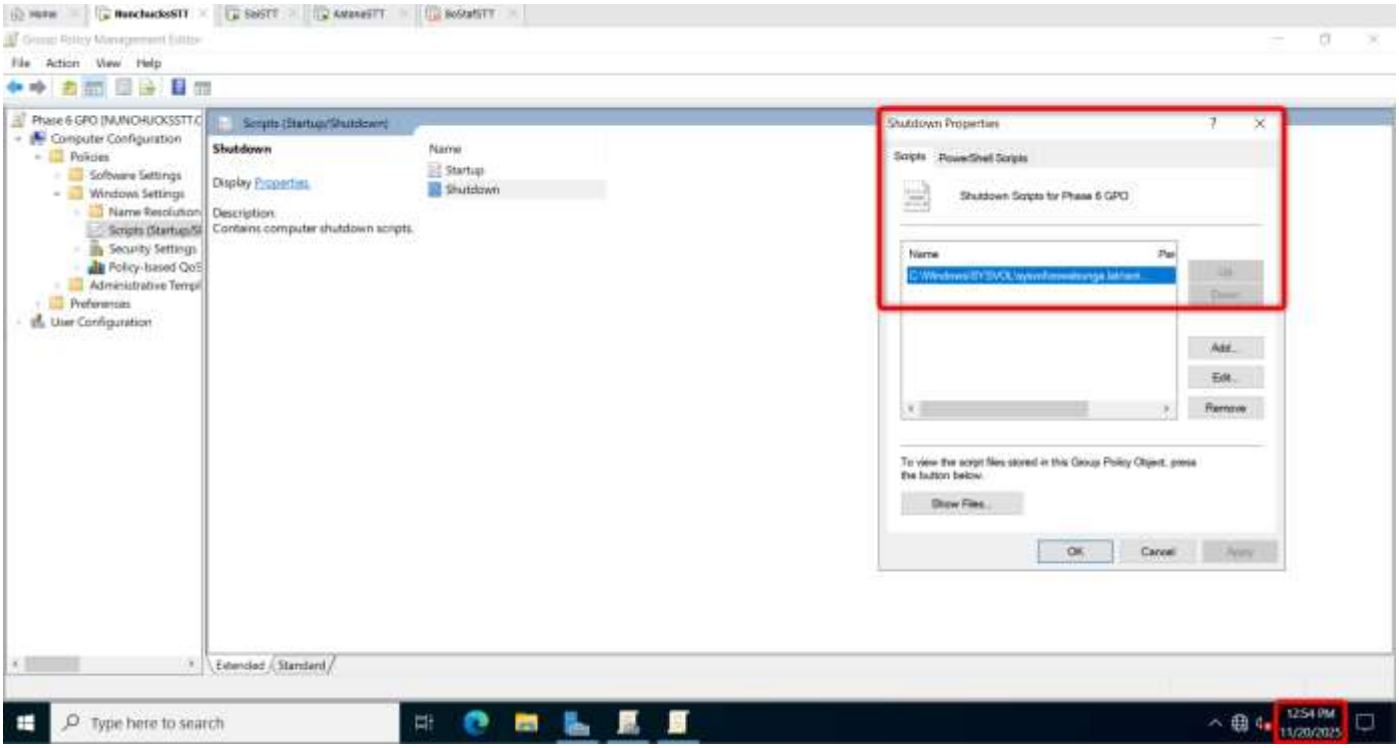
Create a startup script that deletes the contents of the "C:\Users\%username%\AppData\Local\Google\Chrome\User Data\Default\Cache" folder on shutdown



@echo off - is a best practice in batch scripting

rd /s /q – removes directory, subdirectory in q – quiet mode

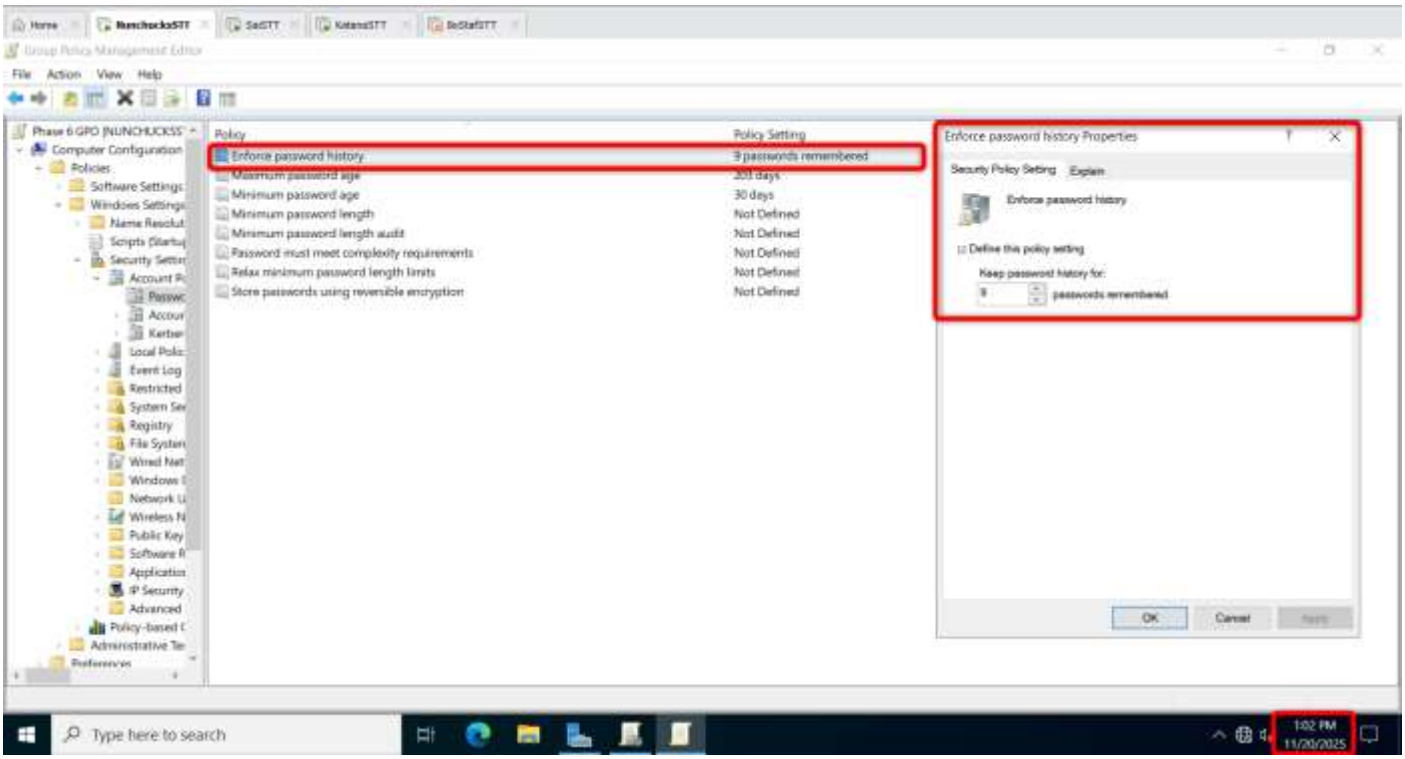




Set password history to 9

Computer Configuration → Windows Settings → Account Policies → Password Policy→ Enforce password history

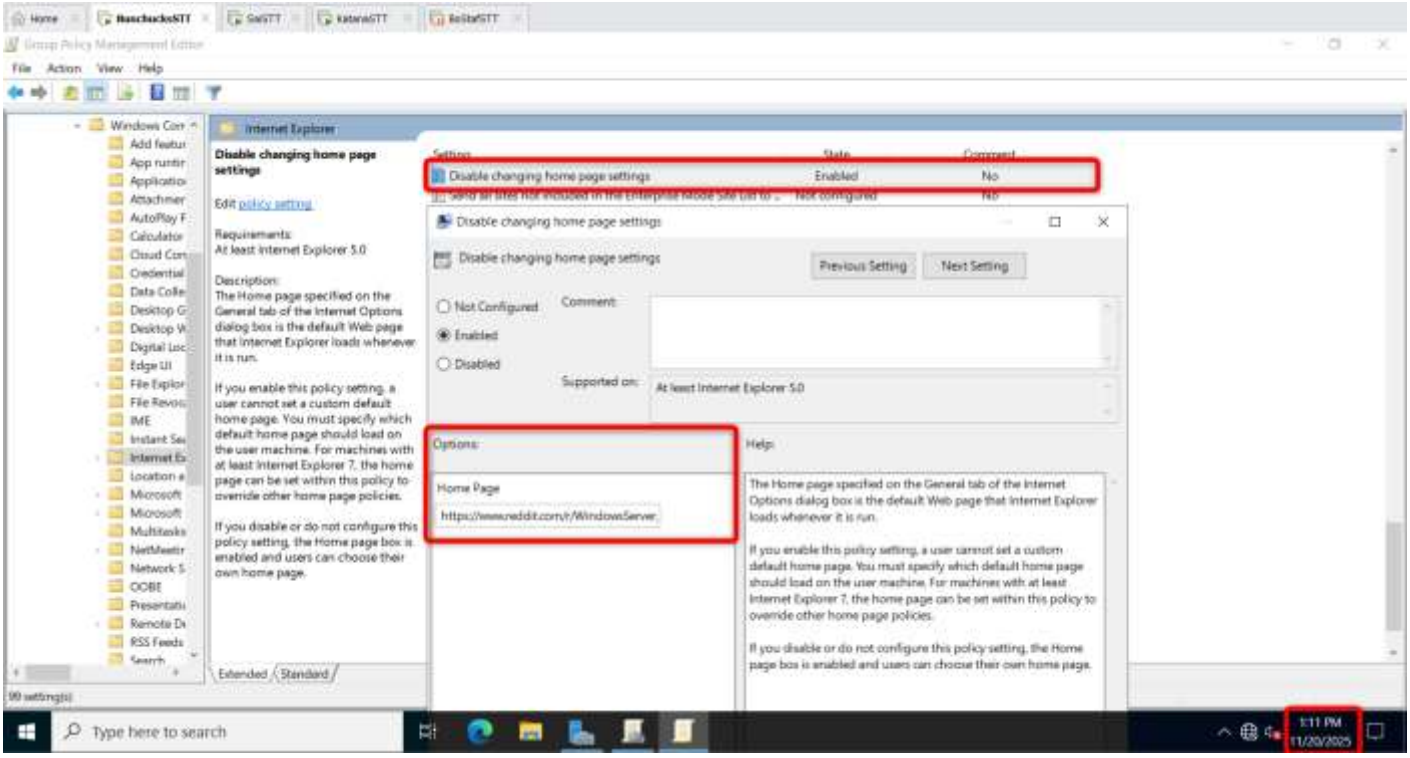
9 passwords remembered → Apply → OK



Set the default webpage to Reddit

User Configuration → Policies → Administrative Templates → Windows Components → Internet Explorer → Disable changing home page settings → Enabled

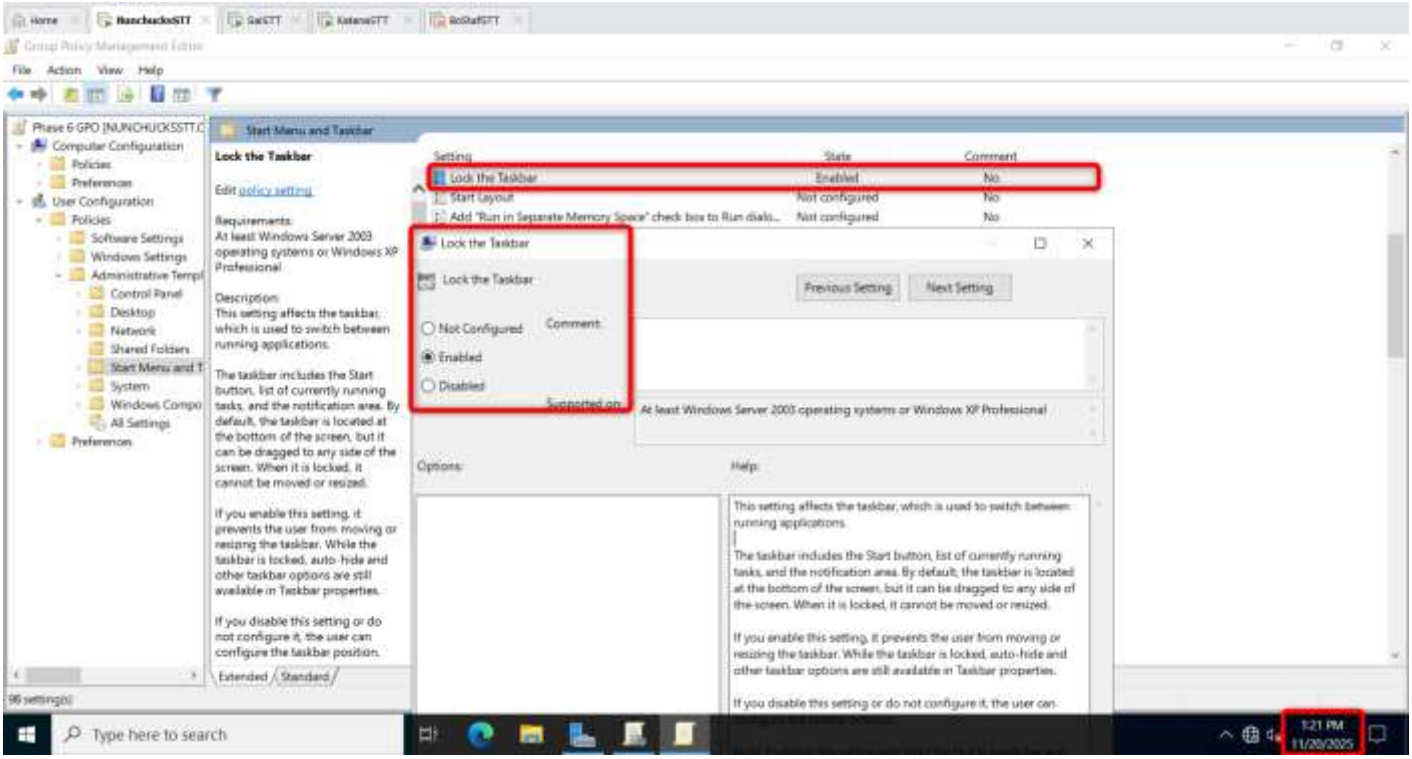
set the home page to:
https://www.reddit.com/r/WindowsServer/ → Apply → OK



Lock the taskbar so users cannot change its location

User Configuration → Policies → Administrative Templates → Start Menu and Taskbar→ Lock the Taskbar

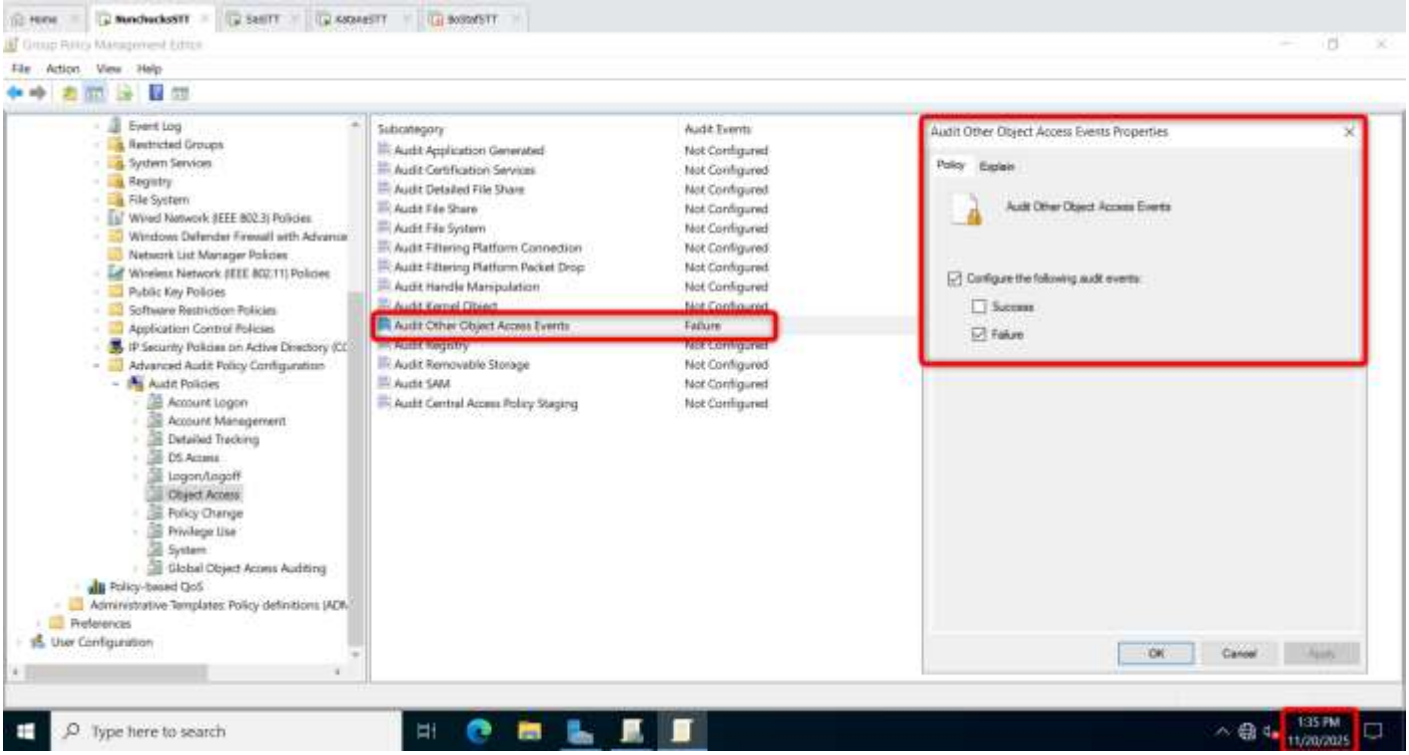
Enabled → Apply → OK



Audit object Access for Failure

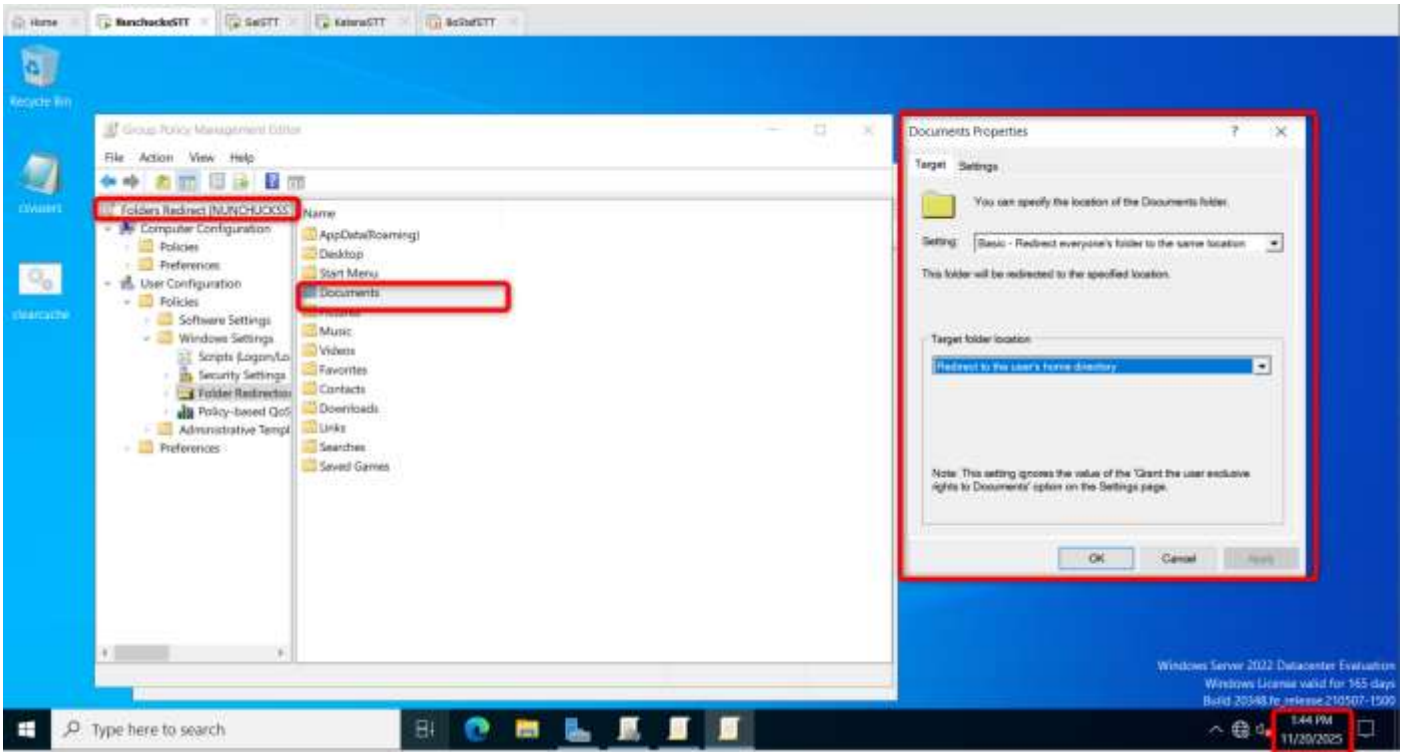
Computer Configuration → Policies → Windows Settings → Security Settings → Advanced Audit Policy Configuration → Audit Policies → Object Access

Audit Other Object Access Events → Check Failure → Apply → OK



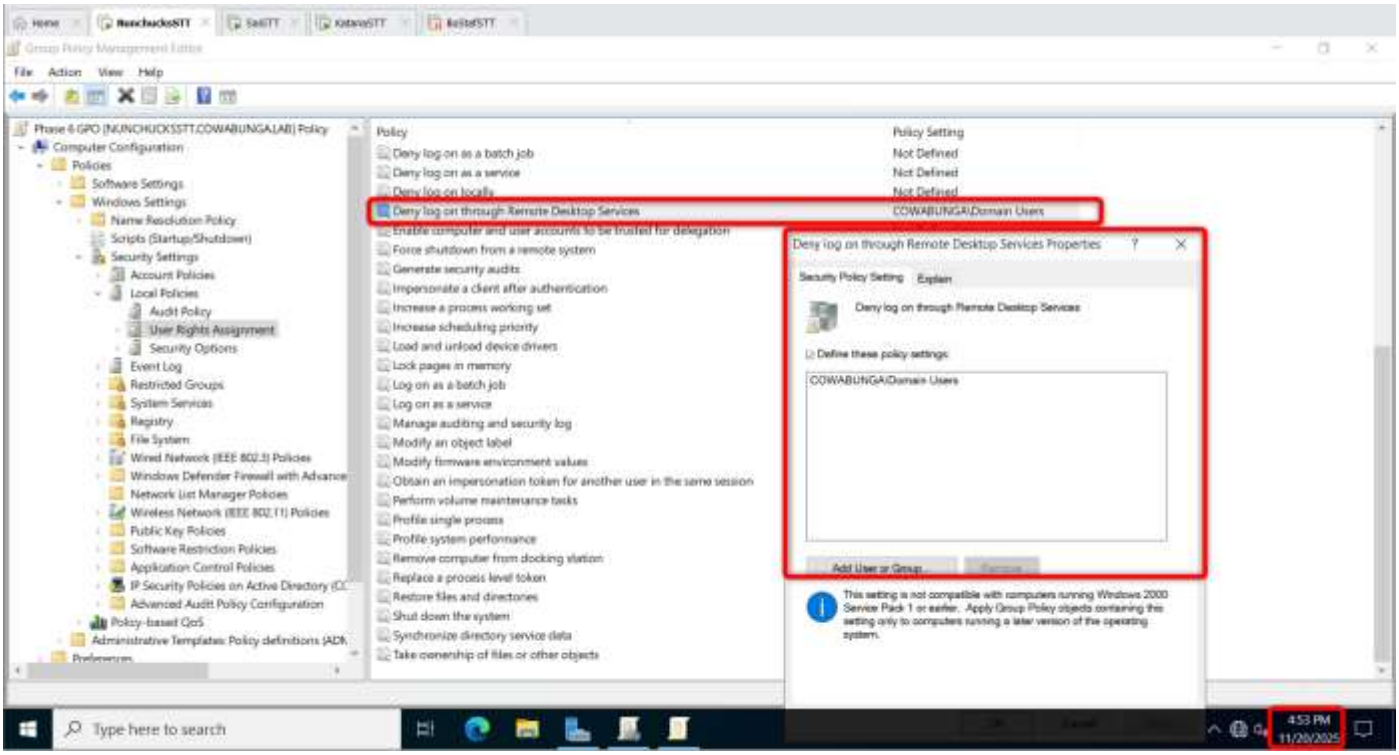
Redirect the documents folder to the File server (the GPO with Folder Redirect)

User Configuration → Policies → Windows Settings → Folder Redirection → Documents



Deny log on through remote desktop services (RDS)

Computer Configuration → Policies → Windows Settings → Security Settings → Local Policies → User Rights Assignment → Deny log on through Remote Desktop Services

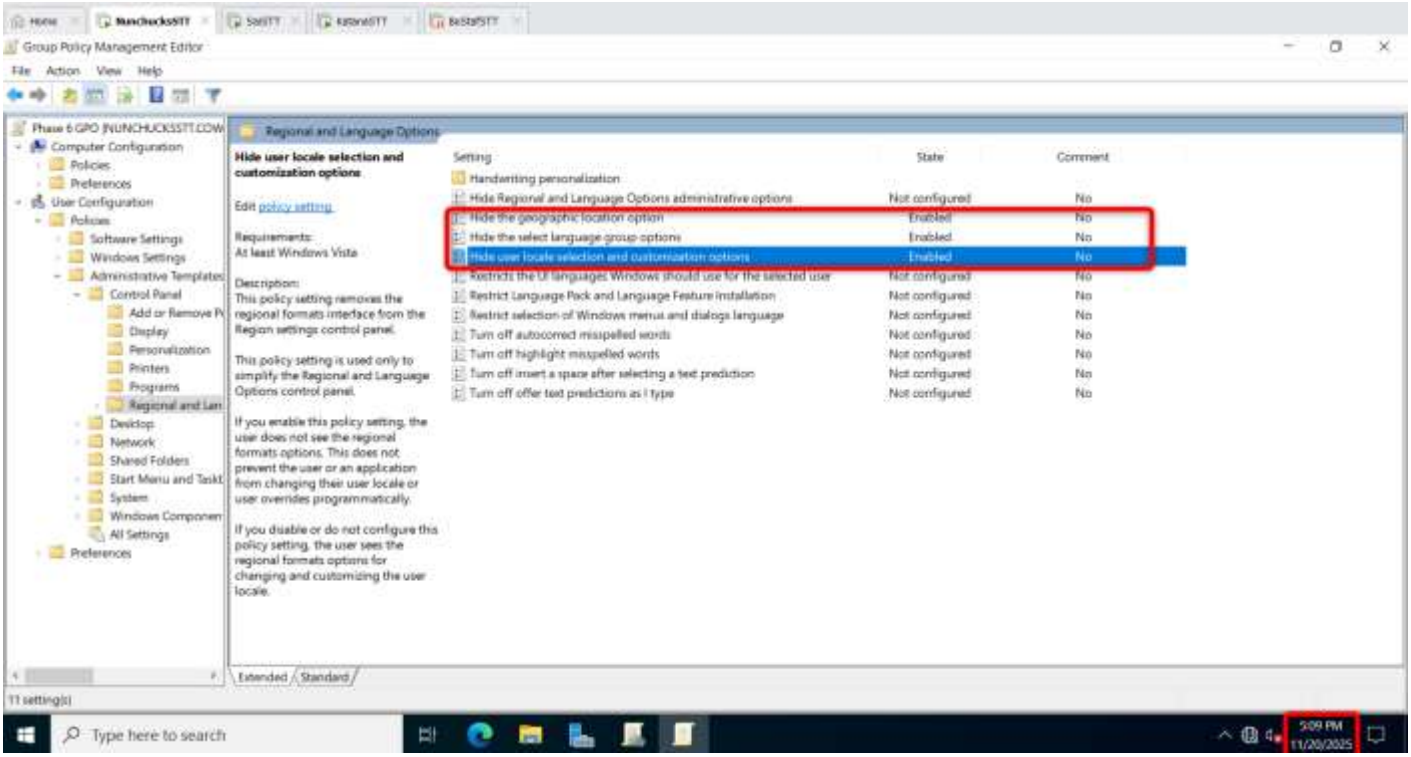


Prevent users from changing their geographic location

User Configuration → Policies → Administrative Templates → Control Panel → Regional and Language Options → Hide the geographic location option

User Configuration → Policies → Administrative Templates → Control Panel → Regional and Language Options → Hide the select language group option

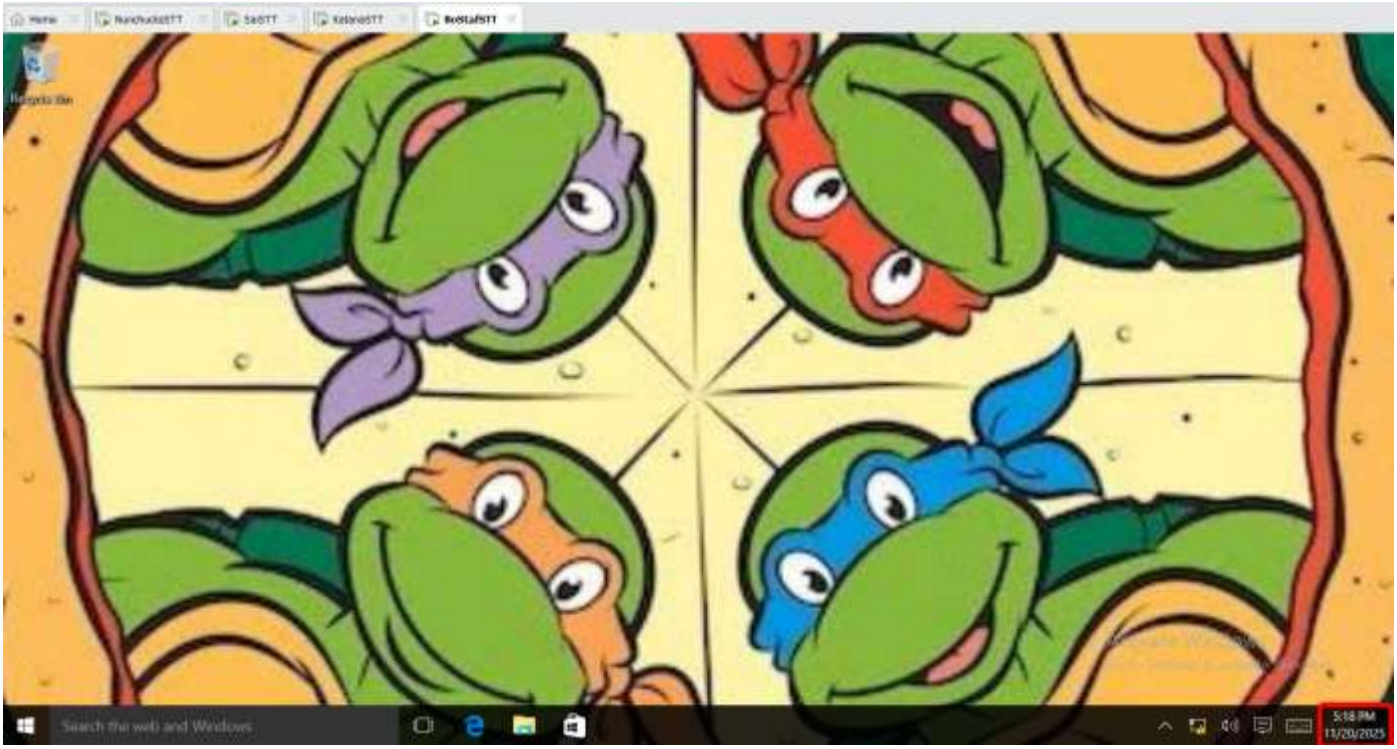
User Configuration → Policies → Administrative Templates → Control Panel → Regional and Language Options → Hide user locale selection and customization option



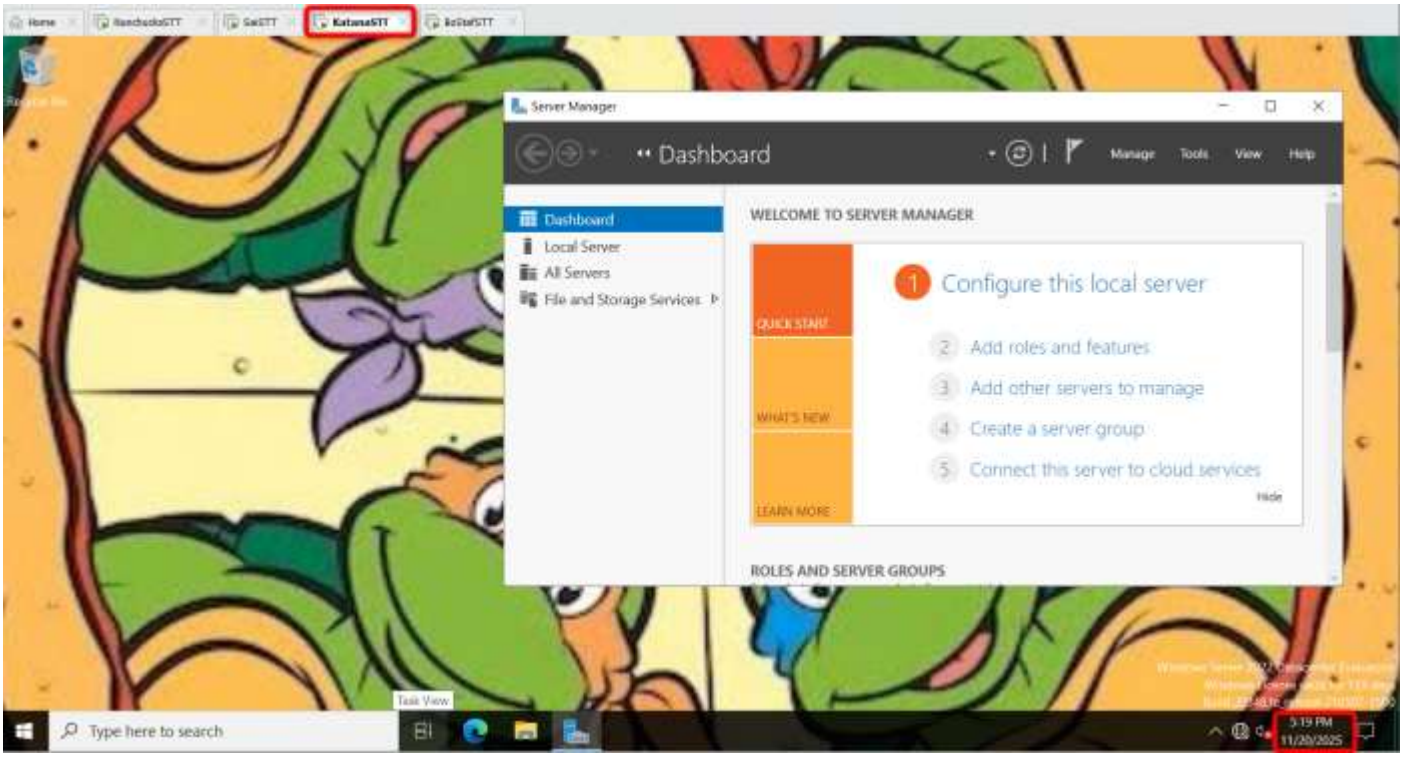
Testing Policies

Change the default background of all computers

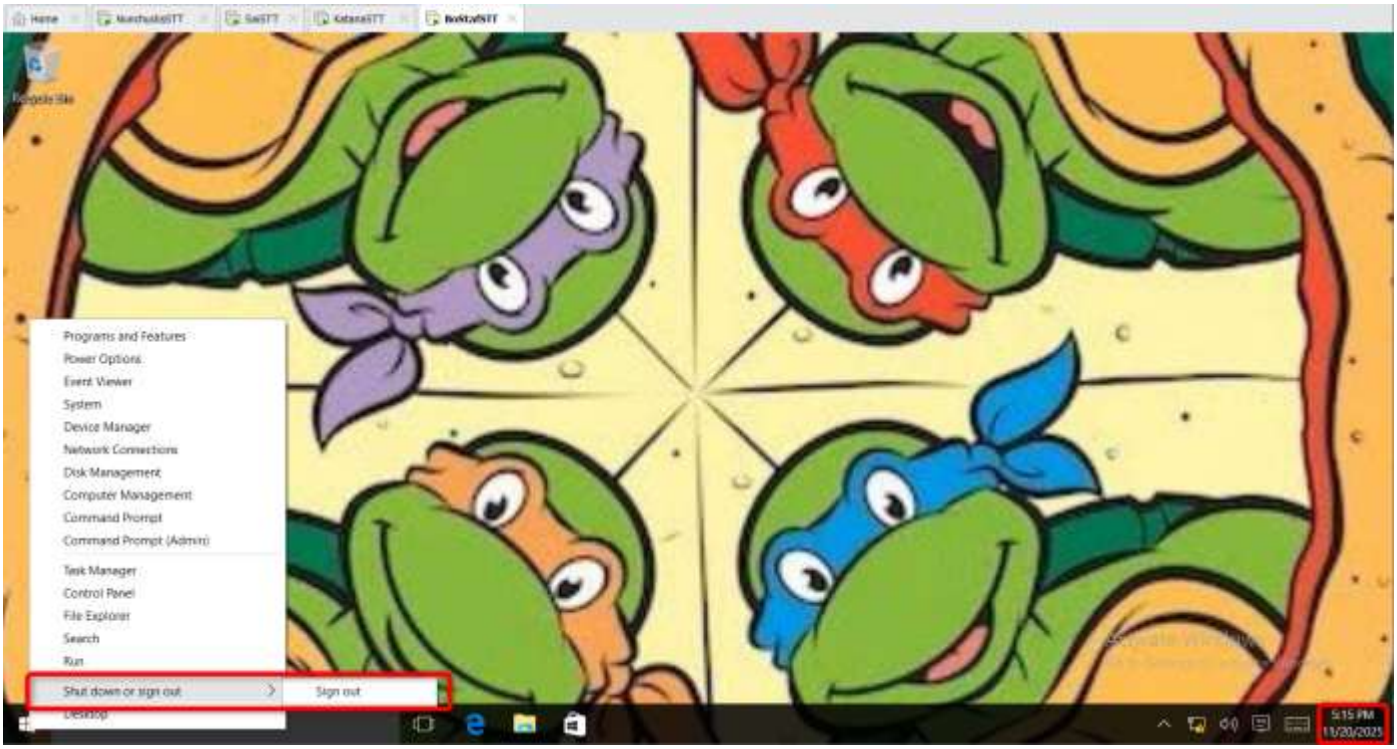
Client machine



Server KatanaSTT

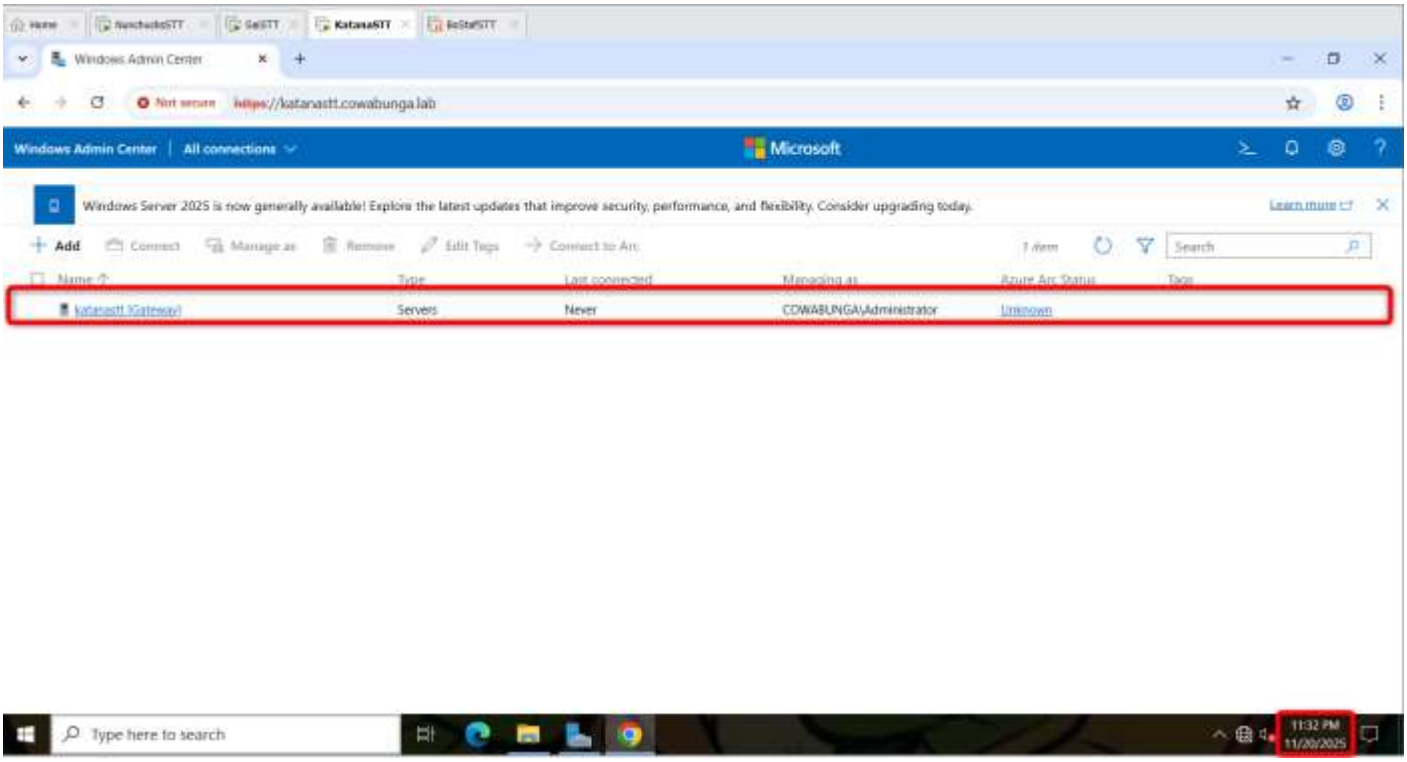


Do not allow users to shut down the system

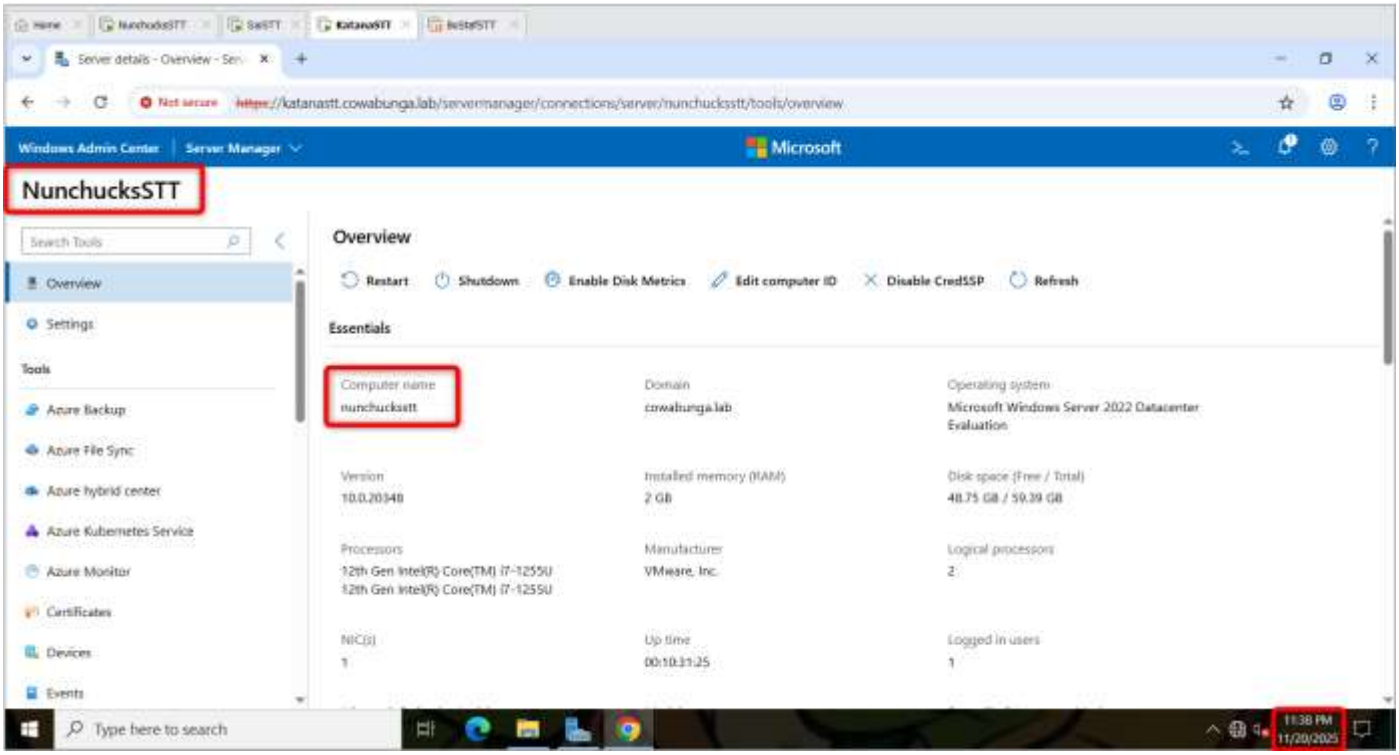


Windows Admin Center

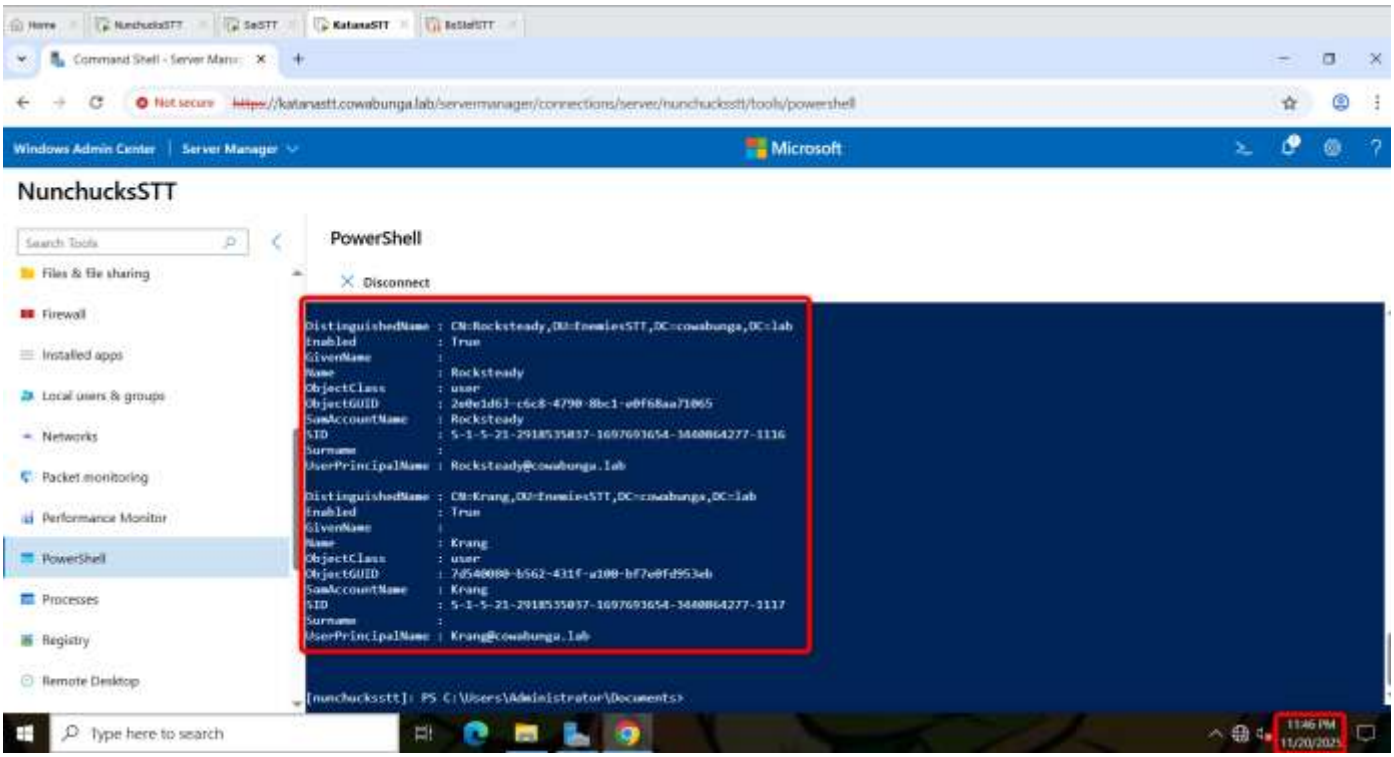
Admin Center Dashboard



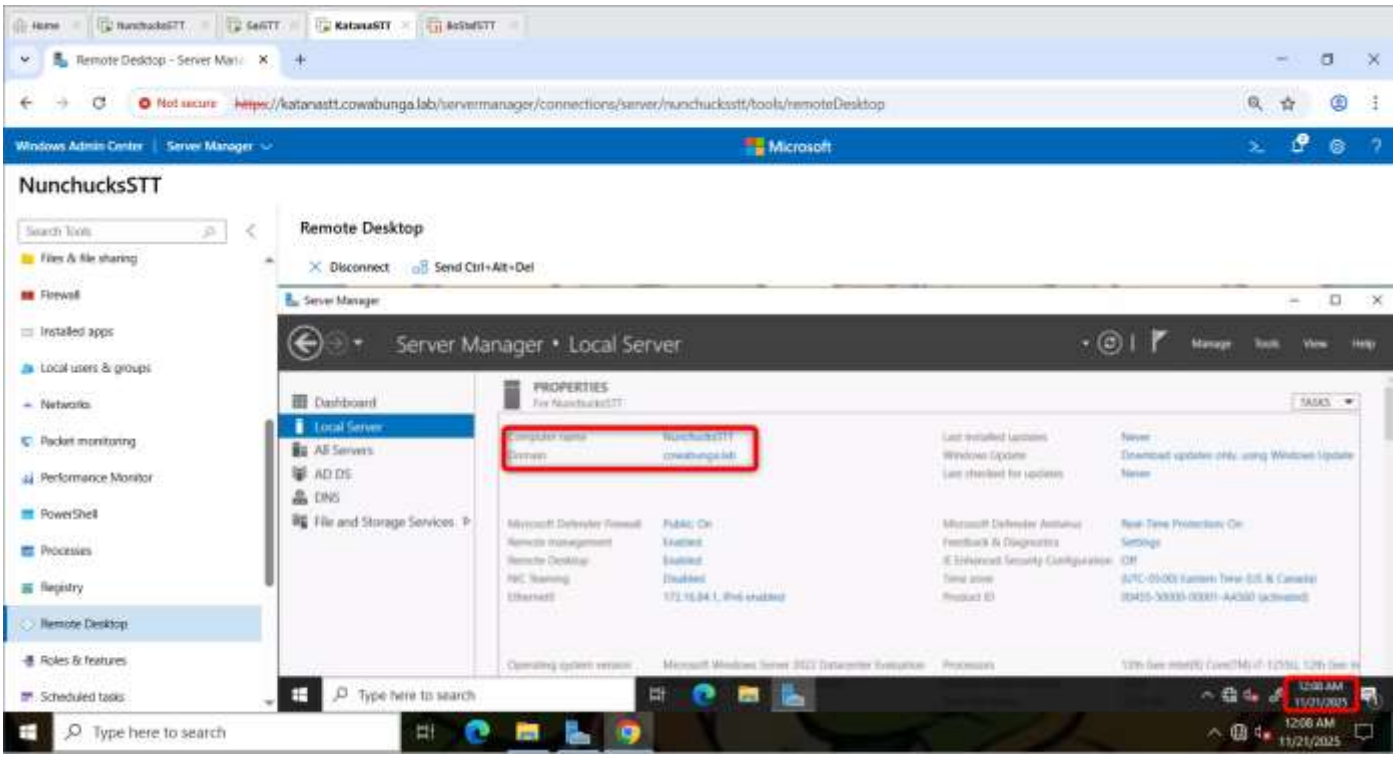
NunchucksSTT (server 1) Dashboard Page (in Admin Center)



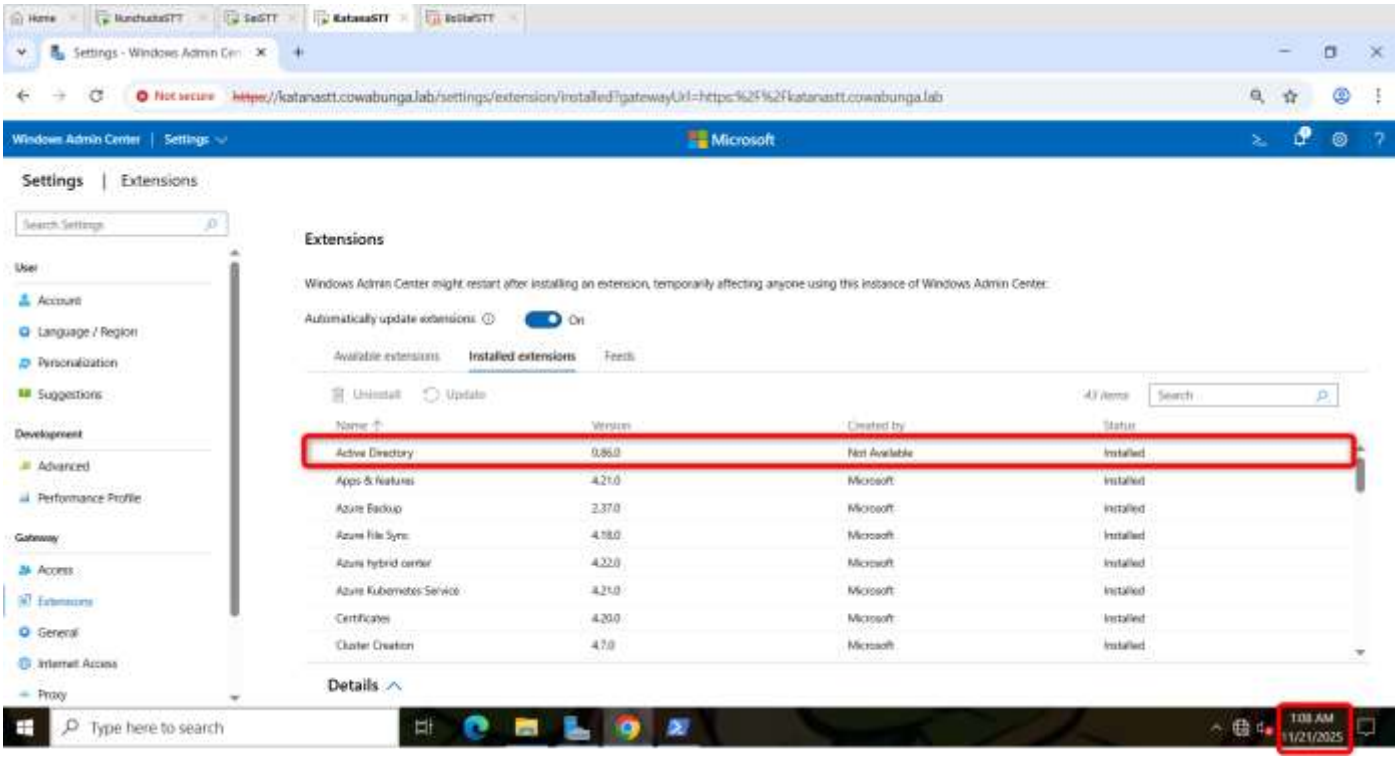
PowerShell command has been run and retrieved ADDS users



Remote Desktop is configured and connected to NunchuksSTT (Server1)

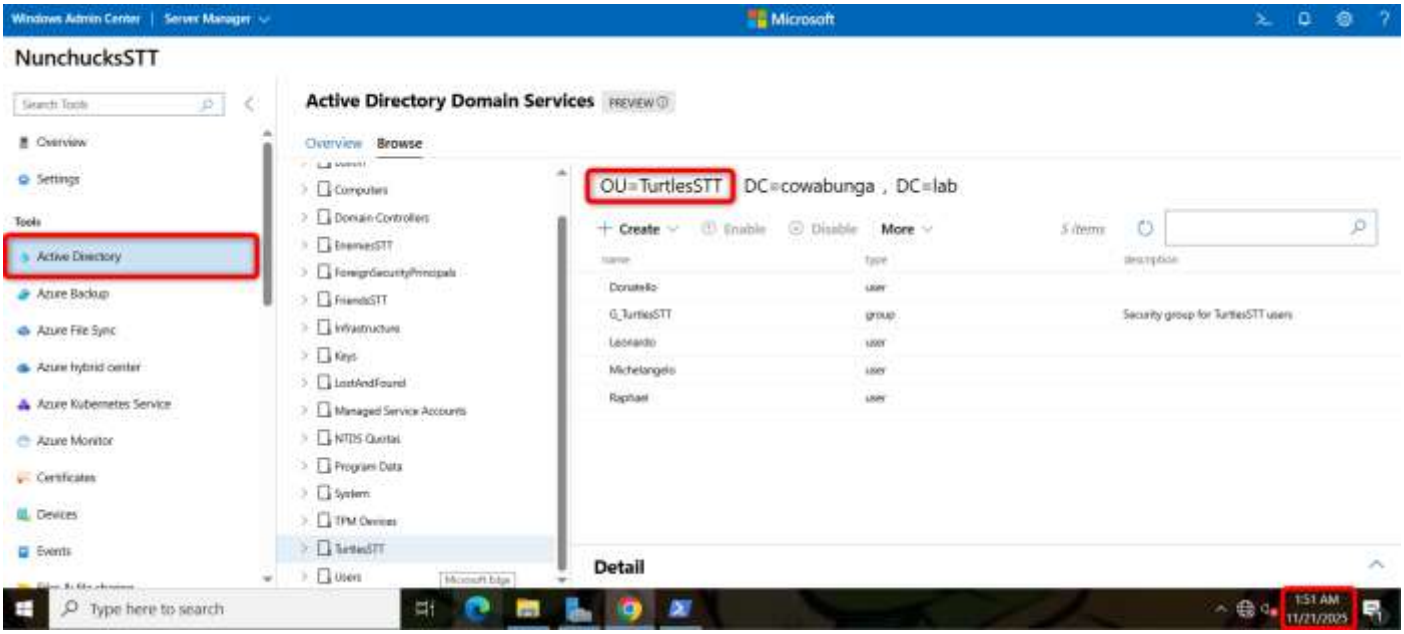


Install an Extension Locally

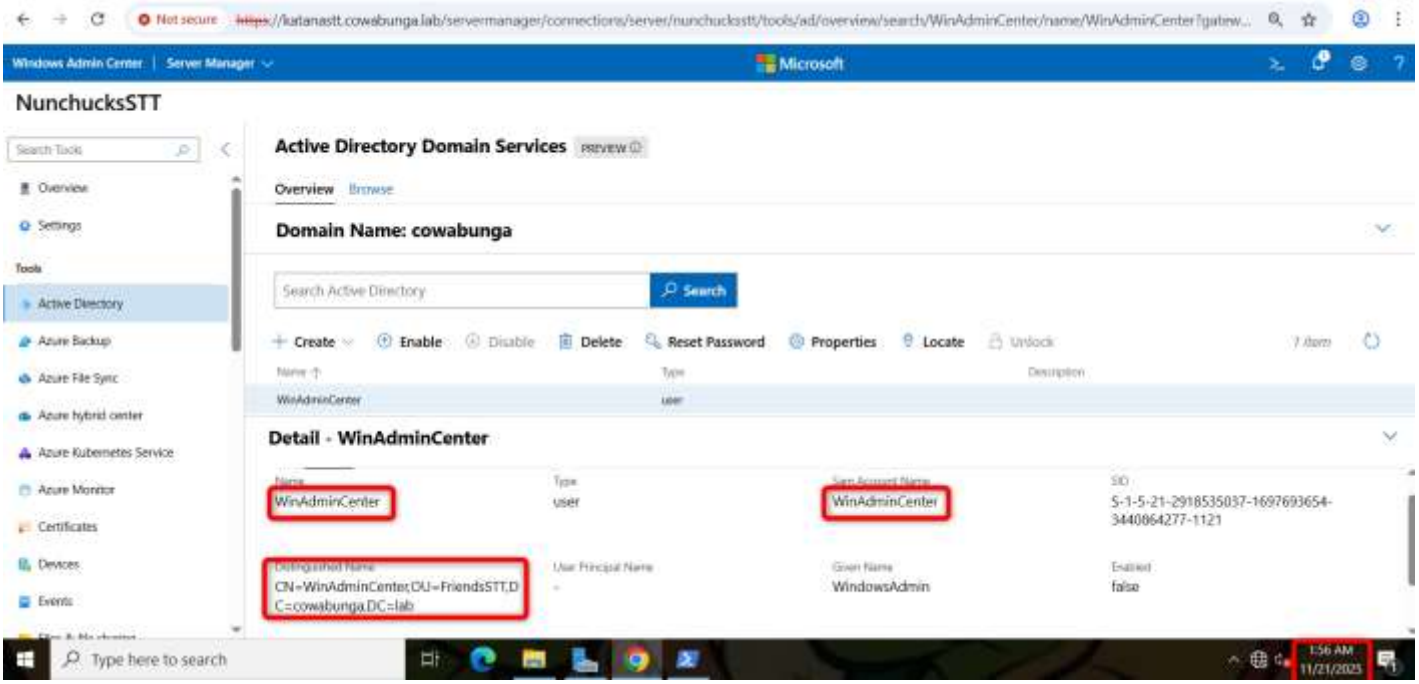


Use the Active Directory Extension

OU



New User in the Overview Page



Lessons Learned

This project helped me understand how to deploy and manage a centralized Windows Server environment for a growing organization. I learned how important it is to configure Active Directory correctly, especially when creating users, groups, and setting up access controls. Working with Group Policies showed me how small settings can affect the whole network, so testing and verifying every change is necessary. I also gained more confidence troubleshooting issues related to authentication, DNS, and workstation communication with the domain. Overall, the project showed me that a stable server environment depends on good planning, clear organization, and careful step-by-step configuration.